

LABUMAT 2 SERVICE MANUAL

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&
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1 Installation and setup

1.1 Site requirements

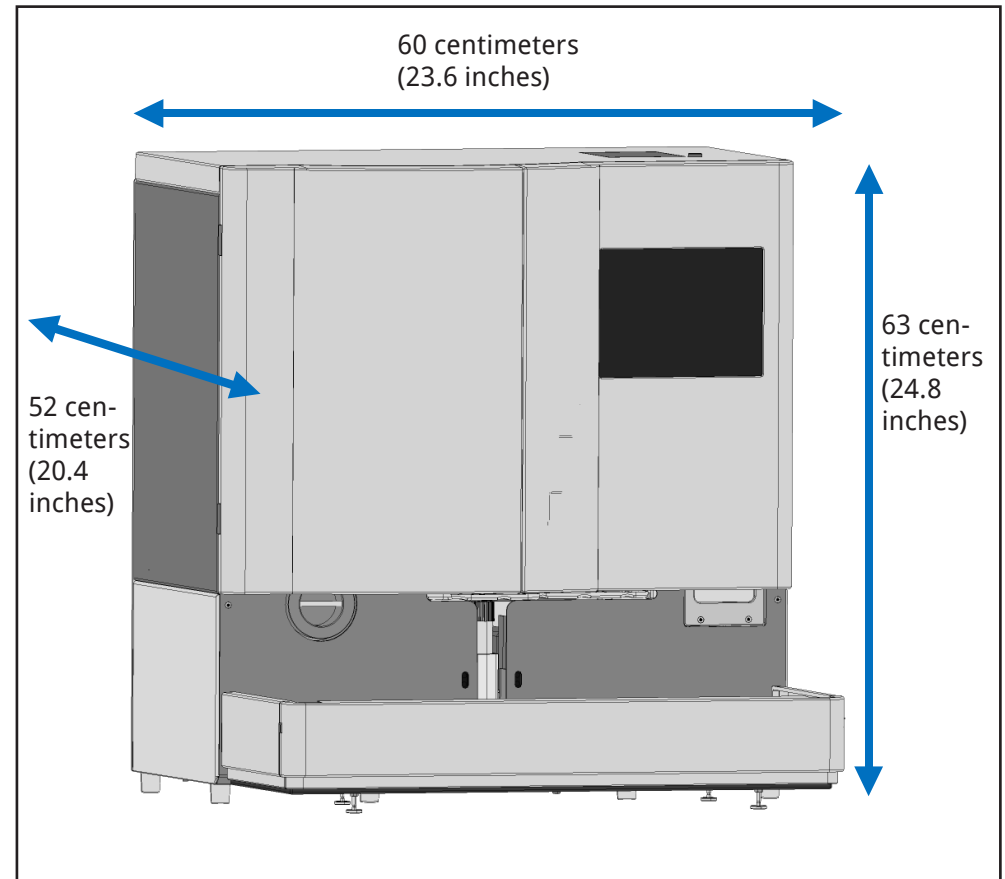
⚠ The device must only be used indoors.

- Install and operate the device on a solid, level surface that can reliably bear the weight of the device: 50 kilograms (110 pounds).
 - Do not install or operate the device in temperatures below 10°C (50°F) or above 35°C (95°F) or in environments with relative humidity levels below 20% or above 85%.
 - Do not install or operate the device in environments with sources of vibration. Vibrations may interfere with smooth rack movement, proper pipetting, and test strip pad reading, and reduce the useful life of the whole system.
 - During operation, do not expose the reading head to intense light such as direct sunlight.
 - The mains requirements of the device are AC 100–250 V at 50–60 Hz.
- The device has a grounded CEE 7/7 power plug.

⚠ Do not operate the device in close proximity to sources of intense electromagnetic radiation (such as an unshielded intentional radio frequency source).

1.2 Clearance limits

⚠ Make sure that the level of the table you install the de-



vice on is appropriate for the operators. The elements that the operator interacts with (the standby and power switches, the touch screen, the rack conveyor, the feeding tube, the unused strip bin, and the strip waste bin) should be easy and comfortable to reach.

⚠ Make sure that there is enough room behind the device for the tubes of the liquid tanks and the easy operation of the power switch. The recommended safety clearance is 20

centimeters (7.8 inches) so that the rigid tubes of the liquid tanks are not strained.

⚠ Make sure that there is enough room around the stand-by switch on the side of the device so that the switch can be reached and operated easily and safely.

1.3 Unpacking the device

⚠ LabUMat 2 is shipped in a custom wooden crate. Always wear protective gloves when handling the crate and its elements.

1 Cut the straps that hold the package together. Remove the lid (①) and the padding, and lift off the folding wooden cover (②).

2 Remove the lid of the cardboard packing sheath (③), the padding around the main unit inside, and the other wrappers. Lift off the cardboard packing sheath (④).

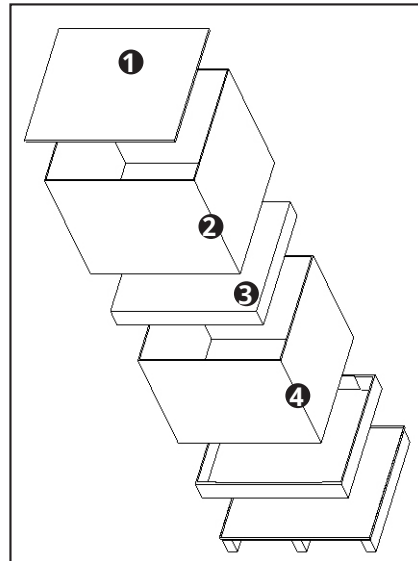
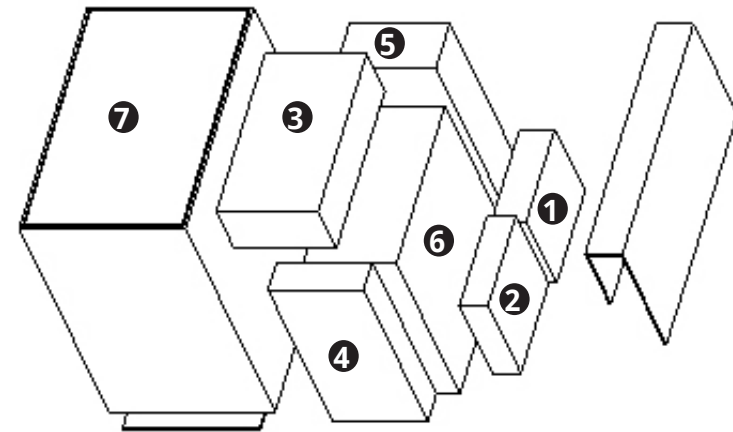
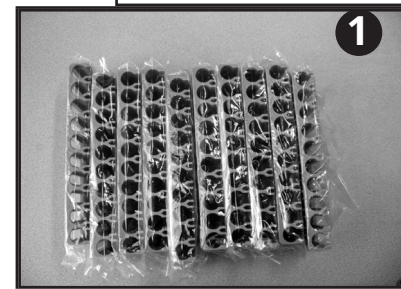
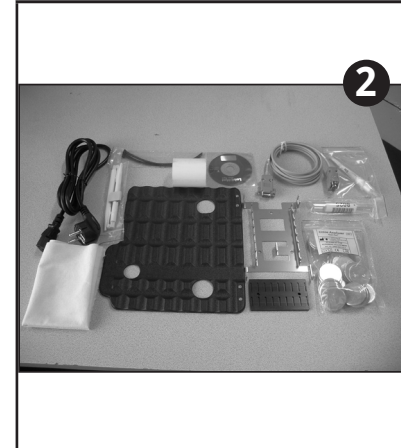


Figure 1: The LabUMat 2 delivery

3 Unpack the boxes inside the cardboard packing sheath in the order specified. Check that you have received all the items, that none of the items is damaged, and that the types and serial numbers match the following packing list. In case of any discrepancy contact 77Elektronika.

LabUMat 2 packing list



	1 Ten (10) racks with a capacity of ten (10) test tubes each [UAZ-4439-3]
	2 - power cable [352HAL01] - microfiber cleaning wipe [90494TORK] - touchscreen stylus [61100001] - roll of thermal printer paper [612LT245] - LabUMat 2 user's manual on CD [UPA-9250-1] - comb conveyor [UPA-033511-1] - pipetting stage [UPA-251159-2] - leveling disks [URM-980133-1] - water filter [55FV2B10] - cable ties [3C100002] - serial UriSed-to-LabUMat interface cable [ALT-466001-1] 12-milliliter graduated sample test tube [9E110X17] - drainboard [UPA-051177-2]

LabUMat 2 packing list

	3 LabUMat 2-to-UriSed rack conveyor connection element [UPA-440600-1] LabUMat 2-to-UriSed spreader plate [UPA-441900-2] rack conveyor side panel [UPA-441414-1]
	4 One hundred (100) 12-milliliter test tubes [9E110X17] test tube stopper caps [9E065793]
	5 LabUMat 2 rack conveyor [UPA-441400-2]
	6 5-liter waste liquid tank [URS-447700-1] 5-liter distilled water tank [URS-447800-1] waste line tube (DIR1) [URS-1722-1] gravity line tube (GRA1) [URS-1721-1] distilled water tube (W3) [URS-1723-1] liquid tank stand [URS-3004-1]

LabUMat 2 packing list

	LabUMat 2 main unit [UPA-810100-4] ⚠ Check that the retainer plate holding the pipetting probe is in place.
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i It is recommended that you save the boxes for future use.

1.4 Installing the device



Figure 2: An assembled LabUMat 2 ready for use

1.4.1 Installing the main unit

⚠ LabUMat 2 must be set up and operated on a table with a structural load capacity of at least 100 kilograms (220 pounds).

⚠ It takes two adult men to lift the LabUMat 2 main unit onto the table.

- 1 Position the table intended to support the device in your preferred location. Keep in mind the [1.1 Site requirements](#) and the final layout as illustrated in [Figure 2](#).
- 2 With the help of at least one of your colleagues, place the main unit on the table.
- 3 Open the doors of the device and unscrew the bolt securing the pipetting probe retainer plate (See Figure 3). Open the top cover of the device and screw the bolt into the device chassis crossbeam (it is the screw on the left in [Figure 4](#)). Save the retainer plate.

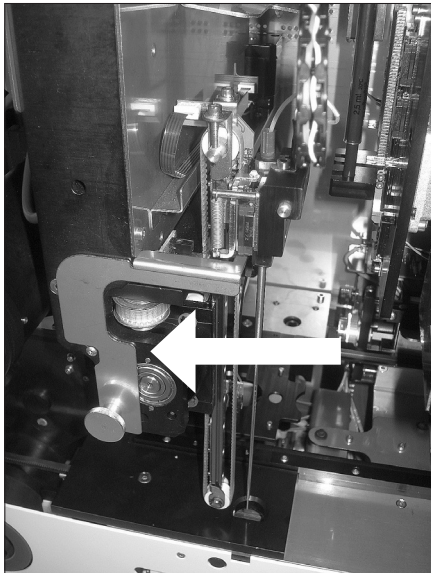
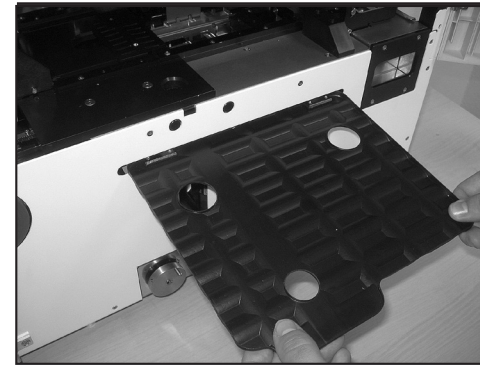


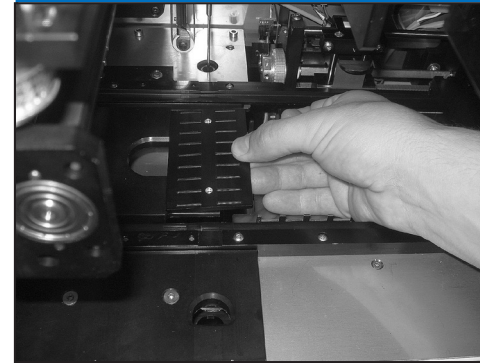
Figure 3: The initial location of the pipetting probe retainer plate

- 4 Push the linear robot all the way to the back of the device and insert the drainboard, the pipetting stage, and the comb conveyor:

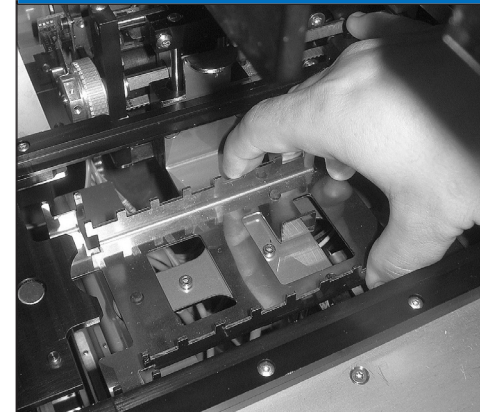


When inserting the drainboard, make sure that the magnetic catches face up and that the two larger holes in the board are towards the back of the device.

Push the board until only its handle in the bottom left corner is sticking out of the device.



When inserting the pipetting stage, make sure that its two holes align with the pins on the platform frame and that the magnetic connection makes it sit tightly in place.



When inserting the comb conveyor, make sure that its holes align with the three rubber pins in the comb conveyor or bottom plate. Two of the pins are to the right- and one is to the left-hand side if you are facing the device.

5 Load the roll of paper into the built-in printer. Position the loose end of the roll so that it sits towards the printer head, not towards the rear of the device. Let a few centimeters (about an inch) of paper hang out of the paper compartment.

6 See [Adjusting the linear robot alignment](#) to adjust the pipetting probe alignment and parallelism.

7 Secure the pipetting probe retainer plate you removed in step 3 to the top crossbeam of the device chassis (see Figure 4) and complete the steps in [1.4.2](#).

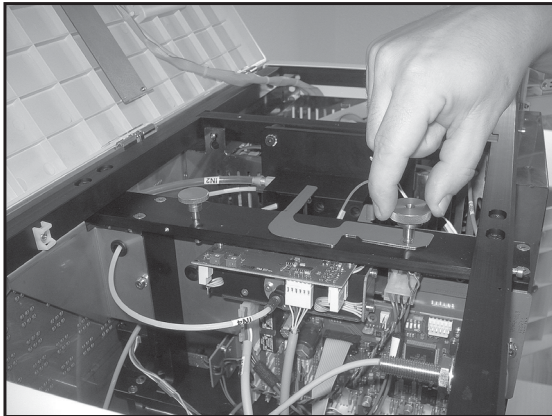


Figure 4: Securing the pipetting probe retainer plate for later re-use

1.4.2 Installing the liquid tanks

1 Place the liquid tank stand on a level surface below the device, preferably on the floor.

2 Fill up the tank labeled 'Wash' with distilled water, and set the liquid tanks into the bowls on the tank stand.

3 Unscrew the flexible tube guide on the cap of the distilled water tank, lead the distilled water tube (W3) through the tube guide, the opening on the tank cap, and the lead clip above the float switch rod (See Figure 5).

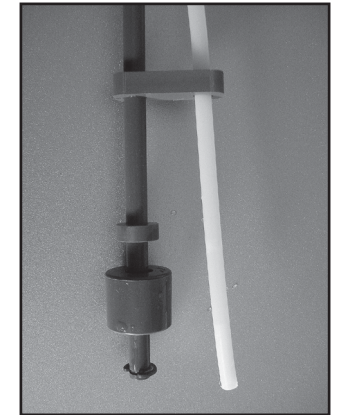


Figure 5: Proper arrangement of the liquid tubes

⚠ Make sure that the end of the tube extends beyond the float switch rod (See Figure 5).

i Push the tube into the tank so that the 'Push until' label on the tube is exactly above the flexible tube guide on the cap of the tank.

4 Repeat the previous step using the two other tubes that came packed with the liquid tanks and the tank labeled 'Waste'.

⚠ The tube labeled 'GRA1' goes into the smaller-diameter opening on the 'Waste' tank.

5 Attach the water filter that came packed in box 2 (see the packing list in [1.3 Unpacking the device](#)) to the end of the distilled water tube. Make sure that the arrow indicating the direction of water-flow points towards the device.

6 Push all three liquid tubes into their respective sockets on the back of the device and connect the 9-pin D-sub plugs of the float switches. Complete the steps in [1.4.3](#).

⚠ Make sure that when the liquid tanks are in their final location, the gravity line tube is dropping all the way to the Waste tank (see Figure 6). Avoid having the gravity tube double back upon itself.

i If you have shortened your liquid tubes, and you need longer liquid tubes later on, for example because you are moving the device into a new location, contact 77Elektronika for new, original-length liquid tubes .



Figure 6: Proper arrangement of the gravity line

1.4.3 Installing the rack conveyor

1 Unscrew the four leveler screws at the bottom of the rack conveyor, then unscrew the two (2) Phillips head screws that secure the rack conveyor's bottom casing panel. Screw the side panel onto the rack conveyor with the two

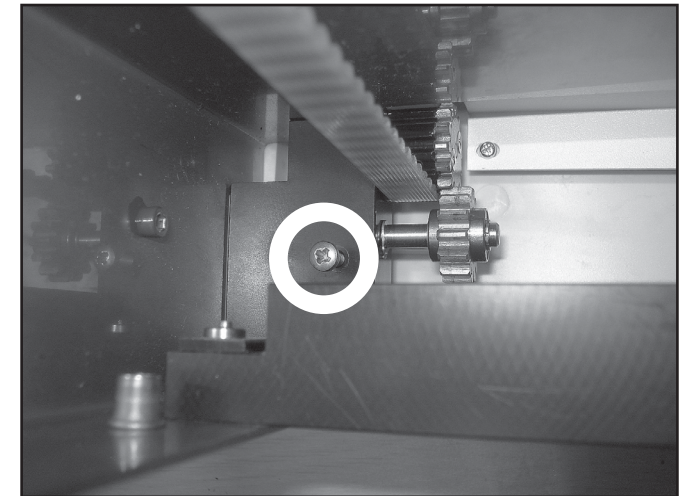


Figure 7: One of the screws securing the side panel to the rack conveyor

(2) Phillips screws from the inside. You will need to hold the rack conveyor upright to access the screw holes under the gear belt (see Figure 7).

2 Snap the assembled rack conveyor to the front of the device. Make sure that the rack conveyor is flush with the front casing panel of the device (see Figure 9). Place a test tube rack on the conveyor belts in the position where it is drawn into the device during normal operation (see Figure 8).

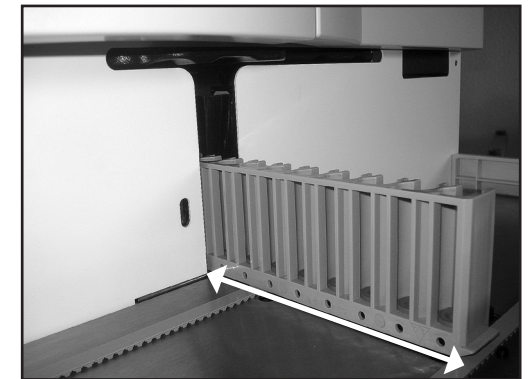


Figure 8: Proper test tube rack-device alignment

3 Adjust the leveler screws at the bottom of the rack conveyor so that the test tube rack sits level with the rack positioning unit bottom plate (see Figure 9). The test tube rack should be able to be drawn into and pushed out of the device without any tilting or jerking. Complete the steps in [1.4.4](#).



Figure 9: The proper position of the rack conveyor

1.4.4 Adjusting the position of the strip clamps

1 To check how the strip clamps interact with the test strip on the pipetting stage, remove the top section of the side casing panel of

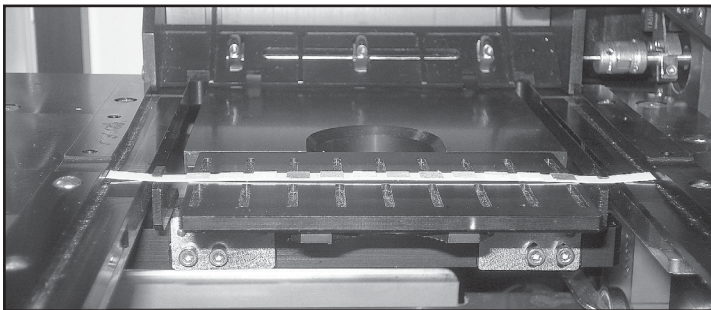


Figure 10: Properly adjusted strip clamps with a test strip over the pipetting stage

the device on the side of the standby switch (see [Removing the top side casing panels](#)).

2 Place a LabStrip U11 Plus GL test strip in the path of the strip clamps and position the test strip across the raised center pipetting stage by manually moving the strip clamps. Looking in from the side of the device, check whether the strip clamps grasp firmly the handles on either end of the test strip (see Figure 10). Gently tug at the test strip: the clamps should be tight enough not to let the test strip move out of position, but should not be so tight that the test strip cannot be dropped onto the comb conveyor.

i The test strip may even bulge up slightly in the middle as the clamps press down on it on either end.

3 If the strip clamps are either too slack or too tight, remove the test strip and the pipetting stage. Loosen the pair of 2-millimeter hex socket head set screws (that you can also see in Figure 10) that secure them to the strip clamp bar, then use the 2.5-millimeter hex socket head spring-loaded leveler

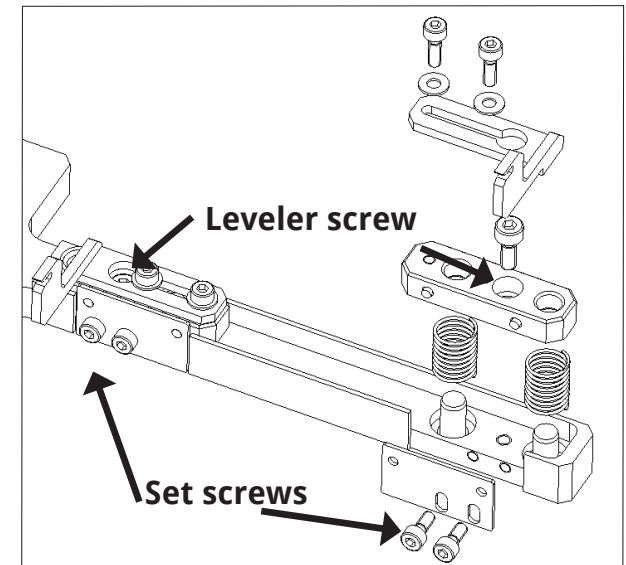


Figure 11: Exploded view of the screws that adjust the strip clamps' vertical position

screws to adjust their height (see Figure 11). Replace the pipetting stage and the test strip and check again. Repeat until the clamp position is optimal.

4 If the strip clamps are positioned properly, complete the steps in [1.4.5](#).

1.4.5 Switching on the device for the first time

1 Make sure that the power cable is properly attached to the device, and plug it into a socket that meets the mains requirements (see [1.1 Site requirements](#)).

2 Toggle the rotary power switch at the back of the device into the on (I) position, and press the standby switch on the side of the device to switch on the analyzer.

3 Complete the steps in [1.5](#).

1.5 Setting up the device

1.5.1 Adjusting the front-back axis alignment of the linear robot horizontal motor

You have already checked and adjusted the right-left axis alignment of the vertical robot in [Adjusting the linear robot alignment](#). You will also need to check and, if necessary, adjust the vertical robot's front-back axis alignment with the rinse well and the test tube apertures. Correct alignment is determined by the rinse well position sensor and the test tube position sensor.

1 Launch Systest (see [2.2.1](#)).

2 Tap **Initialization** on the **Flow tests** menu. Under **Card tests** on the **Robot card** tab, tap **High probe position** in the **Linear robot vertical motor** row.

3 Tap **Test tube position** in the **Linear robot horizontal motor** row. Sink the aligner screw into the test tube aperture (see [Adjusting the linear robot alignment](#)), then tap the **STOP** button in the **Linear robot vertical motor** row but not the **Linear robot horizontal motor** row.

4 Move the pipetting probe manually by pulling at the vertical robot timing belt and check whether the tip of the probe can slide into the aligner screws central hole without any obstruction. If the probe slides into the hole freely, go to step 7.

5 If there is the slightest misalignment between the central hole and the tip of the probe, tap the **Rinse well position** button to gain access to the sensor that determines the motor position. Loosen the two (2) 2-millimeter hex socket button head screws that secure the Test tube position sensor to the top of the horizontal robot bracket, and adjust the position of the sensor as needed.

6 Repeat steps 3–5 as many times as necessary.

7 See [“4.2.1 Removing the back casing panel”](#) and remove the back casing panel. In the Systest software under **Card tests** on the Robot card tab, tap the **High probe position** button in the **Linear robot vertical motor** row.

8 Tap the **Rinse well position** button in the **Linear robot horizontal motor** row. Sink the aligner screw into the rinse well aperture (see [Adjusting the linear robot alignment](#)), then tap the **STOP** button in the **Linear robot vertical motor** row but not the **Linear robot horizontal motor** row.

9 Move the pipetting probe manually by pulling at the vertical robot timing belt and check whether the tip of the probe can slide into the aligner screw’s central hole without any obstruction. If the probe slides into the hole freely, go to step 12.

10 If there is the slightest misalignment between the central hole and the tip of the probe, tap the **Test tube position** button to gain access to the sensor that determines the motor position. Loosen the two (2) 2-millimeter hex socket button head screws that secure the Rinse well position sensor to the top of the horizontal robot bracket, and

adjust the position of the sensor as needed.

11 Repeat steps 8–10 as many times as necessary.

12 Tap **Initialization** on the **Flow tests** menu and complete the steps in [1.5.2](#).

1.5.2 Adjusting the pipetting position

1 Complete the steps in [4.5.4 Setting up the correct pipetting position](#).

2 Complete the steps in [1.5.3](#).

1.5.3 Adjusting the pad-reading position

The optimal position for the reading head to measure the discoloration of a test strip pad is when the LEDs light up directly above the test zone of the pad. Setting up this optimal position has two components: One is along the X (left-right) axis, adjusting the comb conveyor motor position in which the conveyor deposits the test strips below the reading head. The other is along the Y (fore-aft) axis, fine-tuning the positions of the photometer itself. You can adjust both these positions as well as the photometer reference position in the Systest software under **Position setup**.

1 Launch Systest (see [2.2.1](#)).

- 2 Tap **Initialization** on the **Flow tests** menu.
- 3 Open the device doors. Unscrew the four (4) 2.5-millimeter hex socket head screws that secure the protective plate over the reading head bracket.
- 4 Place a LabStrip U11 Plus GL test strip to the right of the crossbeam of the platform support frame (as viewed from the front of the device), on the left edge of the strip reading platform (see Figure 12).

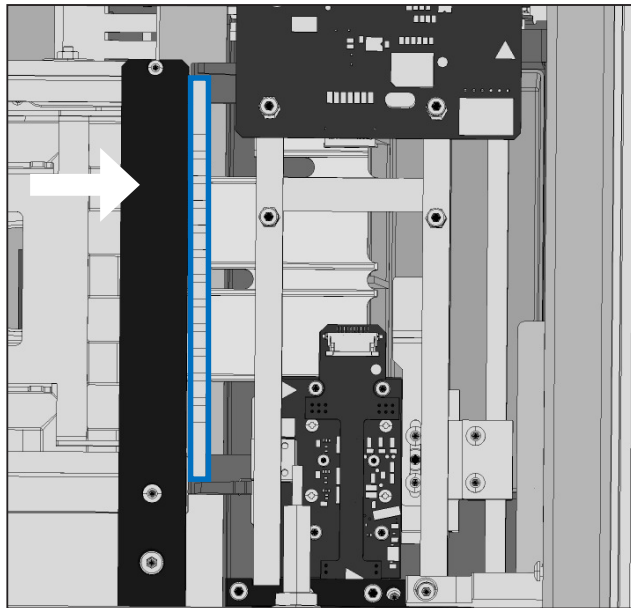


Figure 12: Partial overhead view cut-away of the device. The platform support frame crossbeam is marked with an arrow. The test strip in place for the adjustment is indicated in blue.

- 5 In the Systest software under **Position setup** on the **Head strip position** tab, tap **Position setup** to start the comb conveyor position setup process. This moves the Horizontal strip timer motor into its **Pull to head** motor position, which moves the test strip below the reading head. The reading head is simultaneously moved above the Bilirubin pad of the test strip and one of its LEDs lights up.
- 6 If the light of the LED does not fall in the center of the Bilirubin pad, tap the **Left** or **Right** buttons in the **Head pos. correction** row to modify the Horizontal strip timer motor's **Pull to head** motor position away from or towards the reading head, respectively.

i The comb conveyor's prong can push but not pull the test strip. Because of this, when you tap the **Left** button to adjust the position away from the reading head, you will not be able to assess the results of your adjustment directly. In this case, you will need to repeat steps 3–5 to check that there is no need for further adjustment.

- 7 Look in at the test strip pad under the lit-up LED from the side of the device to check the front-back axis alignment. If the light of the LED does not fall in the center of the Bilirubin pad, tap the **Front** or **Back** buttons in the **Strip start position** row to adjust the reading head position over the test strip pad towards the front or the rear of the device respectively, one motor step at a time.

- 8 Tap the **Store the position** button to store the adjusted values. The system does not confirm successful storage of the adjustments directly but displays a confirmation message under **Messages** (see 2.2.3).
- 9 Tap the **Initialization** button to move the comb conveyor and the reading head back to their home positions. The test strip under

the reading head will be dropped into the Strip bin.

10 To start setting up the optimal reading head position above the white reference pad, tap the **Head reference position** button.

11 Tap the **Position setup** button to move the reading head above the white reference pad towards the back of the device. One of the LEDs of the reading head will also light up.

12 If the light of the LED does not fall in the center of the white reference pad, tap the **Front** or **Back** buttons to adjust the reading head position towards the front or the rear of the device respectively, one motor step at a time.

13 Tap the **Store the position** button to store the adjusted value, then **Initialization** to end the motor position setup process. Complete the steps in [1.5.4](#).

1.5.4 Adjusting the rack catch corrections

The rack catch is a slider with a bent end that enables the Rack In/Out module to draw in racks from the rack conveyor and push them back out

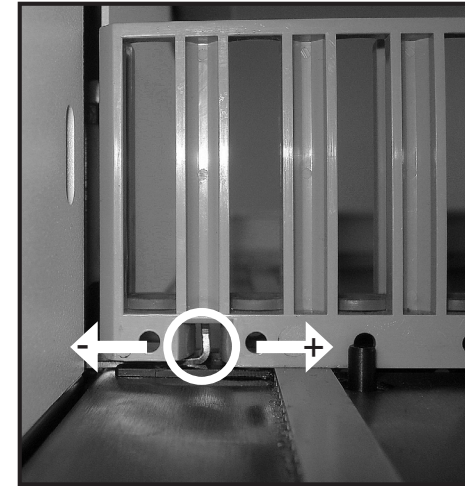


Figure 13: The directions of rack catch correction values

again after sample aspiration. The movement of the racks as they are pushed and pulled should be as smooth as possible. In order that the racks are not jerked and that they do not get jammed, the rack catch is moved a few motor steps each time after pushing out a rack, then again to prepare to draw in the next rack. These short movements are called **Used rack correction** and **New rack correction** respectively, and you can adjust their value if necessary in the Systest software on the **System parameter setup** menu on the **Rack card** tab.

1 Place a rack with some empty test tubes on the arrival (right) side of the rack conveyor.

2 Launch Systest (see [2.2.1](#)), and tap **Initialization** on the **Flow tests** menu.

3 On the **Flow tests** menu, tap the **Rack change test** button. The rack will be drawn into the device and will be stopped when the first test tube in the rack reaches the test tube aperture. Check whether the rack's movements are free of any bumps and jerks.

4 If the rack catch seemed to tug at the rack sharply, tap the **Read parameters**, then the **New rack correction** button on the **Rack card** tab on the **System parameter setup** menu in the Systest software. Adjust the value with the **Increment(+)** or **Decrement(-)** button as necessary (see Figure 13).

 **One or two motor steps in either direction should be enough to fix the problem.**

5 Repeat steps 1–3 to see whether any more adjustment is necessary. Tap the **Save parameters** button to store the adjusted value, then tap **Initialization** on the **Flow tests** menu.

 **6** On the **Flow tests** menu, tap the **Push rack out** button. The rack will be removed from inside the device and the rack conveyor will move the rack along towards the left. Check whether the rack's movements are free of any bumps and jerks.

7 If the rack's movement was uneven, tap the **Read parameters**, then **Used rack correction** button on the **Rack card** tab on the **System parameter setup** menu in the Systest software. Adjust the value with the **Increment (+)** or **Decrement (-)** button as necessary (see Figure 13).

 **One or two motor steps in either direction should be enough to fix the problem.**

8 Repeat steps 6–7 to see whether any more adjustment is necessary. Tap **Save parameters** to store the adjusted values, then tap **Initialization** on the **Flow tests** menu, and complete the steps in [1.5.5](#).

1.5.5 Setting up the Flipped strip sensor

During normal operation, the Flipped strip sensor (see [2.3.2 The Sensor tests menu](#)) detects whether the test strip groove on the feeder drum has caught a test strip and, if yes, whether the pads on the strip are facing up (see [Flipped strip sensor](#)).

To do this, the sensor has a lower and a higher cutoff value for the amount of light it detects, and three states:

- Below the lower cutoff value, there is no test strip below the sensor and it picks up the light reflected from the gray surface of the feeder drum.
- Between the lower and the higher cutoff value, there is a test strip under the sensor and the pads on it are facing up—the sensor is picking up the light reflected from the blue surface of the Ascorbic acid pad on the test strip.
- Above the higher cutoff value, the test strip below the sensor is facing the wrong way up—the sensor is picking up the light reflected from the white surface of the test strip carrier ribbon

1 In the Systest software under **Card tests** on the **Feeder card** tab, tap **STOP** in the **Feeder motor** row.

2 Open the device doors, undo the buckle on the feeder bracket and open the feeder bracket.

3 In the Systest software, enter the **Sensor tests** screen and check that the **Flipped strip sensor** is inactive (green) and that the real-time value next to it is below 120.

4 Place a test strip with the proper orientation (with the pads facing up and with the dark blue ASC pad towards the front of the device) into the test strip groove. Rotate the feeder drum manually so that the test strip groove is moved below the Flipped strip sensor (see [Figure 20](#)). In the Systest software, enter the **Sensor tests** screen and check that the **Flipped strip sensor** is active (red) and that the real-time value next to it is between 120 and 200.

5 Flip the test strip in the test strip groove so that the white carrier ribbon is facing up and once again rotate the test strip groove below the Flipped strip sensor. In the Systest software, enter the **Sensor tests** screen and check that the **Flipped strip sensor** is active and that the real-time value next to it is above 200.

6 If the Flipped strip sensor deviates in any way from the standards detailed in the previous steps, contact 77Elektronika technical support. If the sensor is in order, complete the steps in [1.5.6](#)

⚠ Do not modify any of these System parameter values without prior permission from the 77Elektronika technical support team.

1.5.6 Setting up the test tube minimum sample level

This setting indicates the number of motor steps it takes for the Vertical robot to reach the surface of the liquid inside the test tube when the minimum amount of sample is present. The motor steps are counted off from where the 'High probe position sensor' is activated.

⚠ The minimum urine sample level is 2 milliliters if the LabUMat 2 device is used on its own and 3 milliliters if the samples are transferred to a UriSed or UriSed 2 device for further analysis. The following steps assume that you are using LabUMat 2 on its own—if you have a UriSed device connected, substitute three (3) milliliters for the two (2) milliliters of tap water specified.

1 Launch Systest (see [2.2.1](#)), and tap **Initialization** on the **Flow tests** menu.

2 Measure two (2) milliliters of tap water in the graduated test tube supplied with the device, and pour it into a test tube compatible with the device. Place the tube into the first tube position of a rack and place the rack on the arrival (right) side of the rack conveyor.

⚠ Use tap water for this setup procedure. Distilled water is not suitable for the level setup.

3 Tap Minimum level setup in the Systest software under **Position setup**. The pipetting probe will be moved inside the tube as far as the level of the tap water, then moved back inside the rinse well.

i The system displays the number of motor steps it took the Vertical robot to move the pipetting probe from its topmost position down to the surface of the liquid in the tube in a system message under Messages.

4 The minimum sample level that you set in the previous steps will be stored in the Systest software under **System parameter setup** on the **Robot card** tab as **Level limit**.

⚠ Do not modify the Level limit parameter directly.

Your LabUMat 2 device is now ready for use.

INSTALLATION CHECKLIST

Laboratory		Serial number of device	
Service engineer		Date	
Adjustment	Service manual reference	Comments	✓
Linear robot front-back axis alignment	1.5.1		
Adjusting the pi-petting position	1.5.2		
Adjusting the pad-reading position	1.5.3		
Adjusting the rack catch correction	1.5.4		
Setting up the Flipped strip sensor	1.5.5		
Setting up the test tube minimum sample level	1.5.6		

2 Troubleshooting

2.1 Mechanical system messages

 System messages are displayed in the [Messages window](#). Error messages indicate that the device requires user intervention to continue operating. Warning messages indicate minor problems that do not require immediate corrective action.



2.1.1 Mechanical error messages

Message text	Recommended action
No available serial port	See Disabling the thermal printer and open the comDriver.ini file. Find the line PORTNUM under the SERIAL heading, and change it to 2. Re-initialize the device. If the issue persists, contact 77Elektronika service personnel.
Port open error on instrument	See Disabling the thermal printer and open the comDriver.ini file. Check that the line PORT1 under the SERIAL heading is set to 3. Re-initialize the device. If the issue persists, contact 77Elektronika service personnel.
Port open error on external line	See Disabling the thermal printer and open the comDriver.ini file. Check that the line PORT2 under the SERIAL heading is set to 1. Re-initialize the device. If the issue persists, contact 77Elektronika service personnel.
Too many serial ports defined in .ini file	See Disabling the thermal printer to access the comDriver.ini file, and delete it. Re-initialize the device.  You will need to redefine the PRINTER parameter (see Disabling the thermal printer) after deleting the comDriver.ini file.

Message text	Recommended action
No protocol port defined	See Disabling the thermal printer to access the comDriver.ini file, and delete it. Re-initialize the device.  You will need to redefine the PRINTER parameter (see Disabling the thermal printer) after deleting comDriver.ini.
Memory allocation error in port definition	See 12.3.2 Replacing the PC motherboard and replace the industrial PC motherboard.
Driver thread cannot start	See 12.3.2 Replacing the PC motherboard and replace the industrial PC motherboard.
Port thread cannot start	See 12.3.2 Replacing the PC motherboard and replace the industrial PC motherboard.
Cannot create port class	See 12.3.2 Replacing the PC motherboard and replace the industrial PC motherboard.
Barcode reader link error	Launch the Service software, and tap Initialization on the Flow tests menu. If error persists, check the Ethernet cable or see 7.6.12 Replacing the barcode reader .
Photometer communication error	Launch the Service software, and tap Initialization on the Flow tests menu. Check the Photometer FFC (see Figure 63) and replace the photometer assembly (8.6.13) if necessary.
Sediment analyzer connection error	Check the cable connecting the device to the interfaced sediment analyzer. Launch the Service software, and tap Initialization on the Flow tests menu.

Message text	Recommended action
PMC link error	Launch the Service software, and tap Initialization on the Flow tests menu. If the error persists, check the PMC interface board's connection to Circuit board number 3 (Figure 73)
Circuit board number 1 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 2 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 3 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 4 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 5 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 6 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Circuit board number 7 communication error	Launch the Service software, and tap Initialization on the Flow tests menu. See 2.8 Handling PCB communication errors if error persists.
Rinse well position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Test tube position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages

Message text	Recommended action
High probe position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Low probe position sensor error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Aspirating syringe home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Mixing syringe home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Strip turner wheel home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Strip pusher home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Feeder drum home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Feeder drum drop position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Feeder drum upper empty position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Innermost catch position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages

Message text	Recommended action
Outermost catch position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Rack aligner open sensor error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Rack aligner close sensor error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Rack position sensor error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Horizontal strip timer home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Vertical strip timer home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Photometer home position error	In the User software, tap Initialization on the General menu. If the problem persists, see 2.7 Handling sensor-related error messages
Printer head too hot	Stop the printing process in progress. Launch the Service software, and tap Initialization on the Flow tests menu
System initialization failed	Launch the Service software, and tap Initialization on the Flow tests menu
Measurement error: Strip bin not present	The Strip bin was removed during the measurement process. Launch the Service software, and tap Initialization on the Flow tests menu. Repeat the measurement of the sample.

Message text	Recommended action
Interlock error: Door opened during measurement	One or both of the device doors were opened during the measurement process. Launch the Service software, and tap Initialization on the Flow tests menu. Repeat the measurement of the sample.



2.1.2 Mechanical warning messages

Message text	Recommended action
Feeding tube not present	Insert the feeding tube
Unused strip bin not present	Insert the unused strip bin
Feeder empty	Load test strips into the feeder tube
Water tank warning	Refill the distilled water tank
Water tank empty	The device will not respond until the distilled water tank is refilled.
Waste tank warning	Empty the waste tank
Waste tank full	The device will not respond until the waste tank is emptied
Rack conveyor at maximum capacity	Clear the rack conveyor
No rack	Place a rack with test tubes on the rack conveyor
Left door open	Close the left door
Right door open	Close the right door
Strip bin is out	Insert the strip bin
Printer out of paper	Load paper into the thermal printer
Printer cover open	Close the printer cover
Strip timer comb not present	Insert or adjust the strip timer comb
Pipetting stage not present	Insert or adjust the pipetting stage
Excess light at photometer	Check that there is no external light source affecting the photometer, and eliminate it. If the error persists, complete the steps in 8.6.13 to replace the photometer assembly.

Message text	Recommended action
LED error in photometer	Complete the steps in 8.6.13 to replace the photometer assembly.

2.2 Software system messages



2.2.1 Software error messages

Message text	Recommended action
LIS connection error	The connection between the device and the LIS was broken. See Disabling the thermal printer to access the comDriver.ini file, and verify that the PORT2 parameter is set to 1. Check the LIS system, the connecting cables as well.
Error at LIS connection open	The connection between the device and the LIS could not be established. See Disabling the thermal printer to access the comDriver.ini file, and verify that the PORT2 parameter is set to 1. Check the LIS system, the connecting cables as well.
Upgrade command error	There was a communication error during the upgrade of a module PCB. Reboot the application. If reboot is unsuccessful, complete the steps in 3.3.5.2 Upgrading the module PCBs .
Flash erase error	Reboot the application. If reboot is unsuccessful, complete the steps in 3.3.5.2 Upgrading the module PCBs .
Upgrade flash write error	Reboot the application. If reboot is unsuccessful, complete the steps in 3.3.5.2 Upgrading the module PCBs .
Upgrade file (mhx) error	See Reinstalling the device software and reinstall the device software.
Memory allocation error during upgrade	See Reinstalling the device software and reinstall the device software.
Upgrade file (mhx) not found	See Reinstalling the device software and reinstall the device software.

Message text	Recommended action
No processor	See Reinstalling the device software and reinstall the device software.
File IO error	Make sure that the removable device is working properly and is not write-protected.
Login failed. Incorrect user name or password	Log in using the correct user name and password
Database full	Free up disk space in the database or enable circular memory
Cannot export next sample	Make sure that the connected USB device is recognized and that there is available disk space on it
User name and password must differ	Choose another password
Incorrect previous password	Enter the correct previous password
Re-entered password does not match new password	Retype the new passwords
Minimum password length is five (5) characters	Choose a password of at least 5 characters for higher security
This user name is already in use	Choose another user name
User name too short	User names must be at least 2 characters
You may not delete your own account	Log in using a different user account and try again.
LIS connection error	Check the connection to the LIS
Quality Control database full	Delete QC data
File IO error during logs copying	Check the connected USB device
Wastebin full	Empty the wastebi
Diagnostic error	Retry the diagnostic
This worklist bar code already in use	Enter a new bar code
Strip registration code has expired	Register a new code

Message text	Recommended action
Foreign registration code	The registration code entered is compatible only with a different version of the device software. Enter a registration code compatible with the device software version that you are currently using.
Registration code not detected	Retry the registration
Invalid registration code	Registration code already in use. Use a different code



2.2.2 Software exception messages

Message text	Recommended action
Software exception	Reboot the system.
Database compact failed	Reboot the system.
Card upgrade error	Reboot the system.



2.2.3 Software information messages

Message text
Printing in progress
There are no logs available
Data transfer successful
Password successfully modified
You can open the unused strip bin
Stopping printer
Hardware diagnostic successful
Strip registration successful
Insert statim sample or samples
Level 1 QC measurement passed
Level 2 QC measurement passed



2.2.4 Software warning messages

Message text	Recommended action
Measurement rejected by the sediment analyzer	Make sure that the number of parallel measurements set for the urine sample is the same for both devices
Sediment analyzer not ready	Wait for the sediment analyzer to finish initializing.
Invalid start parameter delivered to the sediment analyzer	Check the interfaced sediment analyzer
Sediment analyzer busy	Wait for the measurement cycle underway on the sediment analyzer to end.
Sediment analyzer waste tank full	Empty the sediment analyzer's waste tank.
Sediment analyzer out of distilled water	Refill the sediment analyzer's distilled water tank.
Sample count has exceeded the sample limit on the sediment analyzer	Check the interfaced sediment analyzer
Cuvette registration required on the sediment analyzer	Enter a valid registration code for the cuvettes in the sediment analyzer.
Sediment analyzer cuvette waste bin full	Empty the sediment analyzer's cuvette waste bin.
All sediment analyzer worklist items completed	Check the interfaced sediment analyzer
The current reflex mode setting does not support parallel measurements	Modify the common reflex mode setting, or set the number of parallel measurements to 1.
When the worklist is enabled, no parallel measurements are allowed on the sediment analyzer	Disable the worklist, or set the number of parallel measurements to 1.
No user is logged in on the interfaced sediment analyzer	Log in to the sediment analyzer.
Sediment analyzer busy with QC measurement	Wait until the sediment analyzer's QC measurement has finished and try again.

Message text	Recommended action
Parallel measurements are not supported when the sediment analyzer is assigning sequential sample IDs	Disable sequential sample ID generation on the sediment analyzer or set the number of parallel measurements to 1
Worklist empty	Populate the worklist or disable its use
All worklist items completed	Populate the worklist or disable its use
Database warning limit reached	Free up disk space in the database or enable circular memory
Not enough disk space on the removable drive	Free up disk space on the removable drive
No removable device present	Connect a USB drive or HDD
Auto export path not selected	Select the path for auto export or disable the feature
Auto export path does not exist	Define a path for auto export or disable the feature
When the worklist is enabled, no parallel measurements are allowed	Disable the worklist feature or set the number of parallel measurements to 1
At least 1 pad must be enabled	Enable one or more pads in the pad sequence setup
Parallel measurements are not supported for generated sample IDs	Set the number of parallels measurements to 1
No LOT activated for every QC level	Activate a LOT number for each QC level
All activated QC LOTs must be of the same solution type	Make sure that the active QC LOT numbers come from the same solution type
No QC LOT activated	Activate a QC LOT number
QC LOT already in use	Set a different QC LOT number
Waste bin warning reached	Empty the waste bin
QC deletion not enabled	One or more selected items contain QC measurement results. Enable QC result deletion in the User software on the Settings/Functions tab

Message text	Recommended action
Low disk space	Free up disk space: Delete database entries (see the User Guide); copy the contents of E:/LabUMat2/Logs to a removable drive, then delete the folder.
Device serial number missing	Set a valid device serial number
Invalid pad reflex condition setting	Set an applicable and valid pad reflex
Strip registration necessary	Register new strips in the User software on the General menu
Level 1 QC measurement failed	See the QC Results for the specific problem. Repeat the QC measurement. If the problem persists, see 1.5.3 Adjusting the pad-reading position to check and adjust the photometer.
Level 2 QC measurement failed	See the QC Results for the specific problem. Repeat the QC measurement. If the problem persists, see 1.5.3 Adjusting the pad-reading position to check and adjust the photometer.

2.3 Database status column error messages

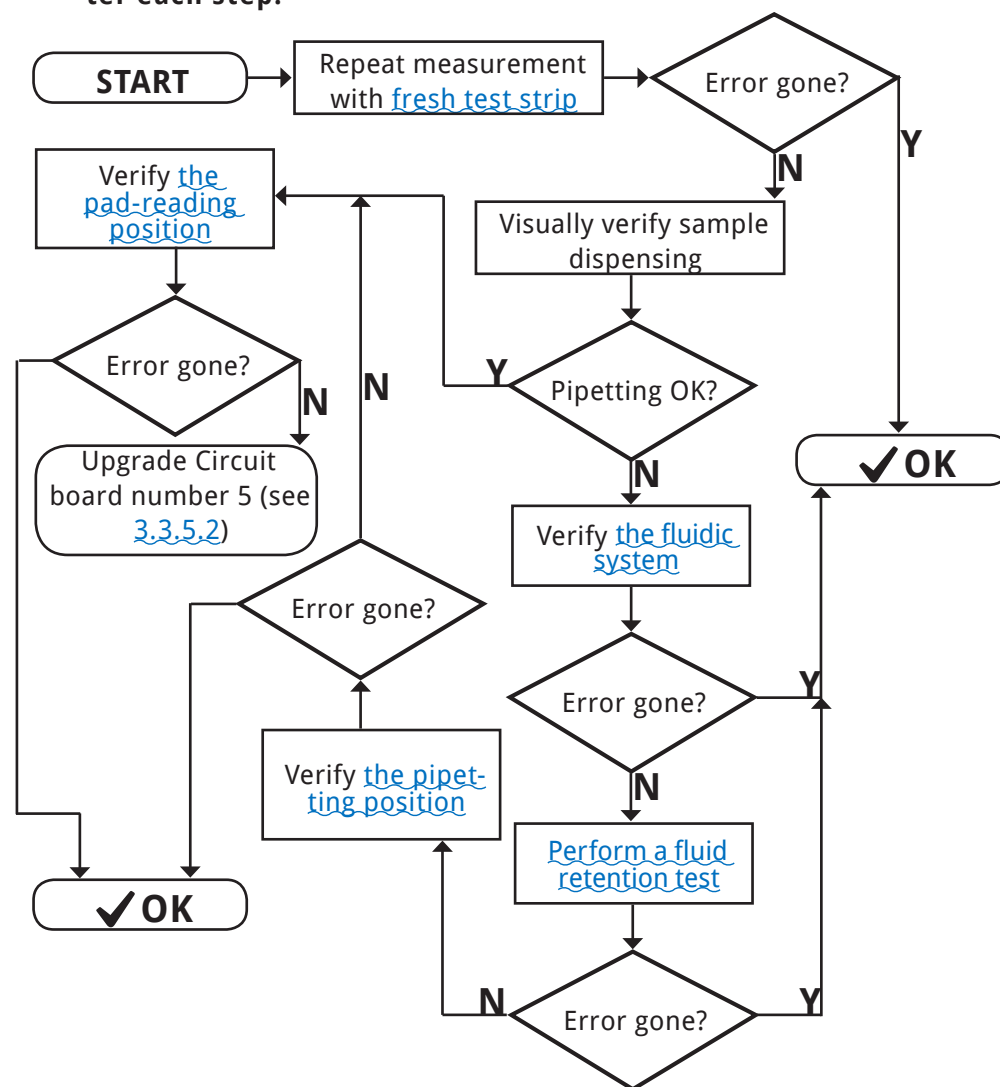
In the User software on the **Measure** menu, the device displays additional information on each of the records in the current measurement cycle. A red X mark and an error code is displayed in the **Status** column of the measurements that could not be processed for some reason. No test result data will be saved in the database for records marked with a red X.

Error code	Description & Recommended action
X1	The test strip was loaded into the feeder drum the wrong way around. Repeat the measurement with a properly oriented test strip.
X2	Strip lost after sample dispensing. Check and adjust the strip clamps , the strip timer comb , and the pipetting position .
X4	Dry strip or inadequate strip pad discoloration. See 2.4 Troubleshooting the X4 error .
X5	The strip flipped during the feeding, sample dispensing, or timing process. See 2.5 Troubleshooting the X5 error .
Xt	Quality control measurement failed. Repeat the QC measurement. If the problem persists, see 1.5.3 Adjusting the pad-reading position to check and adjust the photometer.
X	Measurement stopped by user / Hardware or other undefined error  The records for every sample that enters the device after the error is detected will be marked with a red X mark and will not have any saved test result. You will need to repeat the measurement for these samples.

 A 'fresh test strip' is one that you take out of a test strip vial that you open on the spot, preferably one that is from a lot that is different from the one you have used before.

2.4 Troubleshooting the X4 error

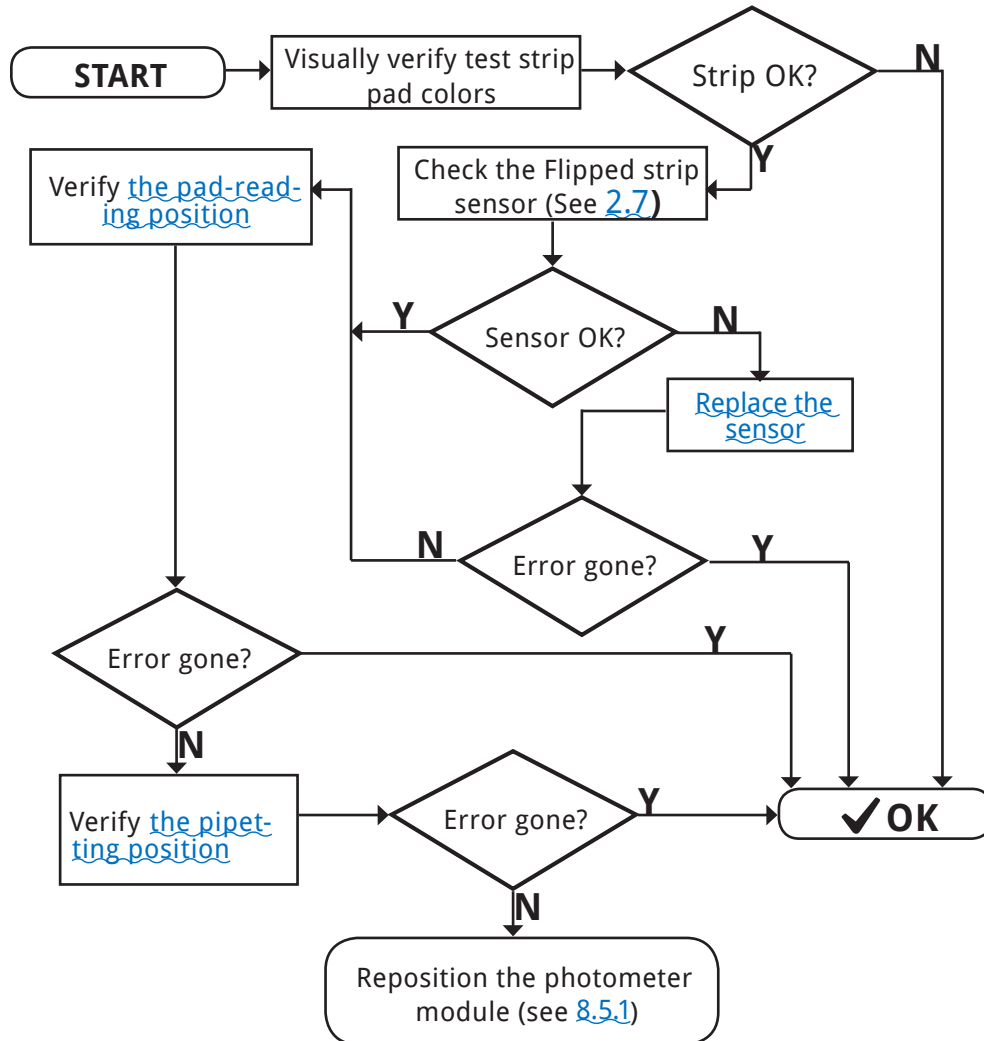
 When troubleshooting, it is assumed that you perform a system re-initialization and a new measurement, where applicable, after each step.



 Contact 77Elektronika service personnel if you cannot resolve your issue using this Troubleshooting Flow Chart.

2.5 Troubleshooting the X5 error

i When troubleshooting, it is assumed that you perform a system re-initialization and a new measurement, where applicable, after each step.



⚠ Contact 77Elektronika service personnel if you cannot resolve your issue using this Troubleshooting Flow Chart.

2.6 Reinstalling the device software

There are some software errors that you can only fix by completely removing and reinstalling the device software. Complete the steps in [12.2.4 Reinstalling the device software](#) to reinstall the user software.

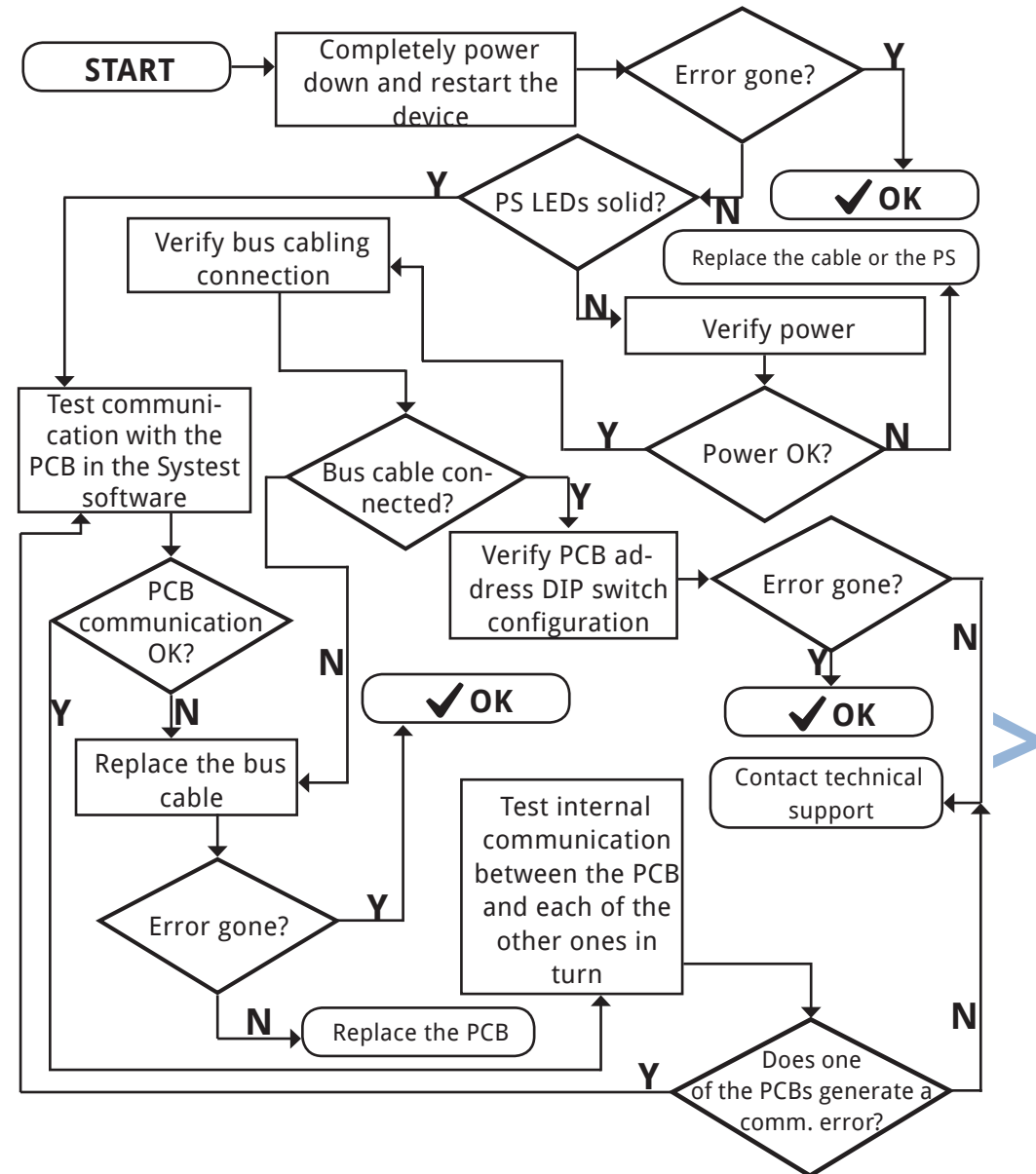
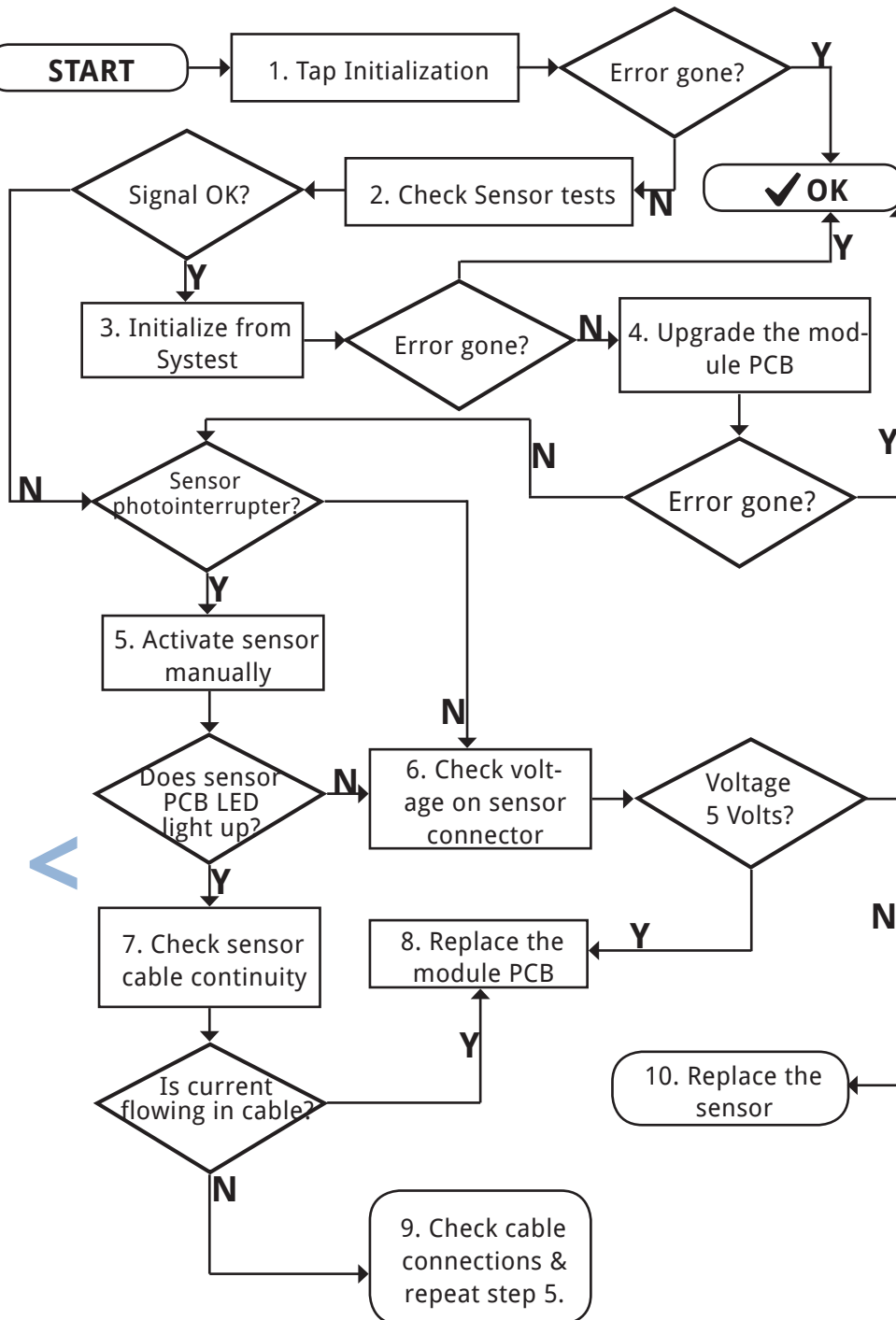
2.7 Handling sensor-related error messages

i When troubleshooting, it is assumed that you perform a system re-initialization after each step, where applicable.

⚠ Contact 77Elektronika service personnel if you cannot resolve your issue using this Troubleshooting Flow Chart.

2.8 Handling PCB communication errors

i When troubleshooting, it is assumed that you perform a system re-initialization after each step, where applicable.



2.8.1 Verifying the power supply and the power cables

1 Power down the device completely and disconnect the power cable from the module PCB that you are troubleshooting (see the board layout diagram in the relevant module overview section).

2 Restart the device and using a multimeter, measure the voltage present at the two outer pins of the power cable connector against one of its two central ground pins. You should get a reading of 5 Volts on one, and 24 Volts on the other outer pin. If you get the correct voltage readings at the PCB connector pin, you have established that both the power supply and the power cable are functioning properly. In this case skip step 3 and go to step 4. If there is no voltage present at the connector pins, go to step 3.

3 Disconnect the power cable from the power supply and using a multimeter, measure the voltage present at the outer pins of the cable connector against the two central ground pins. You should get a reading of 5 Volts on one, and 24 Volts on the other outer pin. If you get the correct voltage readings, you have established that the power supply is functional but the power cable is defective. In this case, replace the power cable. If no voltage is present at the connector pins, see [4.1.10](#) and replace the power supply.

4 Reconnect the power cable and switch on the device. In the Systest software on the **Flow tests** menu, tap **Initialize** to verify that the power supply and the power cable are free of errors.

2.8.2 Testing internal data bus communication

1 Power down the device completely and disconnect every module PCB from the internal data bus cable except for the module PCB you are troubleshooting. Leave only the PCB that you are troubleshooting connected to the PC module via the internal data bus.

2 Launch the Systest software and enter the **Card tests** menu. Tap the tab of the module PCB that you are troubleshooting. Tap the **Communication test** button. Check whether the counter next to the **Stop communication test** button starts running, indicating the number of times the circuit board acknowledges traffic from the embedded PC. If the counter starts running, skip step 3 and go to step 4. If the counter stays at 0 and the **Messages** button in the top right-hand corner turns red after a 15-second timeout, the PC cannot communicate with the module PCB in question. Complete step 3.

3 Replace the internal data bus cable and repeat step 2. If there is still no communication between the PC and the module PCB, see the relevant section in the module overview and replace the module PCB. If the communication error persists, complete step 4.

4 Test the communication between the PCB that you are troubleshooting and all the other module PCBs in turn: Tap the **Stop communication test** button on the **Card tests** menu of the module PCB that you are troubleshooting, and connect a second PCB to the internal data bus. Next, tap the **Communication test** button, check whether the test counter is running or not, then tap the **Stop communication test** button, and disconnect the second module PCB. Repeat the process

with each remaining module PCB. Continue until the communication test between the PCB you are troubleshooting and one of the other module PCBs fails. If this happens, you have successfully established which module PCB is probably defective. Complete steps 1–3 with the suspect module PCB.

⚠ If the module PCB that you are troubleshooting can communicate with all the other module PCBs in turn, but the original communication error message is still active, contact 77Elektronika service personnel for further assistance.

2.9 Troubleshooting Guide



2.9.1 Abnormal analysis results

If the device consistently supplies false positive or false negative results, but none of the status column error messages listed in [2.3](#), you need to take action as soon as possible.

1 Using the same lot of LabStrip U11 Plus GL test strips that you were using when encountering the suspicious results, run a two-level quality check using Quantimetrix Dip and Spin or Bio-Rad Liquichek control solutions. Record the results you get with the control solutions for further reference.

i If using Liquichek, check the control solution's lot number. 77Elektronika supports Bio-rad Liquichek solutions only from lot numbers 61470 and upwards.

2 Regardless of whether the test strips passed or failed the QC measurement, open a new vial of LabStrip U11 Plus GL test strips from a

different lot than the one you used previously, and, to rule out the test strips as a cause for the inaccurate results completely, repeat the quality control measurement with the new lot of strips. Record the results you get for further reference.

3 Repeat the measurement of the urine samples that produced the suspicious results with the new lot of test strips you opened in step 2. If the measurement results are different from the results you suspected to be inaccurate, you are done: the defective lot of test strips you were using was responsible for the inaccurate test results. If the measurement results are still not satisfactory using the new lot of test strips, complete the troubleshooting steps in [Troubleshooting the X4 error](#).



2.9.2 The device does not respond to the power switch

When you cannot power up the device, and you have ruled out obvious causes:

- power failure;
- the device is not connected to the mains;
- the fuses in the power switch have blown;

there are still several possible causes left to investigate.

2.9.2.1 The 'power-on' cable (see [Figure 17](#)) is disconnected or broken

Symptom	Recommended action
The barcode scanner starts beeping after you press the power switch, but the device powers down almost immediately.	Verify the power-on cable connection and replace the cable if it is broken.

2.9.2.2 The standby switch has malfunctioned

Symptom	Recommended action
The see-through plastic ring around the standby switch does not light up.	Measure the continuity between the switch's two pins out of three that have cables connected to them and replace the switch (see 14.3) if there is none.

2.9.2.3 The power switch has malfunctioned

Symptom	Recommended action
There is no continuity between the two pins of the power switch.	See 14.4 and replace the power switch.

2.9.2.4 The power supply module has malfunctioned

Symptom	Recommended action
There is no voltage present at the outer pins of the power supply cable connector (see step 3 in 2.8.1), and the device powers up without any error after you replace the power supply.	See 14.1 and replace the power supply.

2.9.2.5 The 12V cable between the power supply and the PC module (see [Figure 17](#)) is disconnected or broken

Symptom	Recommended action
There is no voltage present at the pins at the end of the 12-Volt connecting cable towards the PC module (see steps 1–2 of 2.8.1 for the procedure).	Replace the 12-Volt cable connecting the power supply and the PC module.

2.9.2.6 The embedded PC motherboard has malfunctioned

Symptom	Recommended action
The device powers up, the PS LEDs on the module PCBs light up and stay solid, but the touch screen stays black.	Verify that it is not the touch screen that has malfunctioned by connecting an external monitor to the VGA-connector at the back of the device, and make the necessary replacement (see 12.3.2 Replacing the PC motherboard or 12.3.5 Replacing the LCD touch-screen).



2.9.3 The device software crashes, freezes, or hangs

1 When the device software crashes, does not respond to any touch screen input, or only partially responds to user input, shut down the software: connect an external keyboard to the device, press **Ctrl+Alt+Del** on it, and end all system processes in the **Task Manager** window that pops up.



If the system does not respond to the **Ctrl+Alt+Del** keyboard shortcut, force the device to power off by switching the device off with the power switch.

2 Switch on the device. [Enter the User software Service menu](#). Insert a removable disk and tap the **Copy logs** button.

⚠ The Copy logs command only works if there is at least one previously created folder on the removable disk that you use.

⚠ If system logs are disabled, turn them on in the User software ([2.1.4 Log files](#)), save the setting, then try to reproduce the error so that a log file of the incident is saved on the disk.

⚠ The E:/LabUMat2/logs folder automatically resets itself once its size reaches 5 gigabytes. Clear the folder from time to time to make room for newer log files.

3 Send the log files to 77Elektronika service personnel for evaluation. Make sure that you include as many details about the circumstances of the software error (date, time, bar code of samples tested, etc.) as possible for easier identification of the relevant log files.



2.9.4 The device is making excessive noise during operation

Symptom	Probable cause	Recommended action
The feeder drum is extremely noisy during operation	The feeder blade is set too low. Test strips get jammed underneath it, causing the feeder drum to rattle.	Adjust the feeder blade
The waste pump is extremely noisy during operation	Debris has accumulated in the waste pump's flap valve.	See 6.7.4.1 and clean the waste pump flap valve.

Symptom	Probable cause	Recommended action
The rack conveyor is extremely noisy during operation	The light silicone grease on the timing belts has worn off, and dust and dirt has accumulated between the timing belt teeth.	See 7.6.15 and clean the rack conveyor timing belt gears and rollers. Replace one or both rack conveyor timing belts if necessary.



2.9.5 The pipetting probe vibrates and splatters

Symptom	Probable cause	Recommended action
The pipetting probe trembles so much when it moves that it sprays the urine sample inside it inside the analyzer	The linear robot vertical timing belt is too tight, and increases the backlash of the cylindrical roller bearing that helps the linear robot movement glide along the spline shaft.	Switch off the analyzer, and try to move the linear robot by pinching together the linear robot vertical timing belt. If you can move the pipetting probe in this way, then the timing belt is too tight. See 6.6.3 and loosen the timing belt to eliminate the backlash.
	The support for the the linear robot horizontal movement's cylindrical roller bearing has become loose, and increases the backlash of the bearing.	Loosen the two 2.5-millimeter hex socket screws (circled in red in Figure 14) that secure, via a plate, two of the cup rollers of the supporting shaft to the cylindrical roller bearing housing. Push the securing plate upwards, in the direction indicated in Figure 14 to make the cup rollers press firmly against the supporting shaft. Tighten the hex socket screws in the new position.

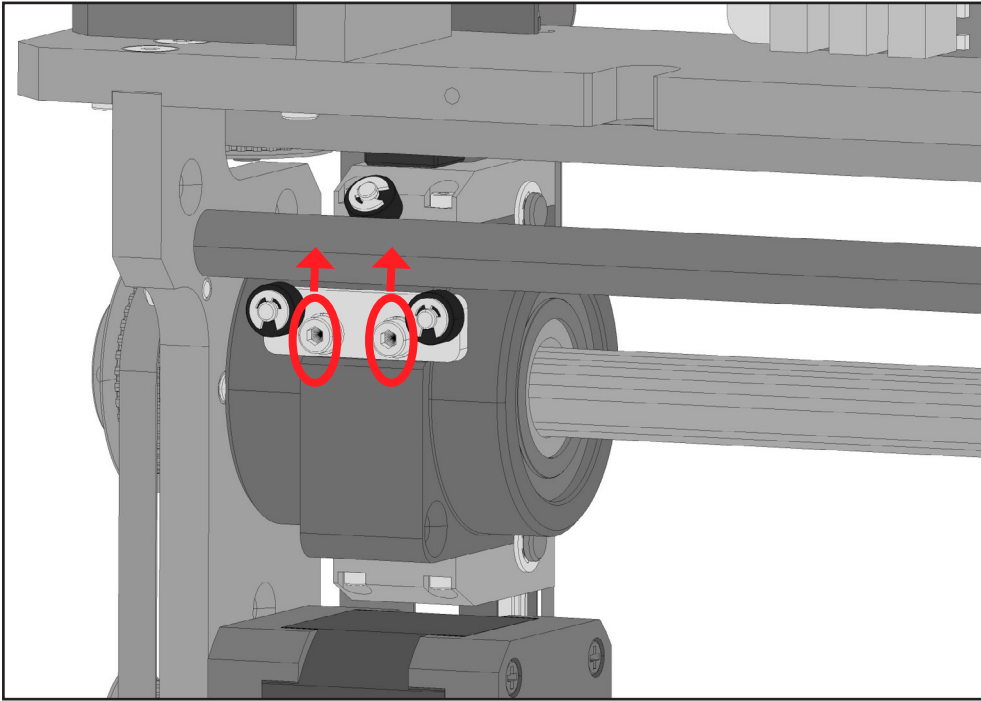


Figure 14: Close-up view of the cylindrical roller bearing on the spline shaft of the linear robot movement



2.9.6 The touch screen does not respond

Verify that it is the touch screen that has malfunctioned by connecting an external monitor to the VGA-connector at the back of the device, and make the necessary replacement (see [12.3.2 Replacing the PC motherboard](#) or [12.3.5 Replacing the LCD touchscreen](#)).



2.9.7 Printout issues

Symptom	Probable cause	Recommended action
Results are not printed or the printout is very faint	The loaded paper is not compatible with the printer.	Make sure that the right size thermal paper is loaded into the printer.
	The thermal paper is too old; the heat sensitive layer has deteriorated.	Load the printer with a fresh roll of thermal paper.
	The printer is defective.	Launch the Systest software and enter the Flow tests menu. Tap Print character table and verify that the printer is not fully functional. Replace the thermal printer (10.5.1).
There are white spots or streaks on the printout where resu	Grease and dirt have accumulated on the printer head.	Open the printer cover. Hold a lint-free wipe dipped in distilled water to the surface of the roller, and use the roller's cog wheel to turn it. Make sure that you wipe every part of the roller surface.

3 The Systest software and the Service menu

3.1 Service-related aspects of the User software



3.1.1 Upgrading the device software



You will need the latest version of the device software (an .msi-extension file named "LabUMat2[latest version number]") in the root folder of a removable drive that preferably does not have any other content saved on it.

- 1 Insert the removable drive with the latest software version into the device, and switch on the device.
- 2 In the User software, tap the **Settings** button, then the **Login** button. A log-in screen will pop up. Tap inside the **Name** text box to make a virtual keyboard appear on the display, or use a computer keyboard connected to the device to enter 'service' (without the inverted commas) as both name and password. Tap **OK** or type Enter on the keyboard to log in as a service-level operator.
- 3 On the **Main** tab, which only becomes available once you have logged in as a service operator, tap the **Upgrade** button. Tap **OK** in the confirmation window that pops up. The system will detect the software upgrade file on the removable drive and upgrade the device software automatically.
- 4 The system will restart to complete the upgrade process. Boot will take longer than usual, and progress bars on the software splash screen will display the upgrade process of the firmware on each of the module PCBs.



3.1.2 Accessing the User software Service menu

Authorized service personnel can use the User software Service menu to quickly access the Systest software and perform a number of service-related data export processes.

- 1 Switch on the device. In the User software, tap the **Settings** button, then the **Login** button. A log-in screen will pop up.
- 2 Tap inside the **Name** text box to make a virtual keyboard appear on the display, or use a computer keyboard connected to the device to enter 'service' (without the inverted commas) as both name and password. Tap **OK** or type Enter on the keyboard to log in as a service-level operator.
- 3 Tap the Service tab, which only becomes available once you have logged in as a service operator.



3.1.3 Organization of the Service menu

Disable interlock	See Device door sensors
Login settings	See the relevant section of the User guide
SERVICE SW	Tap to access the Systest software
DIAGNOSTIC	Tap to save a complete report of the status and settings of the device hardware on the connected removable disk.
LOG OFF SYSTEM	Tap to exit the User software. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.
COPY LOGS	If log files are enabled, tap this button to save the log files on the connected removable disk.
RAW EXPORT	Tap this button to save raw photometry data recorded by the measuring head on the connected removable disk.
PMC RAW EXPORT	Tap this button to save raw PMC measurement data on the connected removable disk.
CHANGE PWD.	Tap this button to change the password of the currently logged in operator. The software will prompt you to enter the current password.
LOAD DEFAULTS	Tap this button to revert to the default settings. ⚠ This is a global setting. Tapping this button will reset every setting you have modified.
SAVE SETTINGS	Tap this button to save the settings you modified before exiting the software. ⚠ This is a global setting. Tapping this button will save every setting you have modified.



3.1.4 Log files

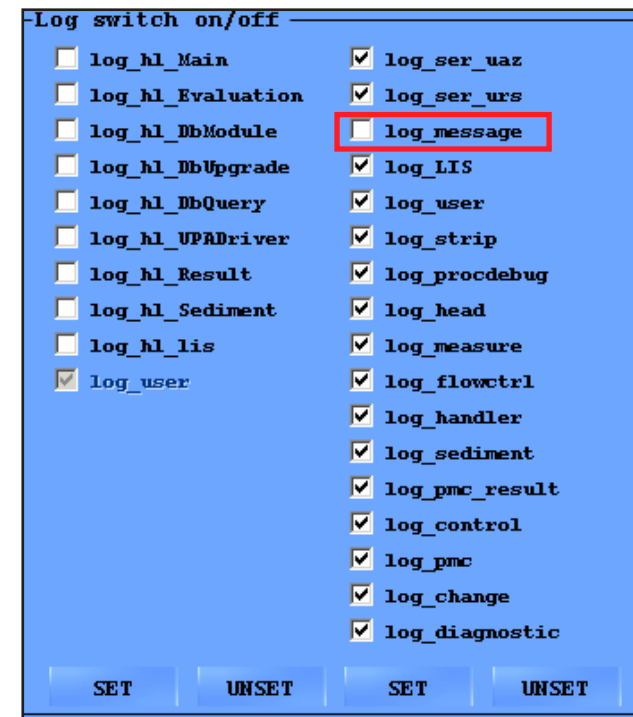


Figure 15: The log file check boxes

Log files function as a sort of 'flight recorder' for LabUMat 2, recording all the system processes and sub-processes during device operation. Logs are often helpful when trying to establish the source of a system error (see [14.9.3](#)). Log files in the left column in Figure 15 record user software-related data; log files in the right column record driver-related data. You can check and uncheck the check boxes individually or use the **SET** and **UNSET** buttons to check and uncheck all of the check boxes in a column with a single click.

⚠ Make sure that the log_message box, indicated in red

in Figure 15, is not checked, as it bloats log file size and the system will soon run out of available hard disk space if it is running.

3.2 The Systest software



3.2.1 Accessing the Systest software

The Systest software is a tool for authorized service personnel to test the functionality of the subsystems of the device, and to adjust the preset parameters that control the operation of the systems and subsystems.

⚠ No setting should ever be modified in the Systest software without first checking that the given subsystem is not working correctly. Always consult the relevant section of this service manual before attempting to modify preset parameters. When in doubt, feel free to contact 77Elektronika service personnel.

⚠ Many parts of the Systest software are not relevant to service personnel. Do not modify any value on any screen, setting, or element in the Systest software not explicitly referred to as adjustable in this service manual.

1 Enter the **Measure** screen in the User software and tap the **Log in** button. A log-in screen will pop up. Tap inside the **Name** text box to make a virtual keyboard appear on the display, or use a computer keyboard connected to the device to enter 'service' (without the inverted commas) as both name and password. Tap **OK** or type Enter on the keyboard to log in.

2 When logged in as a service user, a **Service** screen becomes available. Enter the **Service** screen and tap **Service software**. The user software will be closed, and another log-in screen will pop up. Enter the same name and password as before to enter the Systest software.

LabUMat 2: the systest software and the service menu



3.2.2 Exiting the Systest software

Tap the **Exit** button to close the Systest software and re-open the User software.



3.2.3 Organization of the Systest graphical interface

Control elements are organized into rows of buttons in the graphical interface (see Figure 16). The topmost row (the **menu bar**) contains

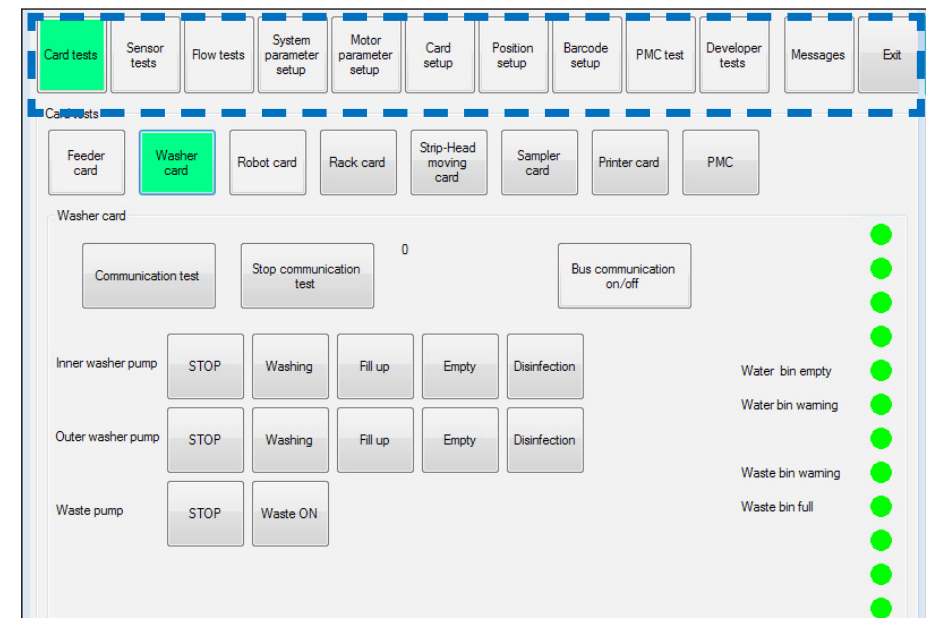


Figure 16: The Systest software graphical interface with the menu bar marked in blue. The green highlighting indicates that the user is currently on the Card tests menu on the Washer card tab.

elements, mostly menus, that are visible no matter where you have navigated in the software. Most of the menus contain one or more additional rows of buttons below the top row (the **modules bar**). The software has a single-window interface and no breadcrumb trail is used; your location within the software is indicated by the menu and tab button or buttons that are highlighted in green. The active state of two-state toggle buttons is also indicated by green highlighting.

 This section will deal only with menus, tabs, and elements that are relevant to several or all of the modules. For further information on module-specific Systest software elements, see the section on the particular module.

3.3 The Systest software—General functions



3.3.1 The Card tests menu

The Card tests menu allows you to test the functionality of all the motors and actuators in the system, broken down by module (more precisely, by the circuit board that controls each module). Consequently, it has eight (8) tabs, one for each of the circuit boards, arranged in the modules bar by circuit board number in ascending order from left to right.

The tabs are highlighted in green when selected to help navigation.

 For the buttons and elements not covered in this section, see the relevant section in the overview of each module for module-specific details.

Buttons and elements shared by all of the module tabs:

- Communication test

Tapping this button will start a series of communication tests between

LabUMat 2: the systest software and the service menu

the embedded computer and the PCB of the module. The details of the signals sent and received is displayed in real time in the **Messages** window. The number of signals the PC sends out is displayed as a real-time counter next to the **Stop communication** button.

- Stop communication test

Tapping this button will stop an ongoing communication test.

- Bus communication on/off

A toggle button to start and stop a global communication test that sends signals from the PC to all of the module PCBs through the bus ribbon cable.

 The PMC module tab does not have a Bus communication on/off button since its controller circuit board has no 16-pin bus ribbon cable connected to it.

- STOP

Tapping this button will stop the particular motor it is associated with. In the case of stepper motors, tapping **STOP** disables the motor so that you will not need to disconnect its cable to move it by hand.

- Sensor column

This is a non-interactive extract from the **Sensor tests** menu. The sensor signals associated with the particular module are displayed in real time to the right of the module-specific motor test buttons.



3.3.2 The Sensor tests menu

The real-time readings for all the sensors of the six (6) circuit boards that have connected sensors are displayed on this menu arranged by circuit board number in ascending order from left to right.

 Red sensor signals (●) indicate an active sensor. Green

sensor signals (●) indicate an inactive sensor. Unused sensor slots display no sensor name and a static green sensor signal.

The following table summarizes all the sensor signals as they should appear on a properly set up and fully operational device after Initialization.

Feeder card		Washer card		Robot card		Rack card		Strip-Head moving card		Sampler card			
Flipped strip sensor				Rinse well pos.		Inner-most catch				Mixing home			
Strip turner home				Test tube pos.		Outer-most catch				Horizontal str. timer home		Asp. pump home	
Pusher home								Liquid level		Vertical str. timer home			
Pipetting pos.								Rack position		Head mover home			
Strip progress		Water tank empty		High probe position		Tube presence		Str. timer comb present					
Empty box		Water tank warning		Low probe position		Conv. max. capacity		Strip bin micro-switch					
Pipetting stage				Left door		Rack delivery							
Loading box		Waste tank warning		Right door									
Feeder home		Waste tank full				Aligner open							
Feeder drop						Aligner close							
Upper empty pos.													
F. drum safe area						Liquid level							



3.3.3 The Flow tests menu

The Flow tests menu brings together in a 5x6 button grid the tests and device processes that involve more than one motor movement or that involve more than one module circuit board.

 **This section only describes global buttons and processes. See the relevant section in the overview of each module for module-specific Flow tests–processes not covered here.**

 **Most of the buttons on the Flow tests menu are operational only after a system initialization.**

Global processes and settings not associated specifically with any of the modules:

- Start polling

Tapping this button starts an internal data bus circle.

- Stop polling

Tapping this button stops the bus communication started with the button above.

- Initialization

Tapping this button starts a system initialization. The specific processes the individual modules perform during initialization are detailed in a dedicated section in each module overview. The sequence in which these initialization subprocesses are performed is detailed in the relevant section of the Principle of operation overview.

 **Make sure that the distilled water and the waste tanks are connected to the device and that the fluidic system is intact before tapping the Start auto measure, Start manual measure, or Start STAT measurement buttons.**

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- Start auto measure

Tapping this button starts an automatic measurement cycle as detailed in the LabUMat 2 user's manual. The **Nr. of measurements** counter keeps track of the samples in the measurement cycle.

- Start manual measure

Tapping this button starts a manual measurement cycle. You can set the number of samples you want to analyze in the manual measurement cycle using the (+) and (-) buttons next to the **Number of samples** counter. You can set the number of samples aspirated for measurement from each of the test tubes in the manual measurement cycle using the (+) and (-) buttons next to the **Number of parallels** counter.

 **The default setting for both the number of samples and the number of repeated measurements using the same test tube is 1.**

The **Nr. of measurements** counter keeps track of the samples in the manual measurement cycle.

- Send to STAT position

If you tap this button while a measurement cycle is under way, the device will process the test tube it is currently handling, then prepare to receive a statim sample on a new rack: the Rack positioning motor is moved into its Push out position to remove the current rack, and the Rack mover motor is moved into its STAT position to make room for the rack with a statim sample in front of the rack passage.

- Start STAT measurement

After preparing the device for a statim sample, tap this button to start the statim measurement. The device will sample and process all the test

tubes in the rack you placed in front of the passage.

- RESET

Tapping this button resets all hardware elements of the system.

- Disable Interlock

Tapping this button toggles disabling and enabling the left and right inductive door sensors. See [Device door sensors](#). The button is highlighted in green when disabling is active.

- Messages window

The **Messages** window, also accessible from the menu bar, takes up the right side of the **Flow tests** menu.



3.3.4 The System parameter setup menu

The System parameter setup menu allows you to adjust preset software settings that control certain processes. The menu is broken down by module (more precisely, by the circuit board that controls each module). Consequently, it has seven (7) tabs, one for each of the circuit boards that have adjustable settings, arranged in the modules bar by circuit board number in ascending order from left to right.

The tabs and parameters are highlighted in green when selected to help navigation.

 For the buttons and elements not covered in this section, see the relevant section in the overview of each module for module-specific details.

Buttons shared by all of the module tabs:

- Minus and plus values (+10, -100, etc.)

Tapping these buttons modifies the selected parameter by the number

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on the button. Positive values increase negative values lower the parameter value.

- Increment (+)

Tapping this button adds 1 to the selected parameter's value.

- Decrement (-)

Tapping this button subtracts 1 from the selected parameter's value.

- Read parameters

Tapping this button makes all parameters on the tab display their current values.

 **Whenever you wish to modify any parameter values, start by tapping Read parameters to make sure that you are modifying the current values.**

- Save parameters

Tapping this button saves any modification you have made to the parameter values.

 **If you leave the tab without saving the modified parameters, all modifications will be lost. Always make sure that you save your modifications before you leave the tab.**

- Load defaults

Tapping this button after you have tapped the **Read parameters** button restores the default value to all the parameters.



3.3.5 The Card setup menu

The Card setup menu allows the configuration of newly installed module circuit boards and the upgrading of existing ones. There are eight (8) tabs on the menu, one for each module circuit board in the system,

listed in the modules bar by circuit board number in ascending order from left to right.

⚠ Whenever you replace a module circuit board, you will have to set it up as detailed in this section for it to be fully operational.

3.3.5.1 Setting up a newly installed circuit board

1 Complete the steps in the relevant module overview to replace the circuit board.

2 Switch on the device and launch the Systest software. On the **Card setup** menu, find the tab for the module whose PCB you have replaced.

3 Tap the **Module ID** text box if it is not already highlighted and using the on-screen keyboard, enter the ID of the particular module as it appears on the sticker stuck on the module. Tap **Store module ID** when you are done.

< i You will find the Module ID on a white sticker in a conspicuous place on the module itself. Make sure that you enter the module ID as it appears on the sticker.

4 Tap the **Serial number** text box and using the on-screen keyboard, enter the serial number of your LabUMat 2 device (printed on the back casing panel) continuously. Tap **Save instrument SN** when you are done.

i The **BACK** button functions like a backspace key on a computer keyboard and deletes the most recently typed character. The **CLR** button clears all the characters in the selected text box.

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5 Enter the **Flow tests** menu and tap **RESET**, then **Initialization** to complete the PCB setup.

3.3.5.2 Upgrading the module PCBs

i Whenever a new version of the device software is installed, the upgraded software automatically upgrades the firmware on all the circuit boards in the system. You do not need to manually upgrade any of the circuit boards when upgrading the global device software version.

1 Switch on the device and launch the Systest software. On the **Card setup** menu, find the tab for the module whose PCB you want to individually upgrade.

2 Tap the **Module ID** text box if it is not already highlighted and using the on-screen keyboard, enter the ID of the particular module as it appears on the sticker stuck on the module. Tap **Store module ID** when you are done.

i You will find the Module ID on a white sticker in a conspicuous place on the module itself. Make sure that you enter the module ID as it appears on the sticker.

3 Tap **Upgrade**, then enter the **Flow tests** menu, and tap **Initialization** to complete the process.

⚠ The Flash address, EEPROM read, and Flash read buttons are currently only included on this menu for software development purposes, and have no service application. Do not modify the Flash address.



3.3.6 The Barcode setup menu

You can set advanced parameter options for the barcode reader on this menu. This section gives an overview of the buttons available for barcode reader configuration. Five types of barcodes are supported, each of which has its own tab and associated parameters. The active tabs and active parameter buttons are indicated by green highlighting. You can modify the values of the active parameter buttons with the +1, -1, +10, and -10 buttons on the right.

- Tapping the **Read configuration** button displays the current values for each of the parameters.
- Tapping the **Save configuration** button saves any values you have modified on the menu.

⚠ If you leave the tab without saving the modified parameters, all modifications will be lost. Always make sure that you save your modifications before you leave the tab.

- Tapping the **Load defaults** button resets the barcode settings to their preset defaults. After replacing the barcode reader in the device (see [7.6.12 Replacing the barcode reader](#)), tap the **Download settings** button to load the barcode settings on the new reader.
- The IS-4125 and ICR-803 buttons indicate, by green highlighting, which of the two types of barcode scanners that the device supports you are configuring.

⚠ Previously, LabUMat 2 devices were shipped with either a Metrologic IS4125 or a Sick ICR803 barcode scanner. Currently only the Sick model with the blue plastic housing is in use in the device.

UPC EAN Code tab

- Enable UPC-A numsys transmit

Enter a positive value for this button to transmit the UPC-A number system character.

⚠ Disabling this feature may result in duplicate values in the database.

- UPC-A check digit on

Enter a positive value for this button to transmit the UPC-A check digit.

- Enable conversion of UPC-A to EAN-13

Enter a positive value for this button to convert UPC-A to EAN-13 by transmitting a leading zero before the barcode.

i This option is only available with the Metrologic barcode scanner.

- Enable UPC-E numsys transmit

Enter a positive value for this button to transmit the UPC-E number system character.

i This option is only available with the Sick barcode scanner.

⚠ Disabling this feature may result in duplicate values in the database.

- Expand UPC-E

Enter a positive value for this button to expand UPC-E to 12-digit equivalent UPC-A.

- UPC-E check digit on

Enter a positive value for this button to transmit the UPC-E check digit.

- UPC-E leading 0 on

Check this box to add a zero at the beginning of each UPC-E barcode.

 **This option is only available with the Metrologic barcode scanner.**

- Enable EAN-8 check digit transmit

Enter a positive value for this button to transmit the EAN-8 check digit.

- Enable EAN-13 check digit transmit

Enter a positive value for this button to transmit the EAN-13 check digit.

- Enable convert EAN-8 to EAN-13

Enter a positive value for this button to convert EAN-8 to EAN-13 by transmitting five zeros at the beginning of the barcode.

 **This option is only available with the Metrologic barcode scanner.**

- Enable '\$' prefix

Enter a positive value for this button to transmit a "\$" character at the beginning of the barcode.

 **This option is only available with the Metrologic barcode scanner.**

- 
- Enable 2 digit supps

Enter a positive value for this button to scan 2 digit supplementals.

- Enable 5 digit supps

Enter a positive value for this button to scan 5 digit supplementals.

- Supplement required

Enter a positive value for this button to specify that a supplement is required for all UPC/EAN labels.

Interleaved 2 of 5 code options

- Check digit enable

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Enter a positive value for this button to scan ITF barcodes that have a Modulo 10 check digit.

- Check digit transmit

Enter a positive value for this button to transmit the ITF Modulo 10 check character.

 **This feature works with the ITF check digit option. In order for this feature to work both must be enabled.**

- Minimum code length

You can specify the minimum number of ITF digits in the barcodes.

- Maximum code length

You can specify the maximum number of ITF digits in the barcodes.

Code 39 tab

- Check digit enable

Enter a positive value for this button to scan Code 39 barcodes only if they have a check digit.

- Check digit send

Enter a positive value for this button to transmit to the host the Code 39 barcode check digit. 

- Enable Start/Stop transmit

Enter a positive value for this button to add Code 39 start and stop characters at the end and the beginning of each barcode.

 **This option is only available with the Sick barcode scanner.**

- Enable Full ASCII CODE39

Enter a positive value for this button to scan full ASCII Code 39 barcodes.

- Enable Italian pharmaceutical

Enter a positive value for this button to scan Italian pharmaceutical barcodes.

Codabar tab

- Check digit enable

Enter a positive value for this button to scan Codabar barcodes only if they have a check digit.

 **This option is only available with the Sick barcode scanner.**

- Check digit send

Enter a positive value for this button to transmit to the host the Codabar check digit.

 **This option is only available with the Sick barcode scanner.**

- Enable Start/Stop transmit

Enter a positive value for this button to add the Codabar start and stop characters at the beginning and at the end of each barcode.

- Enable CLSI editing

Enter a positive value for this button to perform CLSI library-type editing before the information is transmitted to the host.

 **This feature is only available with 14-digit Codabar-type labels and with the Sick barcode scanner.**

Code 128 tab

- Minimum code length

You can specify the minimum number of characters in a barcode.

- Maximum code length

You can specify the maximum number of characters in a barcode.

4 Modules overview & Disassembly

4.1 The modules and the modular structure of LabUMat 2

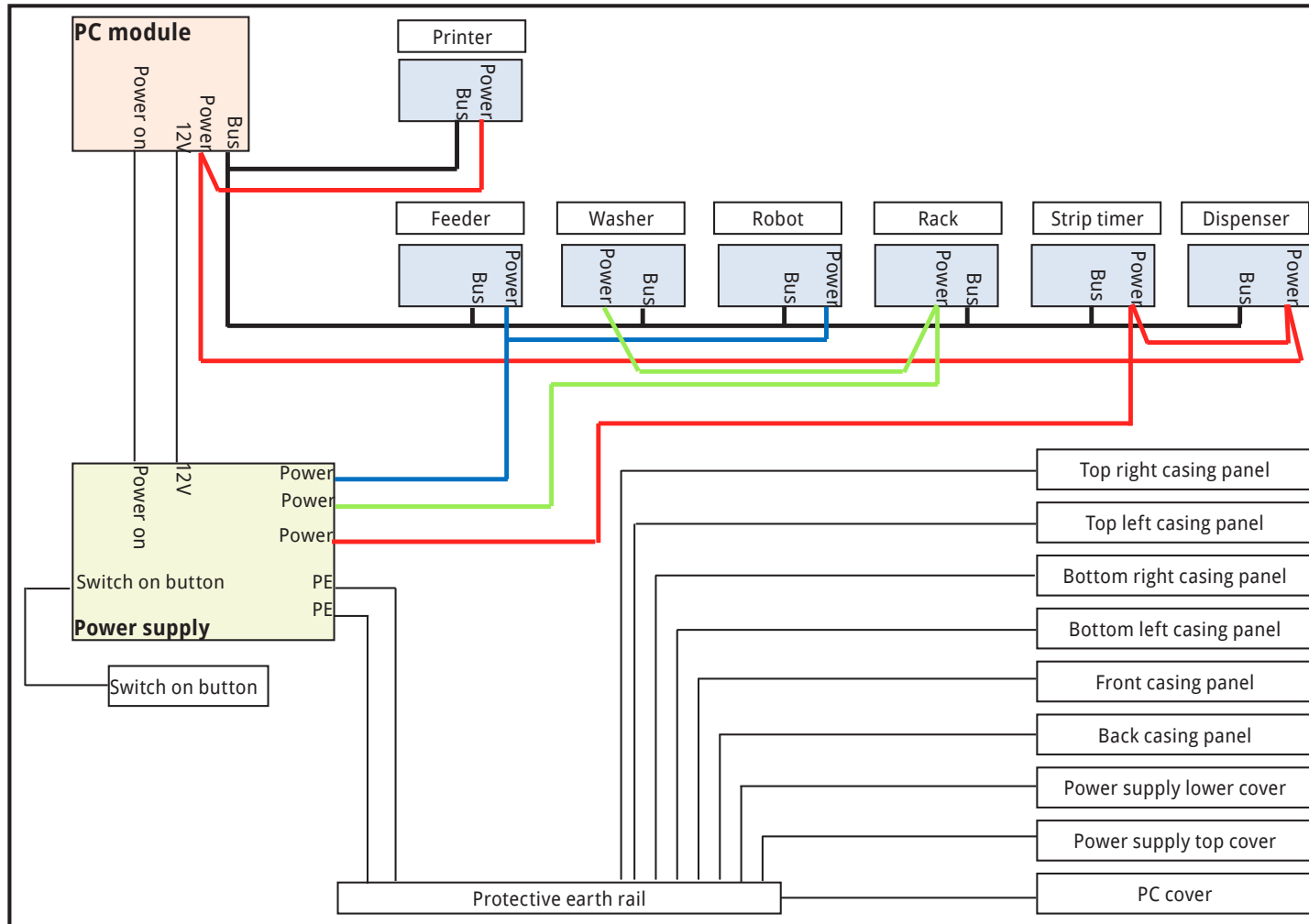


Figure 17: The block diagram of the LabUMat 2 analyzer

Each of the modules in the LabUMat 2 automated urine chemistry analyzer can function separately; the embedded PC synchronizes and controls the modules' operation through an internal data bus interconnecting their control circuit boards (see Figure 17). The module circuit boards that require direct communication with each other instead of routing data through the embedded PC have one or more of the sliders on the double 8-way configuration DIP switches turned on.

 **Parts that are included in the [Product glossary](#) are printed in bold in this section.**

4.1.1 The Feeder and strip pusher module

A system of several cylinders and a set of strip turner wheels that receive fresh LabStrip U11 Plus GL test strips and forward them one at a time and with their pads facing up towards the pipetting probe.

The operator loads test strips into a removable cylinder with a bayonet mount. This **feeding tube** is hermetically housed within another one.

 The openings on the two tubes will align and let the strips drop onto the feeder drum if the operator locks this double feeding tube hermetically into the device.

Test strips drop from the feeding tube onto the **feeder drum**. The rotating feeder drum makes sure that strips are forwarded to the pipetting probe one at a time.

The feeding module bracket houses a second, stationary cylinder below the feeder drum. Test strips that were fed into the feeding tube but are not used in the automated measurement process are collected here.

The **unused strip bin** is accessible separately via a threaded cap.

Three rotating strip turner wheels turning on a pivot intercept the test strips coming from the feeder drum and turn them over if the pads on the strips are facing down. The **strip turner wheels** drop the test strips in the way of a pair of clamps that move horizontally from left to right along a gear rack. The **strip clamps** catch the strips by their handles and forward them first to the pipetting probe then on to the comb conveyor.

4.1.2 The Washer module

A fluidic system of two peristaltic pumps, a membrane pump, two pipetting syringes, and two electromagnetic valves to seal off sections of the system. One peristaltic pump (the '**Inner washer pump**') drives distilled water from the water tank through the water line and into the rinse well through the pipetting probe. It also makes sure that there is an unbroken stream of distilled water in the water line at all times.

The other peristaltic pump (the '**Outer washing pump**') pumps distilled water into the **rinse well**, a chamber where the pipetting probe is lowered into for rinsing and disinfection.

The membrane pump (the '**Waste pump**') drains the rinse well. It is automatically started up even before any of the peristaltic pumps start running and is stopped only after they have stopped.

One of the pipetting syringes aspirates urine from the test tubes and dispenses it onto the test strip pads (the 250µL-volume **Aspirating syringe**). The other mixes the urine in the test tubes (the 2.5 mL-volume **Mixing syringe**). They also make sure that the unbroken stream of

distilled water in the fluidic system extends exactly up to the tip of the pipetting probe.

The valve inside the manifold (the **Aspiration valve**) prevents the movements of the syringe pumps from reversing the flow of distilled water in the Inner washer pump. The **Waste line sealing valve** stops the disinfectant solution from draining away from the rinse well during the daily disinfection and prevents any liquid entering the membrane pump when it is not running.

4.1.3 The Linear robot module

A system of two stepper motors (the **Horizontal motor** and the **Vertical motor**) that move a **pipetting probe** along the horizontal and vertical axes respectively.

The two pipetting syringes (see the Washer and the Sampler modules) aspirate urine samples from the test tubes and dispense the samples onto LabStrip U11 Plus GL test strips via the pipetting probe. The probe shuttles between the test tubes and the rinse well, stopping over each pad on the test strips to dispense a particular amount of urine.

The pipette probe also serves as an antenna for the oscillator's electronic signal that is used to gauge the amount of liquid present in the test tubes.

4.1.4 The Rack transport module

An integrated system of three submodules:

- a **rack conveyor** unit that moves the test-tube-filled racks from right to left, delivering them to and from the opening of the **rack passage**

via a pair of conveyor belts

- a **rack delivery zone** in front of the rack passage with two sets of pins that allow only one rack at a time at the rack passage and that steady arriving racks
- a **rack positioning unit** that draws fresh racks into the device, positions each test tube below the probe, then pushes the processed racks back on the rack conveyor

The integrated **barcode reader** identifies properly labeled test tubes inside the racks.

4.1.5 The Strip timer and photometer module

The module is an integrated system of two submodules:

- The **strip timer** unit that transports test strips from the pipetting stage to the photometer and that ensures that the preset incubation time elapses between urine sample dispensing and the measurement of pad discoloration. The **strip timer comb**, a plate with a series of prongs, is moved by the **Vertical strip timer motor** and the **Horizontal strip timer motor**. The strip timer comb is able to handle up to four test strips at a time, stacking them across the **platform frame** on the way to the photometer.
- The **photometer** unit, moved by the **Head mover motor**, lights up its four preprogrammed-wavelength LEDs above each of the pads of the test strip delivered on the **strip reading platform**. The reflectance values are compared to a **photometry reference pad** (see the relevant section of the overview of the principle of operation).

After a photometer reading cycle, the strip timer comb pushes the strips into the **strip bin**, located below the strip reading platform, where the processed strips are collected, ready for disposal.

4.1.6 The Sample dispenser module

A system of two pipetting syringes (the **Aspirating syringe** and the **Mixing syringe**, see [The Washer module](#)), connected to the fluidic system via a manifold and an electromagnetic valve (the **Aspiration valve**, see [The Washer module](#)).

4.1.7 The Printer module

An optional thermal printer to print analysis results. See [Technical specifications](#) for details.

4.1.8 The Physical Measurement Cell module

< The Physical Measurement Cell (PMC) is the module used to identify three (3) basic physical characteristics of the urine sample (as opposed to the chemical features the test strip pads determine) that test strip-based photometry is unable to determine. (There is a pad on the LabStrip U11 Plus GL test strip to determine specific gravity, but the LabUMat 2 system ignores it during both sample dispensing and photometry.) The physical

- specific gravity (relative density)

The specific gravity measurement is based on the principle that the refractive index of urine (how much the direction of light changes as it

enters it) varies with its specific gravity. Inside the PMC a laser beam is passed through the urine sample and the angle of its refraction is measured. The specific gravity of the sample is calculated based on the sample's refractive index. Since specific gravity is temperature-specific, the calculations are corrected for 20°C (68°F).

- turbidity (how clear or cloudy the urine is)

For the turbidity measurement, the light from four (4) colored LEDs is passed through the urine sample and the ratio of scattered light to total amount of light is determined at two (2) light detectors, one measuring scattered light, the other measuring direct light. There is a total of ten (10) signals—both detectors receive signals from all four LEDs, and there is a no-signal reading on each detector for reference purposes— that are compiled to determine the ratio of scattered light.

- color

The same array of four (4) LEDs and two (2) detectors used to measure turbidity is utilized to determine the color of the sample. This measurement is also based on how much the sample scatters the light of the yellow, blue, green, and red LEDs. The resulting color coordinates are plotted on a color space to determine the color of the particular sample >

4.1.9 The PC and LED touchscreen module

An embedded industrial PC that runs a Windows operating system. It controls all the device processes and is linked up with every module circuit board via a system of bus cables. The PC shares a module housing plate with a resistive touchscreen that is LabUMat 2's primary user

interface. The display can be operated by tapping its touch-sensitive areas with bare fingers, gloved fingers, pens with retracted tips, or any stylus-like object.

4.1.10 The Power supply

Supplies electric power to the embedded PC and all the module PCBs. Includes two subunits: one generating 12, the other 24 Volts of electric power. Only the PC and Touch screen module is supplied with 12 V power directly—the module PCBs are supplied with 5 Volt electric power (converted from the 12 V power) that powers the transmissive photointerrupter-type sensors, and 24 Volt electric power that powers the motors and inductive-type sensors connected to the module PCBs. A second two-pin cable runs between the power supply and the PC module, that allows the device hardware to be powered down by exiting from the device software.

Replaceable parts: the 2 (two) fuses, the standby switch, and the power switch.

⚠ Always power down the device completely before attempting to service the power supply. The power supply has a stand-by mode, which means that unless the device is powered down completely by pressing the power switch after exiting from the device software, there is electric current present in the module.

4.2 Disassembling the device casing

⚠ Do not attempt to remove the bottom casing panel.

4.2.1 Removing the back casing panel

- 1** In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.
- 2** Unscrew and save the ten (10) 2.5-millimeter hex socket screws with flat washers that secure the back casing panel (see [Figure 18](#)).
- 3** Disconnect the grounding cable from the clamp on the reverse of the casing panel and remove the panel from the back of the device.
- 4** Complete the previous steps in reverse order and direction when replacing the back casing panel.

4.2.2 Removing the front casing panel

- 1** In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.
- 2** Release the rack conveyor from the front of the device.
- 3** Unscrew and save the twelve (12) size T10 Torx screws that secure the front panel to the device chassis around the edges.
- 4** Disconnect the grounding cable from the clamp on the reverse of the casing panel and remove the panel from the front of the device.

5 Complete the previous steps in reverse order and direction when replacing the front casing panel.

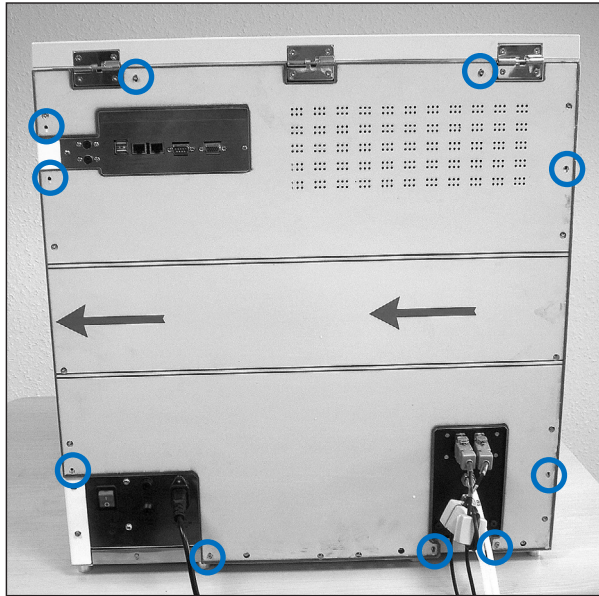


Figure 18: The back casing panel securing screws

4.2.3 Removing the top side casing panels

1 In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.

2 Complete steps 2–3 in [4.2.1](#) to remove the back casing panel.

3 Unscrew and save the three (3) 2-millimeter button head hex socket screw along the front edge of the top side casing panel (see Figure 19).

4 Unscrew and save the three (3) 2.5-millimeter hex socket screws along the back edge of the top side casing panel you wish to remove (see Figure 19).

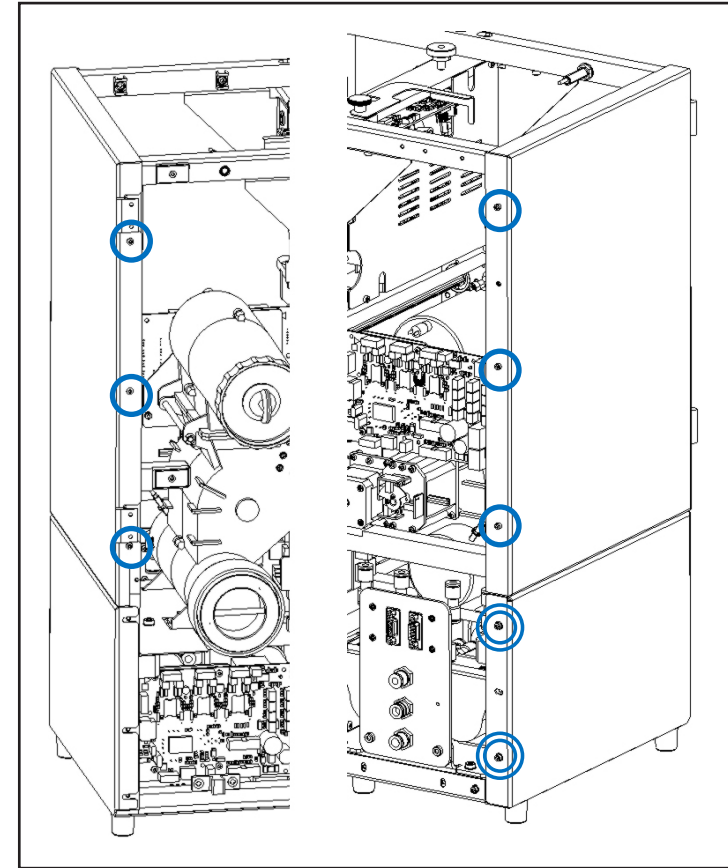


Figure 19: The securing screws of the side casing panels at the front (left) and the back (right) of the device. The screws with flat washers are marked with a concentric circle.

5 Disconnect the grounding cable from the clamp on the reverse of the casing panel and remove the panel from the side of the device.

6 Complete the previous steps in reverse order and direction when replacing the top side casing panel.

4.2.4 Removing the bottom side casing panels

1 In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.

2 Complete steps 2–4 in [4.2.2](#) to remove the front casing panel.

3 Complete steps 2–3 in [4.2.1](#) to remove the back casing panel.

4 Unscrew and save the two (2) 2.5-millimeter hex socket screws with flat washers along the back of the bottom side casing panel you wish to remove (the screws marked with concentric circles in [Figure 19](#)).

5 Disconnect the grounding cable from the clamp on the reverse of the casing panel and remove the panel from the side of the device.

6 Complete the previous steps in reverse order and direction when replacing the bottom side casing panel.

4.2.5 Removing the top casing panel

1 In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.

2 Disconnect the power connector and the bus cable from Circuit board number 7 (see [The printer module control circuit board](#))

3 Lift the lid until it is perpendicular to the top of the device chassis, then slide the top casing panel off the pivot of its hinges.

4 Complete the previous steps in reverse order and direction when replacing the top casing panel.

4.2.6 Removing the doors of the device

1 In the Systest software on the **Flow tests** menu, tap the **Disable Interlock** button to disable the door sensors.

2 Open the doors and unscrew the four (4) 2.5-millimeter hex socket screws (2 each) that secure the door leaves to their metal hinges.

3 Slip the door leaves out of their hinges and lift them off the device.

4 Complete the previous step in reverse order and direction when replacing the device doors.

5 The Feeder and Strip pusher module

i For a detailed overview of the module, see [4.1.1 The Feeder and strip pusher module](#)

5.1 Motors



5.1.1 'Feeder motor' [S-ALT4510101]

stepper motor that rotates the feeder drum

plugged in to connector 29 on Circuit board number 1

home position: identical to the 'lower empty' position

significant positions:

- check position:

Step number-determined motor position.
The test strip groove (1) is directly below the
Flipped strip sensor (3).

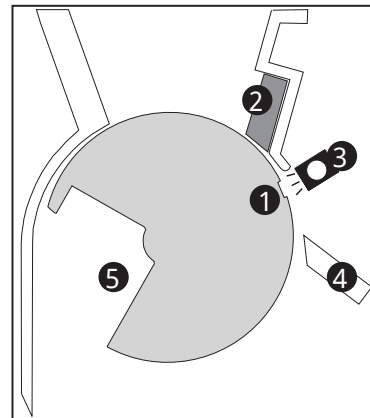


Figure 20: Schematic illustration of the feeder drum check position

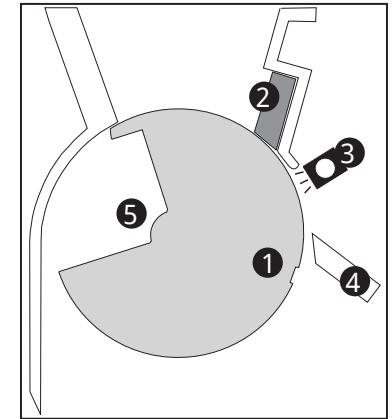
i Tapping the **To check position** button on the **Card tests** menu on the **Feeder** card tab in the **Systest** software will only move the feeder drum if it is not in the home position at the time.

- drop position

The test strip groove (1) is moved clockwise (when seen from the front) below the level of the strip turner ramp (4).

sensor-determined motor position

Figure 21: Schematic illustration of the feeder drum drop position

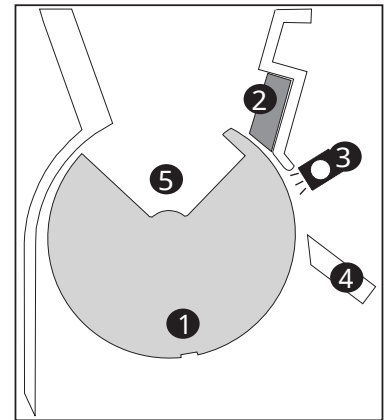


- upper empty

The feeder drum's scoop (5) is turned clockwise (when seen from the front) towards the feeding tube opening, ready to receive strips.

sensor-determined motor position

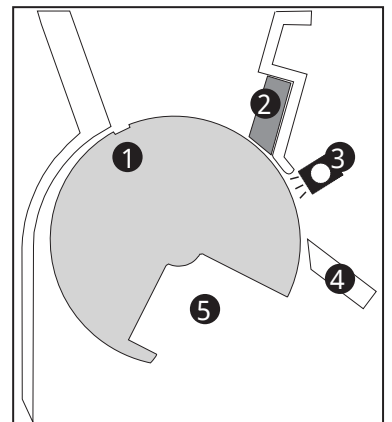
Figure 22: Schematic illustration of the feeder drum upper empty position



- lower empty

The feeder drum's scoop is turned clockwise (when seen from the front) towards the unused strip bin, emptying itself. The test strip groove (1) is aligned with the

Figure 23: Schematic illustration of the feeder drum home and lower empty positions



edge of the feeder module bracket.
sensor-determined motor position

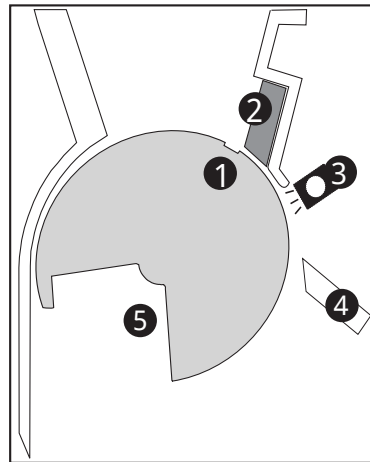
i Although the lower empty position is identical to the feeder drum's home position, the 'lower empty position' is actually a complete motor revolution away from the feeder drum home position.

- feeder blade

The feeder drum moves its test strip groove under the feeder blade (2).

motor step number-determined motor position

Figure 24: The feeder drum feeder blade position



- back from comb

The feeder drum is moved 25 motor steps anti-clockwise (when seen from the front) very fast, then back again. This movement is performed during normal operation each time after the test strip groove is moved under the feeder blade to prevent test strips from jamming under the feeder blade.



5.1.2 'Strip turner motor' [S-ALT4510111]

stepper motor that rotates the strip turner wheels
plugged in to connector 26 on Circuit board number 1

home position: The claws (2) of the strip turner wheels are facing up, ready to intercept and turn over the arriving strip.

sensor-determined motor position

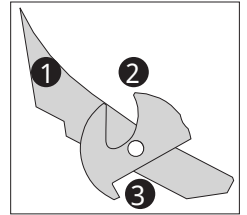


Figure 25: The strip turner wheel home position

significant positions:

- holding position

The spurs (3) opposite the claws of the strip turner wheels completely stand out from the plane of the strip turner ramp (1), ready to catch but not turn

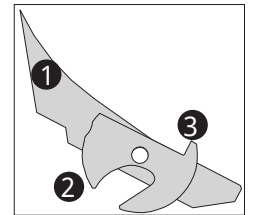


Figure 26: The strip turner wheel holding position

over the arriving strip.

motor step number-determined motor position

- slipping position

The spurs (3) of the strip turner wheels are moved below the plane of the strip turner ramp (1). Arriving strips will freely pass through.

motor step number-determined motor position

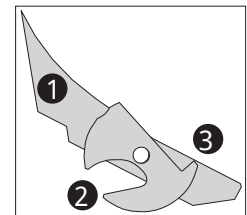
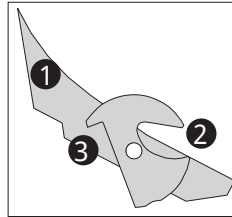


Figure 27: The strip turner wheel slipping position

- turning position

The claws (2) of the strip turner wheels have made a quarter revolution: the lower claw is now below the plane of the strip turner ramp (1). The wrong-side up test strip has been turned over.



motor step number–determined motor position

Figure 28: The strip turner wheel turning position



5.1.3 'Strip pusher motor' [S-ALT4510021]

stepper motor that moves the pair of clamps along a gear rack plugged in to connector 1 on the strip pusher motor interface board

home position: 'Go to feeder'

significant positions:

- 'Go to pipetting'

The clamps forward the test strip directly below the pipetting probe.
motor step number-determined position, although a sensor is also present

- Go to strip timer comb

The clamps forward the freshly pipetted test strip to the strip timer comb.
motor step number–determined motor position

5.2 Sensors



5.2.1 'Feeder motor position sensor/Strip turner home position sensor' [S-ALT4560041]

type: transmissive photointerrupter

connectors (on Circuit board number 1)	corresponding motor position	corresponding systest sensor signal
6	'strip turner home'	'strip turner home'
16	'lower empty'	Feeder drum safe area
12	'upper empty'	'Upper empty pos.'
8	'drop'	'Feeder drop'
5	'home'	'Feeder home'

Systest sensor signal	 	
Strip turner wheel home	The strip turner wheels are in their home position.	Sensor not activated.
Feeder drum safe area	The feeder drum is between the 'lower empty' and the 'drop' motor positions.	Sensor not activated.
Upper empty pos.	The feeder drum is in the 'upper empty' position.	Sensor not activated.

Systest sensor signal		
Feeder drop	The feeder drum is in the 'drop' position.	Sensor not activated.
Feeder home	The feeder drum is in the 'home' position.	Sensor not activated.

i The Feeder drum safe area sensor is included to indicate if the feeder drum should accidentally start emptying itself during the strip feeding process. Because of this, the Feeder drum safe area sensor must be activated all through the feeding process.

The device is properly set up and operational only if the following sensors are active (their signal is red) after system initialization (See [2.3.2 The Sensor tests menu](#)): Strip turner home, Feeder drum safe area, Feeder home.



5.2.2 Flipped strip sensor [S-UPA4402143]

type: customized photoelectric sensor

plugged in to connector 17 on Circuit board number 1

Systest sensor signal		
Flipped strip sensor	There is a strip under the Flipped strip sensor. The real-time reflectance value is also indicated.	Sensor not activated.

The device is properly set up and operational only if the Flipped strip sensor is inactive (its signal is green) and the real-time value next to it is

below 120 after system initialization. (See [2.3.2 The Sensor tests menu](#).)



5.2.3 Strip progress sensor [S-ALT4550031]

type: BOS 2K photoelectric sensor

plugged in to connector 21 on Circuit board number 1

Systest sensor signal		
Strip progress	A strip has passed the sensor. The next strip is ready to be dispensed.	Sensor not activated.

The device is properly set up and operational only if the Strip progress sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)



The Strip presence sensor is activated only when a test strip passes below it. Because of this, during normal operation, its red signal will regularly flash red but will revert to green.



5.2.4 Strip pusher position sensor [S-4560231]

type: transmissive photointerrupter

connectors (on Circuit board number 1)	corresponding motor position	corresponding systest sensor signal
13	'pipetting'	'Pipetting pos.'
9	'home'	'Pusher home'

Systest sensor signal		
Pusher home	The strip clamps are in home position.	Sensor not activated.
Pipetting pos.	The strip clamps are under the pipetting probe.	Sensor not activated.

The device is properly set up and operational only if the Pipetting position sensor is inactive (its signal is green), and the Pusher home sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



5.2.5 Feeding tube microswitch [S-ALT4580021]

type: microswitch

plugged in to connector 14 on Circuit board number 1

Systest sensor signal		
Feeding tube	The feeding tube is in place.	Sensor not activated.

The device is properly set up and operational only if the Feeding tube sensor is active (its signal is red) after system initialization. (See the [Sensor overview](#) chart.)



5.2.6 'Unused strip bin inductive sensor' [S-ALT4540071]

type: inductive

plugged in to connector 22 on Circuit board number 1

Systest sensor signal		
Unused strip bin	The unused strip bin is in place.	Sensor not activated.

The device is properly set up and operational only if the Unused strip bin sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



5.2.7 Pipetting stage presence sensor [S-ALT4550021]

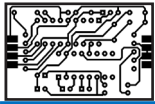
type: BOS 5K photoelectric sensor

plugged in to connector 18 on Circuit board number 1

Systest sensor signal		
Pipetting stage	The pipetting stage [S-UPA2511592] is in place.	Sensor not activated.

The device is properly set up and operational only if the Pipetting stage sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))

5.3 PCBs



5.3.1 Feeder and Strip pusher module control circuit board [S-UPA43011]

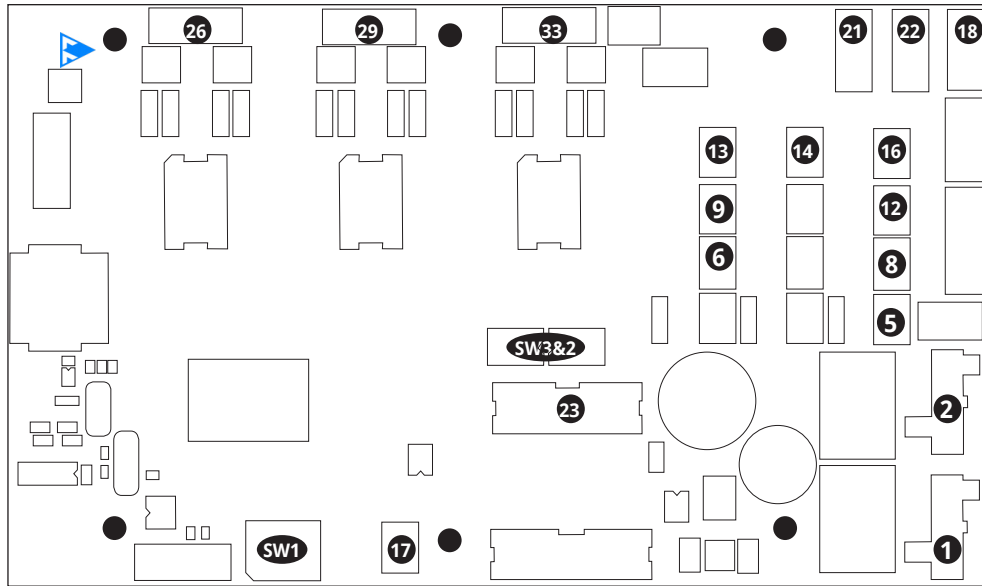


Figure 29: Board layout of Circuit board number 1.

Key to Figure 29 and Figure 30

- 26. Strip turner motor connector
- 29. Feeder motor connector
- 33. Strip pusher motor interface board connector
- 21. Strip progress sensor connector
- 22. Unused strip bin sensor connector
- 18. Pipetting stage presence sensor connector
- 13. Strip pusher pipetting position sensor connector

- 9. Strip pusher home position sensor connector
- 6. Strip turner home position sensor connector
- 14. Strip feeding tube microswitch connector
- 16. Feeder motor 'lower empty' position sensor connector
- 12. Feeder motor 'upper empty' position sensor connector
- 8. Feeder motor 'drop' position sensor connector
- SW3&2. Two 8-way configuration DIP switches (with all sliders off—in the up position)

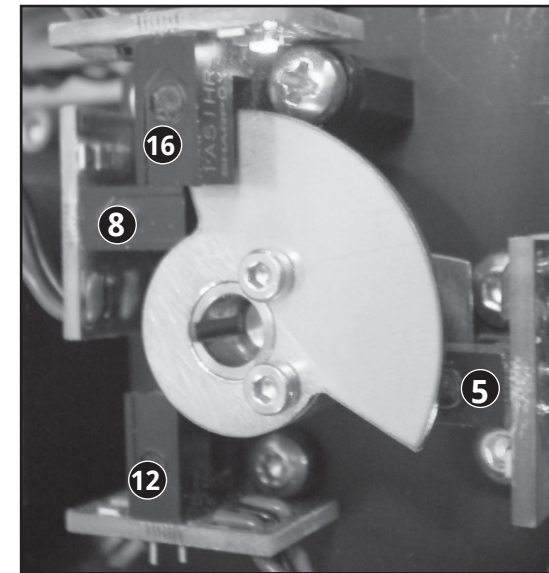
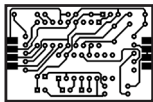


Figure 30: Locations of the feeder motor sensors

- 5. Feeder motor 'home' position sensor connector
- 23. Bus cable connector from Circuit board number 3 (Linear robot)
- 1. Power connector for the power supply*
- SW1. PCB address DIP switch (only slider 1 is on—in the down position)

* Connectors 1 and 2 are interchangeable

- 17. Flipped strip sensor connector
- 2. Power connector from Circuit board number 3*



5.3.2 Strip pusher motor interface board [UW6-4216-1]

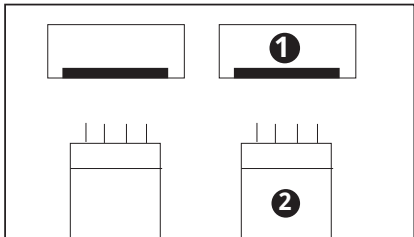


Figure 31: Board layout of the strip pusher motor interface board

Key to Figure 31

1. Strip pusher motor connector 2. Board-to-board connector

5.4 Feeder module Flow tests processes



5.4.1 Initialization

1	The strip turner wheels are first moved into their home position, then moved through two (2) full turning cycles.	
2	The strip forwarder clamps are moved into their home position ('Go to feeder').	
3	The feeder drum is moved into its home position.	The strip forwarder clamps are moved into the 'Go to comb' position, then back into their home position.



5.4.2 Feeding test

- 1 Tapping the **Feeding test** button once starts a movement of the feeder drum that will forward a test strip to directly under the Flipped strip sensor. The sensor should be activated (its signal should turn red).
- 2 Tapping the button a second time makes the feeder drum move into its 'drop' position. The Feeder drop sensor should be activated (its signal should turn red). If you have put test strips into the feeding tube, a test strip should arrive at the strip turner wheels.
- 3 Tapping the button a third time makes the Strip pusher motor move the strip clamps to their 'Go to pipetting' position. The Pipetting pos. sensor should be activated (its signal should turn red). If you have put test strips into the feeding tube, a test strip should arrive at the

pipetting stage.

 **The strip clamps will not move back to their home position unless you tap Go to feeder under Card tests on the Feeder card tab in the Systest software.**



5.4.3 Feeder empty test

Tapping the **Feeder empty test** button makes the Feeder drum move into its 'lower empty' position.



If you exit the User software and choose to perform the daily rinsing procedure, the feeder drum will go to its 'lower empty' position.



5.4.4 Shutdown test

The Feeder and Strip pusher module performs the same operations during Shutdown as during Initialization.

5.5 Service actions—Adjustments



5.5.1 Accessing the preset system parameters

You can review the preset system parameters for the feeder and strip pusher module in the Systest software on the **System parameter setup** menu on the **Feeder card** tab by tapping the **Read parameters** button.

- **Strip detect** indicates the lower cutoff value for the Flipped strip sensor: the minimum difference between the case when no strip is present (the sensor senses the gray surface of the feeder drum), and the case when the pads on the strip face the sensor (the sensor senses the dark blue surface of the ASC pad)
- **Strip turning** indicates the higher cutoff value for the strip orientation sensor: the minimum difference between the case when the pads on the strip under the sensor face the sensor (the sensor senses the dark blue surface of the ASC pad), and when the test strip is turned the wrong way up (the sensor senses the white surface of the plastic carrier ribbon)



Do not modify the preset system parameters.

- **Robot correction** displays a corrective parameter that is added to the strip pusher motor **Go to pipetting** motor position. Negative values shift the clamps towards the comb conveyor; positive values shift the clamps towards the feeder unit.



Do not modify this Robot correction value. See [4.5.4 Setting up the correct pipetting position](#) for information about adjusting the corresponding motor position.



5.5.2 Accessing the preset motor parameter settings

You can review the preset motor parameters for each of the motors in the feeder and strip pusher module in the Sytest software under **Motor parameter setup** on the **Feeder card** tab by entering the **Moving parameter settings** menu and tapping **Read parameters**.

⚠ It is strictly forbidden to modify any of these values without previous permission from the 77Elektronika technical support team.



5.5.3 Adjusting the feeder blade

⚠ The clearance between the feeder drum and the feeder blade should be 0.15 millimeters so that only one test strip sitting in the feeder drum's groove can pass underneath the blade.

1 Align the right edge (the one towards the pipetting probe) of the groove of the feeder drum with the feeder blade.

< 2 Loosen the two (2) hex socket button head set screws securing the feeder blade on the feeder bracket.

i Because thread-locking fluid is applied in manufacturing, you will need to manually jiggle the feeder bar to be able to move it after loosening the screws.

3 Use a feeler gauge to set a clearance of 0.15 millimeters between the feeder blade and the edge of the feeder groove below both set screws, then re-tighten the screws.

4 Use a LabStrip U11 Plus GL test strip to check whether only test strips that sit snugly in the feeder groove can pass under the feeder blade.

5 Reapply thread-locking fluid over the button heads.

i You can use ordinary nail polish instead of commercial thread-locking fluid.



5.5.4 Setting up the correct pipetting position

Setting up the correct pipetting position—where the strip clamps forward the test strips for pipetting, which must be directly below the pipetting probe—has two components. One is fine-tuning the Strip pusher motor **Go to pipetting** motor position. The other is fine-tuning the horizontal and vertical positions where the pipetting probe must be moved before pipetting starts (**Go to strip** and **To above strip** respectively). It is recommended that you adjust all these positions in the Systest software under **Position setup** on the **Robot full strip position** tab.

i You can also adjust the Strip pusher motor's **Go to pipetting** motor position with the **Robot correction** button on **System parameter setup** menu on the **Feeder card** tab. However, this method only adjusts the motor position indirectly (by modifying the step number listed in the Strip pusher motor's **Go to pipetting** box under **Motor parameter setup** on the **Feeder card** tab). You will not be able to immediately check the results of your adjustments and so it is recommended that you use the **Position setup** screen.

i All of the movement buttons on the **Position setup** screen refer to movements of a single motor step in the direction indicated.

1 To check whether the pipetting probe's horizontal movement is perfectly parallel with the test strip, place a test strip in the middle of

the pipetting stage and move the probe manually over it. If the pipetting probe seems to be at an angle with the test strip, see [Adjusting the linear robot alignment](#).

2 Run an **Initialization** in the Systest software on the **Flow tests** menu.

3 Place a test strip with the pads facing up in front of the strip clamps in the position where they would arrive from the strip turner wheels.

4 Under **Position setup** on the **Robot full strip position** tab, tap the **Position setup** button. The strip clamps are moved into their **Go to pipetting** position, and the pipetting probe is moved first into sampling position, then above the frontmost (Bilirubin) pad on the test strip.

5 Check that the tip of the pipetting probe is directly above the center of the pad and that there is a clearance of 0.5 millimeters between the tip and the pad's surface.

- If the pipetting probe deviates from the center of the pad along the left–right axis, tap the **Left** or **Right** buttons in the **Robot correction** row to adjust the pipetting position of the Strip pusher motor

 The strip clamps are supposed to push the test strips but are not designed to pull them. If you need to adjust the clamps to the left, you will need to make sure that the strip stays inside the clamps by moving it with your hands.

- If the pipetting probe deviates from the center of the pad along the backward–forward axis, tap the **Front** or **Back** buttons in the **Strip X correction** row to adjust the Robot X motor's **Go to strip** motor position

- If the clearance between the tip of the probe and the surface of the pad is more or less than 0.5 millimeters, tap the **Down** or **Up** buttons in the **Strip Y front corr.** row to adjust the Robot Y motor's **To above strip** motor position

6 Tap the **Store the position** button to store the adjusted values. The system does not confirm successful storage of the adjustments directly but displays a confirmation message under **Messages**.

 Because of a manufacturing process variance, the pipetting stage on which the test strips sit during pipetting may deviate from a completely horizontal position. The **Robot Y strip rear position** tab has been added to the **Position set-up** screen so that you can compensate for any such deviation, if you encounter it.

7 To adjust the clearance between the pipetting probe and the surface of the hindmost (Specific gravity) pad, tap the **Down** or **Up** buttons in the **Strip Y rear corr.** row in the Systest software under **Position setup** on the **Robot Y strip rear position** tab. The optimal clearance is the same as for the frontmost pad: 0.5 millimeters.

8 Tap the **Store the position** button to store the adjusted values in the system.

 It is recommended that you adjust the clearance between the pipetting probe and the hindmost pad as part of the pipetting position setup, because if you tap the **Robot Y strip rear position** button after tapping **Initialization**, the strip clamps will not be moved into their pipetting position.

9 Tap **Initialization** under **Position setup** to move the strip clamps and the pipetting probe back to their initial positions (**Go**

to **pipetting** and above the frontmost pad respectively). To check whether no further adjustments are needed:

- 1 Tap the **Go to pipetting** button in the **Strip pusher motor** row in the Systest software under **Card tests** on the **Feeder card** tab
- 2 Tap the **Robot test** button on the **Flow tests** menu and check whether the pipetting probe stops above the center of each pad during pipetting

5.6 Service actions—Replacements



5.6.2 Replacing the Feeder and strip pusher module [S-UPA4402005]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect the following cables from Circuit board number 1 (See [Figure 29](#)):

- the power cable from connector 1 or 2
- the bus cable from connector 23
- the strip pusher motor interface board connector cable from connector 33
- the strip pusher motor position sensor cables from connector 13 and 9
- the pipetting tray presence sensor cable from connector 18

3 Unscrew the two (2) 3-millimeter hex socket screws that secure the guard plate next to the feeder bracket base opposite the strip turner wheels. Save the serrated washers.

4 Unscrew the three (3) 3-millimeter hex socket screws that secure the feeder module on the device chassis. Save the serrated washers.

 Some models have four (4) securing screws. With these models, you will need to remove the Washer module (see [Replacing the Washer module](#)) to access the last screw.

5 Firmly take hold of the feeder bracket and gently maneuver the module out of the device chassis. Complete the previous steps in

reverse order when installing a new module.

6 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.3 Replacing the Feeder motor [S-ALT4510101]

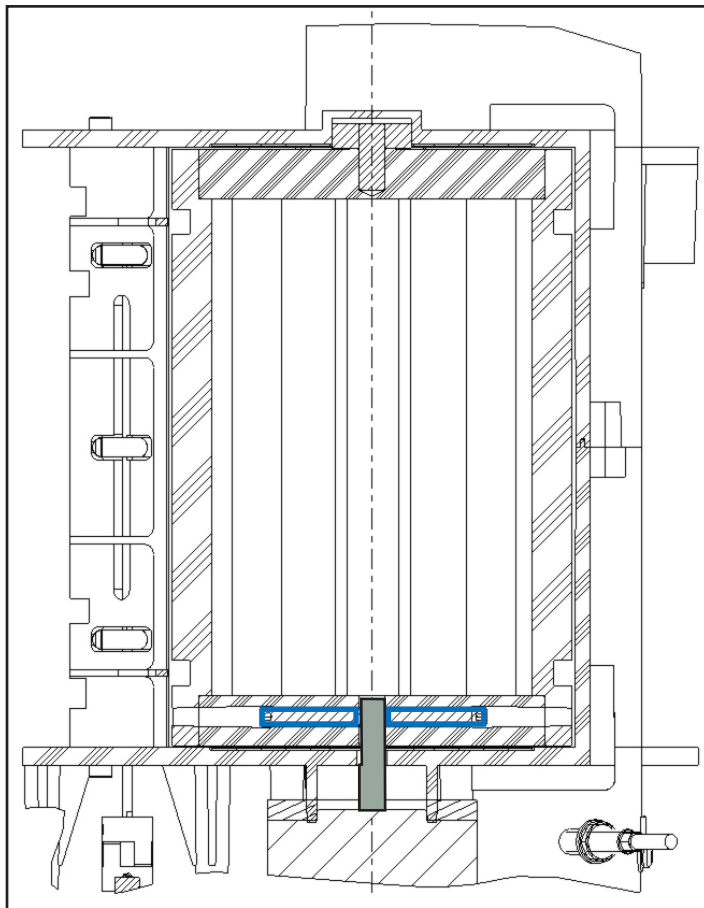


Figure 32: Top view cross section of the feeder drum with the feeder motor axle and its set screws

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect connectors 5, 8, 12, 16, and 29 from Circuit board number 1 (see [Figure 29](#) and [Figure 31](#)).

3 Unscrew the four (4) Pozidriv screws that secure the feeder motor to the spacer bars coming from the feeder bracket, and the 2-millimeter hex socket button head screw with a 5.5-millimeter hex nut that secures the motor to the circuit board plate. Save the hex nut.

4 Unscrew the two (2) 2-millimeter hex socket set screws inside screw shafts drilled into the feeder drum on opposite sides of the cylinder wall, that engage with the feeder motor axle (marked blue in Figure 32).

5 Remove the feeder motor. Unscrew the four (4) 2.5-millimeter hex socket head screws that secure the plate with the feeder motor sensors to the motor you are replacing, and screw the plate with the sensors on the new feeder motor.

6 Secure the new feeder motor to the feeder drum by driving in the screws you removed in steps 3–4, and connect the sensor cables you removed in step 2.

7 Switch on the device and launch the Systest software. Under **Card tests** on the **Feeder card** tab, tap **STOP** in the **Feeder motor** row.

i Ignore any error messages that are displayed during system initialization at this point.

8 Loosen the 1.5-millimeter hex socket head set screw of the feeder motor position sensor cam (see Figure 33) so that you will be able to adjust it manually.

9 Manually position the feeder drum into its home/lower empty position (see [Figure 23](#)). In the Systest software under **Sensor tests**, check that the 'Feeder home' and the 'Sector sensor' sensors are active. If necessary, adjust the sensor cam until both sensors are active.

10 Retighten the set screw of the sensor cam. You may need to rotate the feeder drum to access the screw. If so, make sure that the sensor cam does not shift while you rotate the feeder drum.

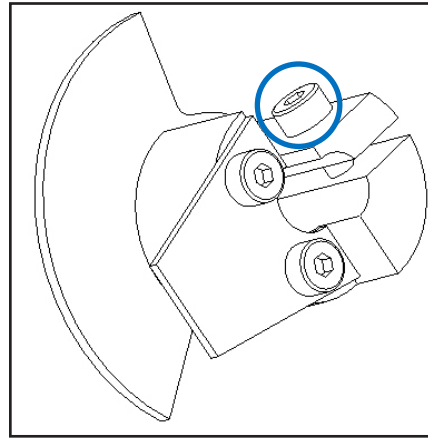


Figure 33: The feeder motor position sensor cam's set screw

< 11 In the Systest software under **Card tests** on the **Feeder card** tab, tap all the motor positions in the **Feeder motor** row and check whether the corresponding sensors light up (see [2.3.2 The Sensor tests menu](#) and [5.2 Sensors](#)). Repeat steps 8–9 as necessary.

12 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.4 Replacing the feeder bracket sealing [S-UPA0500161]

- 1** Switch on the device, launch the Systest software and on the **Flow tests**, menu, tap **Initialization**.
- 2** Undo the buckle of the feeder bracket, and pry off the sealing you wish to replace.
- 3** Clean the surface underneath the sealing with a solvent such as nail polish remover to remove traces of adhesive.
- 4** Carefully put the new self-adhesive sealing in place, pressing down on it all around the bracket.
- 5** In the Systest software on the **Flow tests** menu, run an **Initialization** before resuming normal operation.



5.6.5 Replacing the Strip turner motor [S-ALT4510011]

- 1** See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2** Disconnect connectors 26 and 6 from Circuit board number 1 (see [Figure 29](#)).
- 3** Unscrew the four (4) size Pozidriv screws that secure the strip turner motor to the feeder bracket, and the 2-millimeter hex socket button head screw with a 5.5-millimeter hex nut that secures the motor to the circuit board plate. Save the hex nut.

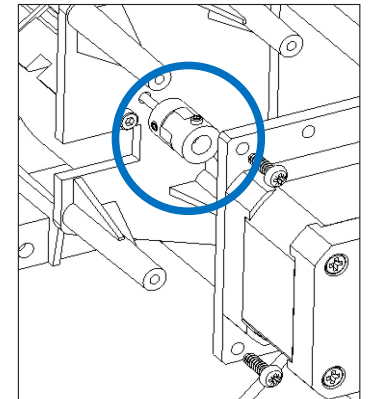


Figure 34: Detail of an exploded diagram of the strip turner unit with the strip turner motor coupler

4 Unscrew the 1.5-millimeter hex socket flat point set screw on the side of the coupler towards the motor (see Figure 34).

5 Remove the strip turner motor and insert a new one. Complete the previous steps in reverse order and direction. In the Systest on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.6 Replacing the Strip turner motor coupler [S-UPA5000651]

1 Complete steps 1–4 of [4.6.5](#).

2 Unscrew both 1.5-millimeter hex socket flat top set screws that hold the strip turner motor coupler in place.

3 Remove the coupler, replace it with a new one, and complete the previous steps in reverse order and direction.

4 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.7 Replacing the Strip pusher motor [S-ALT4510021]

1 See [4.2.2 Removing the front casing panel](#) and remove the front casing panel.

2 Push the strip clamps as far as they go towards the comb conveyor to get them out of the way. To remove the platform support frame, unscrew the following screws:

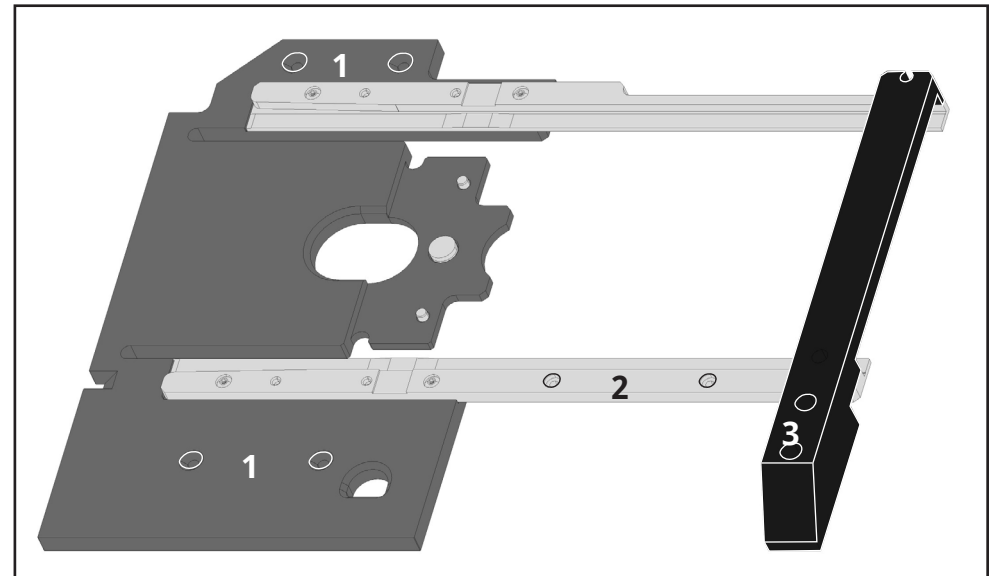


Figure 35: The screws that secure the platform support frame to the chassis

- the four (4) 2.5-millimeter flat head hex socket screws marked '1' in Figure 35
- the two (2) 2-millimeter hex socket button head screws that fix the frame and a spacer bar to the device chassis, marked '2' in Figure 35
- the two (2) 3-millimeter hex socket head screws marked '3' in Figure 35

Remove the platform support frame and the spacer bar.

3 Unscrew the three (3) Torx size T10 screws that secure the bottom plate to the device chassis and remove it from under the platform support frame.

4 Find the cable guard plate (see Figure 36) opposite the bottom plate and unscrew the two (2) 2.5-millimeter hex socket button head screws that secure it to the device chassis. Handle the cables of the Strip pusher position sensor that run under the plate with care.

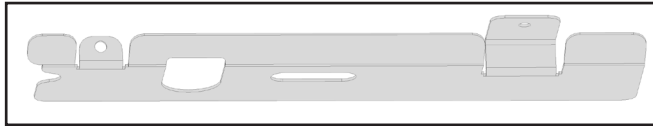


Figure 36: The cable guard plate

5 Unscrew the two (2) 3-millimeter hex socket head screws that secure the strip pusher unit's bracket plate to the device chassis. Disconnect the strip pusher motor's cable from the Strip pusher motor's interface board and the strip pusher unit's sensors from Circuit board number 1 (connectors 13 and 9 in Figure 29).

6 Dislodge the strip pusher unit from the pair of aligner pins beside the two securing screws you unscrewed in step 5 and remove the unit from the device.

7 Unscrew the two (2) 2.5-millimeter hex socket head screws that secure the strip pusher motor to the unit bottom plate, remove the motor and replace with a new one.

8 Complete the previous steps in reverse order and direction. Adjust the strip clamps and the pipetting position (see [1.4.4 Adjusting the position of the strip clamps](#) and [1.5.2 Adjusting the pipetting position](#)),

and tap **Initialization** on the **Flow tests** menu in the Systest software, before resuming normal operation.



5.6.8 Replacing the Feeder motor position and the Strip turner home position sensor

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect the cable connector of the sensor or sensors you wish to replace from Circuit board number 1 (connector 5, 6, 8, 12, or 16 in Figure 29).

3a If you are replacing one or more of the Feeder motor position sensors, unscrew the two (2) Phillips screws (in case of the 'lower empty' motor position sensor) or the two (2) 2.5-millimeter hex socket screws (in case of the other sensors) that secure them to the sensor plate.

3b If you are replacing the Strip turner home position sensor, you will first need to remove the Strip turner motor to gain access to the sensor. Complete steps 2–4 in [5.6.5](#). Unscrew the 1.5-millimeter hex socket head set screw of the Strip turner position sensor cam, and the two (2) 2-millimeter hex socket button head screws that secure the sensor itself.

4a If you have replaced one or more of the Feeder motor sensors, replace the sensor with a new one and complete the previous steps relevant to your task in reverse order and direction. Complete steps 7–11 in [5.6.5](#).

4b If you have replaced the Strip turner motor home position sensor, make sure that the sensor cam is aligned with the central slit of the sensor. In the Systest software on the **Card tests** menu on the **Feeder card** tab, tap the **To home position** button in the **Strip turner motor** row and check whether the corresponding sensor lights up (see [2.3.2 The Sensor tests menu](#)). Re-position the sensor cam if necessary.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.9 Replacing the Flipped strip sensor

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect the Flipped strip sensor's cable from connector 17 on Circuit board number 1.

3 Unscrew the 2-millimeter hex socket head screw that secures the sensor's hinged casing to the feeder bracket on the side towards the back of the device.

4 Slide the sensor, complete with casing and cable, out of its slit on the feeder bracket. Replace with a new sensor and complete the previous steps in reverse order and direction. In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.

i There is usually no need to replace the separate, smaller bit of the sensor hinge opposite the slitted bit. If you need to replace it, it is held in place by a 2-millimeter hex socket head screw

similar to that of its counterpart.



5.6.10 Replacing the Strip presence sensor [S-ALT4550211]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect connector 21 on Circuit board number 1.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the circuit board plate.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.11 Replacing the Strip pusher position sensor [S-ALT4560231]

1 Complete steps 1–4 in [4.6.7](#) to access the Strip pusher position sensors.

2 Disconnect the relevant connectors (13 or 9 in case of the pipetting or the home position sensor respectively) from Circuit board number 1.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor you are replacing (the home position sensor is the one to the left if you are facing the front of the device) to the strip pusher unit's bracket plate.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Adjust the pipetting position (see [1.5.2 Adjusting the pipetting position](#)). In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.12 Replacing the microswitch sensor [S-ALT4580021]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect connector 14 from Circuit board number 1.

3 Unscrew the microswitch from its custom threaded bore in the back of the feeding tube.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

< 5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.13 Replacing the inductive sensor [S-ALT4540071]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect connector 22 from Circuit board number 1.

3 Using a size 17 (11/16) wrench, unscrew the induction sensor from the top of the unused strip bin.

4 Remove the sensor and replace with a new one. Complete the previous steps in reverse order and direction. In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.14 Replacing the pipetting tray presence sensor [S-ALT4550071]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect connector 18 from Circuit board number 1.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the sensor bracket plate.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.15 Replacing Circuit board number 1

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect all the cable connectors from Circuit board number 1.

3 Unscrew the three (3) 2-millimeter hex socket button head screws that secure the circuit board to the circuit board plate. Save the washers, spacers between the board and the circuit board plate, and the hex nuts. Unscrew the 2-millimeter hex socket head screw with a serrated washer that secures the circuit board to the circuit board plate from the side of the plate.

4 Replace the circuit board with a new one and complete the previous steps in reverse order and direction. Refer to [Figure 29](#) when reconnecting the cables and pay attention to the configuration of the DIP switches.

5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.

4 Replace the circuit board with a new one and complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



5.6.16 Replacing the Strip pusher motor interface board

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

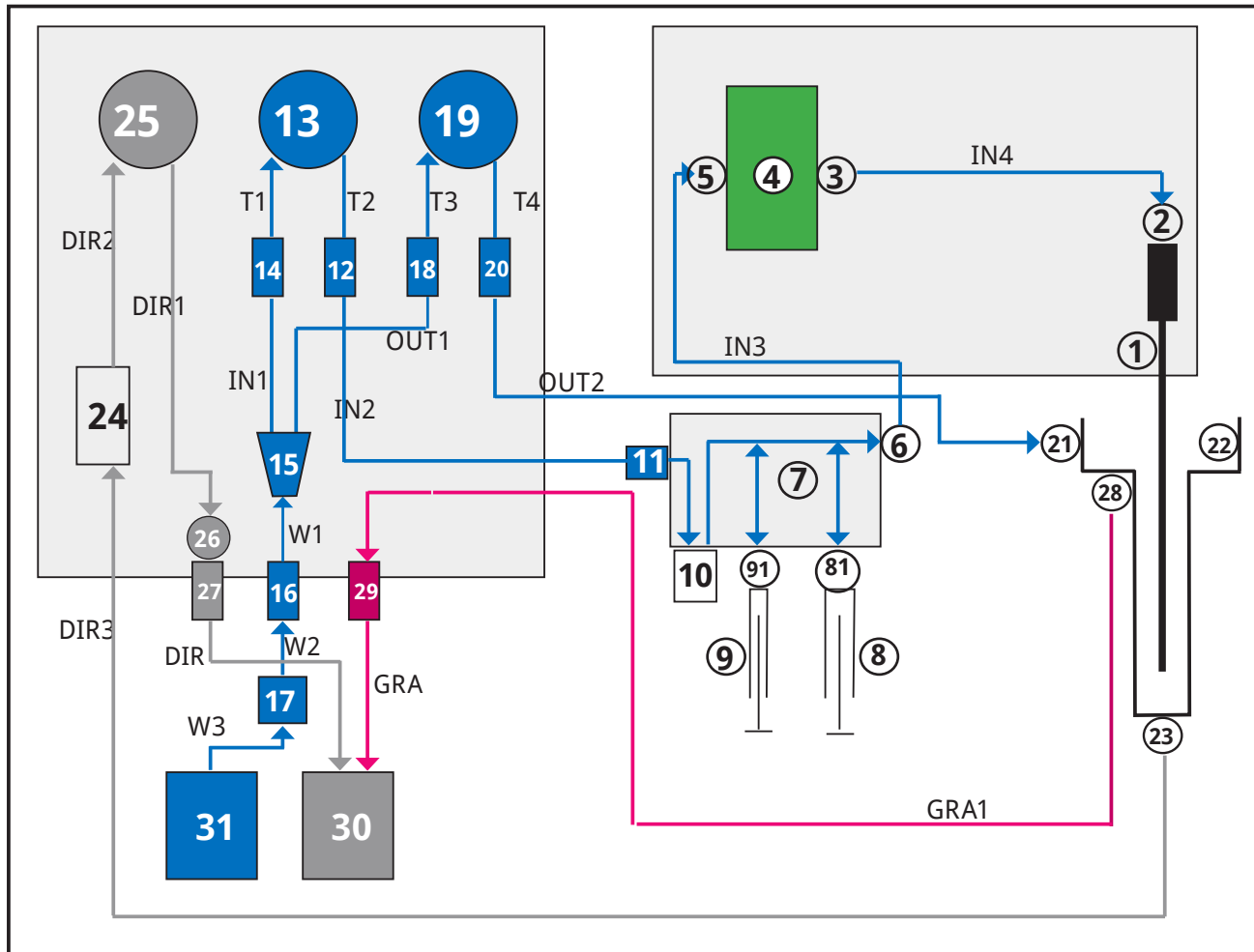
2 Disconnect connector 33 on Circuit board number 1 and the strip pusher motor connector from the interface board.

3 Unscrew the two (2) 2-millimeter hex socket button head screws with a hex nut that secure the board to the interface board plate. Save the hex nuts.

6 Washer Module

i For a detailed overview of the module, see [4.1.2 The Washer module](#)

6.1 Overview of the fluidic system



i Abbreviations and conventions used in Figure 37:

	the gravity line
	the distilled water line
	the waste line
ID	inner diameter
OD	outer diameter

Figure 37: Schematic diagram of the LabUMat 2 fluidic system

Key to Figure 37: Elements and connectors

Number	Designation & Article Number	Description
1	Pipetting probe [S-UPA4403232]	ID 1 mm, length 213 mm, volume 167.3 µL
2	Tube fitted with fitting nut [55OFNF16] and ferrule [55OFFT16]	OD 1/16 inch (1.588 mm) tube
3		
4	PMC	Volume 201µL
5	Tube fitted with fitting nut [55OFFT32] and ferrule [55OFNF32]	OD 1/8 inch (3.1 mm) tube
6		
7	Manifold [S-UPA2635022]	ID 2mm, volume 320 µL
8	Mixing syringe [S-55201250]	Hamilton 1002AD 2.5mL syringe
81	Mixing syringe O-ring	S-56125X10
9	Aspirating syringe [S-55201100]	Hamilton 1725AD 250µL syringe
91	Aspirating syringe O-ring	S-56125X10
10	Electromagnetic valve [S-ALT4590081]	WTA-2R-N4F1
11 21 27	Push-in fitting [55M5A600]	OD 6 mm
12 14 18 20	Push-in connector [S-55EPU600]	OD 6 mm
13 19	Peristaltic pump [S-ALT4520211] 13: 'Inner washer pump' 19: 'Outer washer pump'	With replaceable flexible tubing [3EDCSR25]
15	Push-in Y fitting [S-55EY6660]	OD 6 mm
16	Water in push-in fitting [S-55EAL6X6]	OD 6 mm
17	Water filter [S-55FV2B10]	

Key to Figure 37: Elements and connectors

Number	Designation & Article Number	Description
22	Rinse well with O-ring and fittings [S-UPA4423002]	
23	Push-in fitting [55M5K690]	OD 6 mm, L-shaped
24	Electromagnetic valve [S-ALT4590161]	XTA-2R-MFFH-2
25	Membrane pump [S-ALT4520181]	Thomas 5002F
26	Push-in fitting [S-55]G0876]	OD 6-to-8 mm
27 29	Push-in fittings for the Waste pump and the Gravity line [S-55EAL8X8]	OD 8 mm
30	Waste liquid tank [S-55130970]	5-liter
31	Distilled water tank [S-55130870]	5-liter

Key to Figure 37: Tubing

Number and article number	Tube material	OD (mm)	ID (mm)	Length (mm)	Volume (µL)
IN4 [UPA-170028-1]	PTFE	1.6	1	480	377
IN3 [UPA-170027-1]	PTFE	3.2	1.5	380	672
IN2 [UPA-170030-1]	Polyurethane	6	4	750	9425
T2 [UPA-170031-1]	Polyurethane	6	4	30	377
T1 [466PU6X4]	Polyurethane	6	4	30	377
IN1 [UPA-170025-1]	Polyurethane	6	4	230	289
W3 [S-URS17231]	PTFE	6	4	2000	2500

Key to Figure 37: Tubing

Number and article number	Tube material	OD (mm)	ID (mm)	Length (mm)	Volume (μL)
W2 [S-URS17291]	PTFE	6	4	70	880
W1 [UPA-170017-1]	Polyurethane	6	4	90	1131
OUT1 [UPA-170023-1]	Polyurethane	6	4	230	2890
T3 [466PU6X4]	Polyurethane	6	4	30	377
T4 [466PU6X4g]	Polyurethane	6	4	30	377
OUT2 [UPA-170024-1]	Polyurethane	6	4	160	2011
DIR3 [UPA-170022-1]	Polyurethane	6	4	160	2011
DIR2 [UPA-170029-1]	Polyurethane	6	4	70	880
DIR1 [UPA-170021-1]	Polyurethane	6	4	210	2639
DIR [S-URS17221]	PTFE	8	6	2000	
GRA [S-URS17211]	PTFE	8	6	2000	
GRA1 [UPA-170019-1]	Polyurethane	6	4	320	4021

6.2 Motors



6.2.1 'Inner washer pump' [S-ALT4520211]

DC motor that pumps distilled water from the water tank to and from the pipetting probe

⚠ If the rinsing system is connected to the distilled water tank, you must make sure that the pipetting probe is inside the rinse well, and that you turn the Waste pump on before running the Inner washer pump. Another solution to safely run the Inner washer pump cycles is if you first drain the system by running a Shutdown test on the Flow tests tab in the Systest software, then disconnect the tubes that connect the device to the liquid tanks.

plugged in to connector 27 on Circuit board number 2

preprogrammed motor cycles:

- washing

The Inner washer pump performs the timed movements required during a Washing test cycle.

- fill up

The Inner washer pump performs the timed movements required during a Filling up test cycle.

- empty

The Inner washer pump rotates backwards for a motor cycle.

- disinfection

The Inner washer pump performs the timed movements required during a Disinfection test cycle.



6.2.2 'Outer washing pump' [S-ALT4520211]

DC motor that pumps distilled water from the water tank to and from the rinse well

⚠ If the rinsing system is connected to the distilled water tank, you must make sure that you turn the Waste pump on in the Systest software under Card tests on the Washer card tab before running the Outer washing pump. Another solution to safely run the Outer washing pump cycles is if you first drain the system by running a Shutdown test on the Flow tests tab in the Systest software, then disconnect the tubes that connect the device to the liquid tanks.

plugged in to connector 25 on Circuit board number 2

preprogrammed actions:

- washing

The Outer washing pump performs the timed movements required during a Washing test cycle.

- fill up

The Outer washing pump performs the timed movements required during a Filling up test cycle.

- empty

The Outer washing pump rotates backwards for a motor cycle.

- disinfection

The Outer washing pump performs the timed movements required during a Disinfection test cycle.



6.2.3 'Waste pump' [S-ALT4520181]

membrane pump that drains the rinse well

plugged in to connector 29 on Circuit board number 2
preprogrammed action:

- waste on

The Waste pump is turned on.



In the Systest software on the **System parameter setup** menu on the **Washer card** tab, the **Waste pump delay** parameter indicates how long the the membrane pump will go on running, in seconds, after both peristaltic pumps have been stopped.



Do not modify this value.



6.2.4 Waste line sealing valve [S-ALT4590081]

The electromagnetic valve that seals the waste line.

plugged in to connector 28 on Circuit board number 2

The valve is open when it is turned on.



Because the valve and the membrane pump are interlinked, you can turn on the Waste line sealing valve by tapping **Waste ON** under **Card tests** on the **Washer card** tab in the Systest software.



See [9.2.3 Aspiration valve](#) for details on the Aspiration valve.



The **Disinfection time** parameter in the Systest software on the **System parameter setup** on the **Washer card** tab indicates the length of the daily rinsing procedure. The parameter determines how long the disinfectant solution will be allowed to stay in the rinse well, in seconds.



Do not modify this value.

6.3 Sensors



6.3.1 Waste sensor [S-URS4532002]

type: float switch

plugged in to connector 1 on the Liquid tanks interface board

Systest sensor signal		
Waste tank warning	The waste tank is almost full	Sensor not activated
'Waste tank full'	The waste tank has reached its maximum capacity	Sensor not activated

The device is properly set up and operational only if the Waste tank full sensor is inactive (its signal is green) after system initialization. If the Waste tank warning sensor is active (its signal is red), there is room in a 5-liter tank for the liquid waste of approximately thirty (30) further measurements. (See [2.3.2 The Sensor tests menu](#).)



See Figure 38 for the pinout of the Waste sensor's D-sub plug. Be aware that in case of the Waste sensor, none of the pins need to be shorted if you wish to operate the device without liquid tanks.



6.3.2 Wash sensor [S-URS4531002]

type: float switch

plugged in to connector 3 on the Liquid tanks interface board



The sensor signals for the wash sensor are reversed compared to the other sensors. A red signal indicates an inactive sensor.

Systest sensor signal		
Water tank warning	Sensor not activated	The distilled water tank is almost empty
Water tank empty	Sensor not activated	There is no more distilled water in the distilled water tank

The device is properly set up and operational only if the Water tank empty sensor is inactive (its signal is red) after system initialization. If the Water tank warning sensor is active (its signal is green), there is enough distilled water in a 5-liter tank for the rinsing needs of approximately thirty (30) further measurements. (See [2.3.2 The Sensor tests menu](#).)



The pins indicated in black in Figure 36 need to be shorted when you wish to operate the analyzer without its liquid tanks connected (for demonstration purposes for example.)

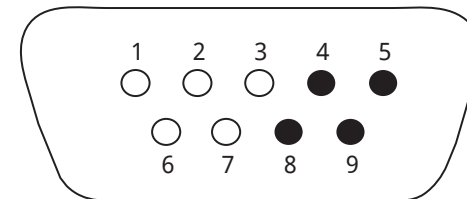
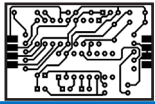


Figure 38: Pinout of the Wash sensor D-sub plug

6.4 PCBs



6.4.1 Washer module control circuit board [S-UPA43021]

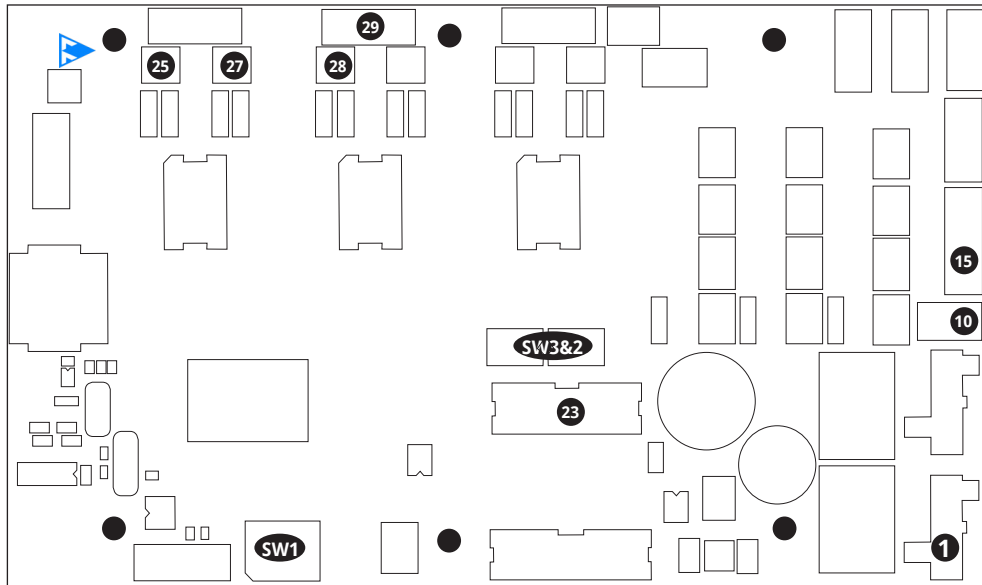


Figure 39: Board layout of Circuit board number 2

Key to Figure 39

- 25. Outer washer peristaltic pump motor connector
- 27. Inner washer peristaltic pump motor connector
- 28. Waste line sealing valve connector
- 29. Membrane pump motor connector
- 15 & 10. Liquid tank sensor interface board connectors
- SW3&2. Two 8-way configuration DIP switches (slider 5 of SW2 is on—in the down position, all other sliders are off) (☞ [Figure 40](#))
- SW1. DIP switch (only slider 2 is on—in the down position)

23. Bus cable connector from and to the Feeder and the Rack mover modules

1. Power connector coming from the Rack mover module PCB

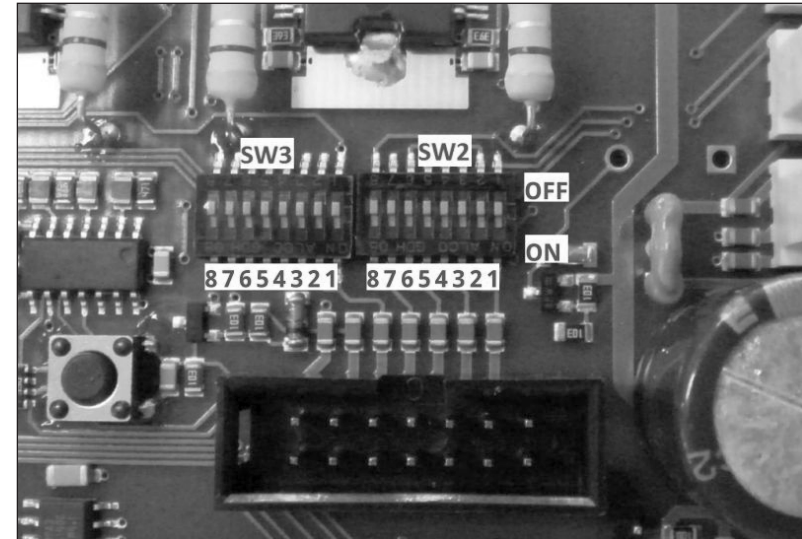
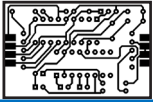


Figure 40: A close-up of the configuration DIP switches with the layout of the sliders indicated



6.4.2 Liquid tank sensor interface board [S-UPA42062]

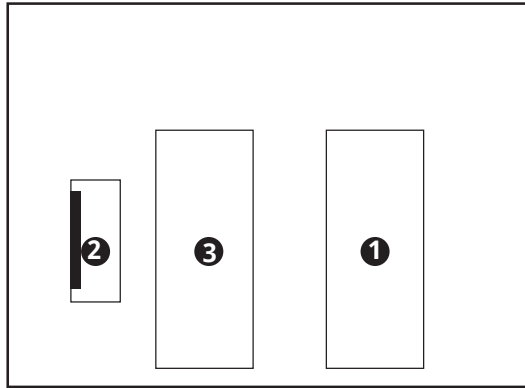


Figure 41: Board layout of the liquid tank sensor interface board

Key to Figure 41

1. Waste sensor connector
2. Board-to-board connector
3. Wash sensor connector

6.5 Washer module Flow tests processes



6.5.1 Initialization

i Given that the pipetting syringes are also activated, the Washing module's initialization is not specific to the Washer module, but involves the fluidic system as a whole.

1

The Waste pump is turned on. A 'Fill-up' cycle is started in the Outer washer pump.

The aspirating and the mixing syringe plungers are moved into their home positions.

2

The air is removed from the fluidic system: The Aspiration valve is opened; the Inner washer pump is turned on; the aspirating syringe plunger is moved into **To drag position** while the mixing syringe plunger is moved into **First mixing** position; the Inner washer pump is stopped; the Aspiration valve is closed; the aspirating and mixing syringe plungers are moved into their home positions.



The Waste pump is turned on automatically whenever any of the pumps start running. It is turned off two (2) seconds after all the pumps have stopped running.



6.5.2 Washing test



If the rinsing system is connected to the distilled water tank, make sure that the waste tank is also connected and that the pipetting probe is inside the rinse well before you perform a Washing test.



Given that the pipetting syringes are also activated, the Washing test is not specific to the Washer module, but involves the fluidic system as a whole.

1

The Aspiration valve is opened.

2

The Inner washer and the Outer washing pumps are turned on.

3

Both pipetting syringe plungers are moved into their home positions

- 4 The rinse well and pipetting probe are rinsed for about 3–4 seconds.
- 5 The Aspirating syringe plunger is moved into **To drag position** while the mixing syringe plunger is moved into **First mixing position**.
- 6 The Inner washer and Outer washing pumps are turned off.
- 7 The Aspiration valve is closed.
- 8 Both pipetting syringe plungers are moved into their home positions

6.5.3 Filling up test

The Washer module performs the same operations during the Filling up test as it does during Initialization.

6.5.4 Disinfection test

⚠ Make sure that both the distilled water and the waste tanks are connected to the rinsing system and that there is a rack on the rack conveyor with a single test tube filled with six (6) milliliters of a 2:100 NaOCl (Sodium hypochlorite) solution before you perform a Disinfection test.

ℹ Given that the pipetting syringes are also activated, the Disinfection test is not specific to the Washer module, but involves the fluidic system as a whole.

- 1 The rack with the disinfectant is pulled inside the device (the movements necessary for a **Tube change test/Rack change test** are performed)

- 2 The pipetting probe is moved above and inside the test tube.
- 3 The Mixing syringe aspirates 1.4 milliliters of disinfectant solution from the test tube (it is moved into its **Disinfection** motor position).
- The Waste line sealing valve is closed.
- The pipetting probe is moved above and inside the rinse well.
- The aspirated disinfectant is discharged into the rinse well (the Mixing syringe is moved into the **Mixing home** position).
- 4 Steps 2–3 are repeated with the difference that the Mixing syringe does not discharge the disinfectant when inside the rinse well.
- 5 The disinfectant solution inside the pipetting probe and inside the rinse well it is submerged in is allowed to sit for three (3) minutes to take effect.
- 6 The rinse well and the pipetting probe are rinsed thoroughly with distilled water, as in a [Washing test](#), but for ten (10) seconds.

5.5.5 Shutdown test

ℹ Given that the pipetting syringes are also activated, the Washing module Shutdown test is not specific to the Washer module, but involves the fluidic system as a whole.

The Washer module performs the same operations during the Shutdown test as it does during a Washing test, in reverse order and direction.

6.6 Service actions—Adjustments



6.6.1 Checking the fluidic system

Whenever you modify or replace any of the tubes, fittings, or any other part of the fluidic system, you must make sure that all three functions of the system are performed properly.

⚠ Drain the fluidic system and disconnect the waste tank and the distilled water tank before checking any of these functions.

6.6.1.1 Checking waste drainage

Switch on the device and launch the Systest software. Under **Card tests** on the **Washer card** tab, turn on the Waste pump and press your finger against the rinse well aperture. If the pump is functioning properly, you should be able to feel suction from the waste line.

6.6.1.2 Checking rinse well fill-up

Switch on the device and launch the Systest software. Under **Card tests** on the **Washer card** tab, tap the **Fill up** button in the **Outer washing pump** row, and press your finger against the rinse well aperture. If the pump is functioning properly, you should be able to feel the pressure in the rinse well.

6.6.1.3 Checking water line fill-up

Switch on the device and launch the Systest software. Under **Card tests**

on the **Sampler card** tab, turn on the Aspiration valve. Under **Card tests** on the **Washer card** tab, tap the **Fill up** button in the **Inner washer pump** row and press your finger against the tip of the pipetting probe. If the pump is functioning properly, you should be able to feel pressure in the water line.

i You can also use a small container of liquid (such as a bottle cap with a few drops of water) to test the Inner washer pump. If you immerse the tip of the pipetting probe in liquid while the Fill up cycle is going on, air bubbles should form in the liquid.



6.6.2 Aligning the rinse well with the linear robot

Complete steps 7–10 in [Adjusting the linear robot alignment](#) to make the adjustment.

6.7 Service actions—Replacements



6.7.1 Replacing the Washer module [S-UPA4405004]

- 1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2 To remove all liquids from the fluidic system, run a **Shutdown test** on the **Flow tests** menu in the Systest software, and disconnect the liquid tanks and their sensors from the device.
- 3 Disconnect:
 - the tube between the Inner washer pump and the manifold (labeled IN2)
 - the tube between the Outer washing pump and the rinse well

(labeled OUT2)

- the tube between the rinse well and the Waste line sealing valve (labeled DIR3)
- the gravity line tube (labeled GRA1)

4 Unscrew the two (2) 3-millimeter hex socket bolts that secure the module housing to the chassis. Save the serrated washers.

5 Gently pull the module outward to access Circuit board number 2 on the side towards the front of the device. Reach in and disconnect the power cable and the bus cable from Circuit board number 2. See [Figure 39](#).

i If you have trouble accessing the circuit board or want to avoid accidentally damaging the cables or the PCB, you can remove the front casing panel to easily access Circuit board number 2. See [4.2.2](#).

6 Slide the module out of the device, insert a new one, and complete the previous steps in reverse order and direction.

7 Perform all the checks listed in [5.6.1](#) before resuming normal operation.

⚠ Make sure that the pin on the bottom of the new module's housing plate aligns with the tail on the chassis towards the front of the device, to lock the module in place.



6.7.2 Replacing the Inner washer pump

1 To remove the Washer module, complete steps 1–5 in [5.7.1](#).

2 Disconnect the tubes labeled IN2 and IN1.

3 Unscrew the two (2) 3-millimeter hex screws that secure the pump to the module housing plate.

4 Replace the pump with a new one and complete the previous steps in reverse order and direction.

5 Perform all the checks listed in [5.6.1](#) before resuming normal operation

6.7.2.1 Replacing the Inner washer pump flexible silicone tubing

i The flexible tubing inside peristaltic pumps is prone to degrade over time and requires periodic replacement.

1 Complete steps 1–2 in [5.7.2](#) to access the Inner washer pump.

2 Turn the Z-shaped clasp on top of the pump in the direction labeled 'Auf' to release the see-through clip that holds the flexible silicone tube (See [Figure 42](#)). Slide the clip and the tube inside it out from under the pump bracket.

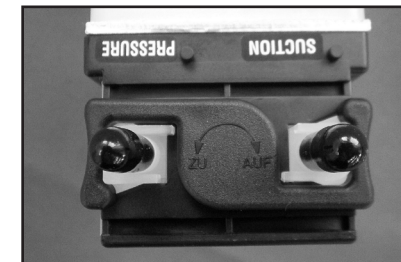


Figure 42: Peristaltic pump S-shaped retain-er clasp

- 3 With a pair of pliers, press together the tabs of the springs clamps on both ends of the flexible silicone tube to release the tube.
- 4 Replace the flexible silicone tube and complete the previous steps in reverse order and direction.
- 5 Perform all the checks listed in [5.6.1](#) before resuming normal operation.



6.7.3 Replacing the Outer washing pump

- 1 To remove the Washer module, complete steps 1–5 in [5.7.1](#).
- 2 Disconnect the tubes labeled T3 and T4 in [Figure 37](#).
- 3 Unscrew the two (2) 3-millimeter hex socket bolts that secure the pump to the module housing plate.
- 4 Replace the pump with a new one and complete the previous steps in reverse order and direction.
- 5 Perform all the checks listed in [5.6.1](#) before resuming normal operation.

6.7.3.1 Replacing the Outer washing pump flexible silicone tubing

- 1 Complete steps 1–2 in 6.7.3 to access the Outer washing pump.
- 2 Turn the Z-shaped clasp on top of the pump in the direction labeled 'Auf' to release the see-through clip that holds the flexible silicone tube (See Figure 42). Slide the clip and the tube out from under the

pump bracket.

- 3 Press the tabs of the springs clamps on both ends of the flexible silicone tube together with pliers to release the tube.
- 4 Replace the flexible silicone tube and complete the previous steps in reverse order and direction.
- 5 Perform all the checks listed in [5.6.1](#) before resuming normal operation.



6.7.4 Replacing the Waste pump

- 1 To remove the Washer module, complete steps 1–5 in [5.7.1](#).
- 2 Disconnect the tubes labeled DIR1 and DIR2 in [Figure 37](#).
- 3 Unscrew the single (1) 3-millimeter hex socket bolt in the clamp that secures the waste pump to the module housing plate.
- 4 Replace the pump with a new one, and complete the previous steps in the reverse order and direction.
- 5 Perform all the checks listed in [5.6.1](#) before resuming normal operation.

6.7.4.1 Cleaning the Waste pump flap valve



During normal operation, tiny debris or burrs from the plastic tubes may be sucked into the Waste pump and clog its valve. See also: [14.9.4](#).

- 1 To remove the Washer module, complete steps 1–5 in [5.7.1](#)
- 2 Unscrew the four (4) size T10 Torx screws securing the top panel of the Waste pump.
- 3 Check and clean the two flap valves, then complete the previous steps in reverse order and direction.
- 4 Perform all the checks listed in [5.6.1](#) before resuming normal operation.



6.7.5 Replacing the Waste line sealing valve

- 1 To remove the Washer module, complete steps 1–5 in [5.7.1](#)
- 2 Disconnect the tubes labeled DIR3 and DIR2 in [Figure 37](#) from the pipe ends of the valve.
- 3 Replace the valve with a new one, and complete the previous steps in reverse order and direction.
- 4 Perform all the checks listed in [5.6.1](#) before resuming normal operation.



6.7.6 Replacing any of the connecting tubes or fittings

- 1 Identify the section of tubing that you need to replace using Figure 37.
- 2 To remove all liquids from the fluidic system, run a **Shutdown test** on the **Flow tests** menu in the Systest software, and disconnect

the liquid tanks from the device.

- 3 Remove the part or parts that the section is connected to, locate the tubing in question, and replace it.

⚠ Make sure that the tubes you install are not twisted, and cannot get caught in moving parts during operation.

- 4 Perform all the checks listed in [5.6.1](#) before resuming normal operation.



6.7.7 Replacing the rinse well

- 1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2 To remove all liquids from the fluidic system, run a **Shutdown test** on the **Flow tests** menu in the Systest software, and disconnect the liquid tanks from the device.
- 3 Disconnect the Gravity line tube (labeled GRA1), the tube between the Waste line sealing valve and the rinse well (labeled DIR3), and the tube coming from the Outer washing pump (labeled OUT2).
- 4 Unscrew the four (4) 2-millimeter button head hex socket screws around the rinse well aperture that secure the rinse well to the bottom plate.
- 5 Remove the rinse well, insert a new one, and complete the previous steps in reverse order and direction.
- 6 Check the alignment of the linear robot with the rinse well aperture. See [Adjusting the linear robot alignment](#).

7 Perform all the checks listed in [5.6.1](#) before resuming normal operation



5.7.8 Replacing the Waste sensor

77Elektronika supplies the waste tank float switch sensor bundled with the sensor rod, the threaded waste tank cap, and the cable and 9-pin RS232 plug (female in case of the waste sensor) with the article number [S-URS4532002]. See [for waste tank installation guidelines](#).



6.7.9 Replacing the Wash sensor

77Elektronika supplies the distilled water tank float switch sensor bundled with the sensor rod, the threaded waste tank cap, and the cable and 9-pin RS232 plug (male in case of the wash sensor) with the article number [S-URS4531002]. See [for distilled water tank installation guidelines](#).



6.7.10 Replacing Circuit board number 2

1 See [4.2.2 Removing the front casing panel](#) and remove the front casing panel.

2 Disconnect all the cable connectors from Circuit board number 2.

3 Unscrew the four (4) 2.5-millimeter button head hex socket screws securing the board to the Washer module housing.

4 Replace the circuit board with a new one and complete the previous steps in reverse order and direction. Refer to Figure 39 when

reconnecting the cables and pay particular attention to the configuration of the DIP switches.

5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.



6.7.11 Replacing the liquid tank sensor interface board

1 To remove the Washer module, complete steps 1–5 in [5.7.1](#).

2 Disconnect all the cable connectors from the liquid sensor interface board.

3 Unscrew the four (4) 2-millimeter button head hex socket screws that secure the interface board to the module housing plate.

4 Replace the interface board with a new one and complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.

7 The Linear Robot module

 For a detailed overview of the module, see [4.1.3 The Linear robot moduleicro](#).

7.1 Motors



7.1.1 Linear robot horizontal motor [S-ALT4510061]

A stepper motor that moves the pipetting probe carriage horizontally via a timing belt.

plugged in to connector 33 on card number 3

home position: Rinse well position

Significant positions:

- Rinse well position

The pipetting probe is moved horizontally above the rinse well aperture.
sensor-determined motor position

- Test tube position

The pipetting probe is moved horizontally above the test tube aperture.
sensor-determined motor position

- Go to strip

The pipetting probe is moved horizontally above the center of the frontmost (Bilirubin) pad on a test strip that is sitting on the pipetting stage.
motor step number-determined motor position (counted off from the Test tube position)



7.1.2 Linear robot vertical motor [S-ALT4510071]

A stepper motor that moves the pipetting probe carriage vertically via two timing belts.

plugged into connector 29 on card number 3

home position: Probe rinse position

significant positions:

- High probe position ('High probe position')

The pipetting probe is moved into its topmost position.

sensor-determined motor position

- To above strip

The pipetting probe is moved to the height where it is above the center of the frontmost (Bilirubin) pad on the test strip that is sitting on the pipetting stage.

motor step number-determined motor position (counted off from the

High probe position)

 In the Systest software under **Position setup**, on the **Robot Y strip rear position** tab, the **Strip Y rear corr. value** modifies this motor position. 

- Probe rinse position

The pipetting probe is moved to the height where, during normal operation, it is rinsed in inside the rinse well.

motor step number-determined motor position (counted off from the High probe position)

- Aspirating position

The pipetting probe is moved to the height where, during normal operation,

urine sample aspiration takes place inside the test tube.

motor step number–determined motor position (counted off from the High probe position)

- Dip position

The pipetting probe is moved to the height where, during normal operation, it is briefly dipped after each rinsing cycle inside the rinse well, before it is moved to aspirate the next urine sample.

motor step number–determined motor position (counted off from the High probe position)

- Leave sample

The pipetting probe is moved into the High probe position at about half speed.

sensor-determined motor position when initiated in the Systest software (identical to the High probe position)

motor step number–determined motor position during normal operation

 **During normal operation, the pipetting probe is moved at half speed only after urine sample aspiration, while it is inside the test tube, and only up to the previously sensed level of the liquid in the tube. From this level upwards, the pipetting probe is moved at normal speed. This is to prevent excess urine from sticking to the probe as it is lifted from the test tube.**

7.2 Sensors



7.2.1 Rinse well position sensor [S-ALT4560041]

type: transmissive photointerrupter

plugged in to connector 3 on Circuit board number 3

Systest sensor signal		
Rinse well pos.	The pipetting probe is aligned with the rinse well aperture.	Sensor not activated.

The device is properly set up and operational only if the Rinse well position sensor is active (its signal is red) after system initialization. (See the [2.3.2 The Sensor tests menu](#).)



7.2.2 Test tube position sensor [S-ALT4560041]

type: transmissive photointerrupter

plugged in to connector 6 on Circuit board number 3

Systest sensor signal		
Test tube pos.	The pipetting probe is aligned with the test tube aperture.	Sensor not activated.

The device is properly set up and operational only if the Test tube position sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)



7.2.3 High probe position sensor [S-UPA42033]

type: transmissive photointerrupter

plugged in to connector 10 on Circuit board number 3 via the FFC (Flexible Flat Cable) connecting the Vertical motor interface board and the Horizontal motor interface board

Systest sensor signal		
High probe pos.	The pipetting probe is in its topmost position.	Sensor not activated.

The device is properly set up and operational only if the High probe position sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



7.2.4 Low probe position sensor [S-ALT4560031]

type: transmissive photointerrupter

plugged in to connector 10 on Circuit board number 3 via the FFC connecting the Vertical motor interface board and the Horizontal motor interface board

Systest sensor signal		
Low probe pos.	The pipetting probe has passed the sensor on its way down the test tube aperture but has not yet entered the test tube.	Sensor not activated.



The Low probe pos. sensor is currently inoperative. In future models it will indicate if the pipetting probe gets caught on the rim of the test tube and cannot enter the liquid inside

for sample aspiration.



7.2.5 Left door inductive sensor [S-ALT4540051]

type: inductive sensor

plugged in to connector 18 on Circuit board number 3

Systest sensor signal		
Left door	The left leaf of the device door is closed properly.	Sensor not activated.



7.2.6 Right door inductive sensor [S-ALT4540051]

type: inductive sensor

plugged in to connector 3 on the Oscillator interface board

Systest sensor signal		
Right door	The right leaf of the device door is closed properly.	Sensor not activated.



You can disable both door sensors in the Systest software on the Flow tests menu by tapping the Disable Interlock button. This may be useful when you are servicing the device and you need to keep the doors open, or to remove them altogether.



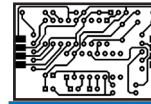
7.2.7 Liquid level sensor [S-UPA42121]



The inclusion of this sensor in the Linear robot module list of sensors is for information purposes only. This sensor is not connected to the Linear robot module circuit board: It is synchronized with the sensor of the same name that is on the Rack module via the Configuration DIP switches (see the key to [Figure 43](#)) on Circuit board number 3. As such, it is activated and inactivated in parallel with the Liquid level sensor on the Rack module, although with a slight delay. If the two signals differ, check the Configuration DIP switch settings and the cables connecting the [Level detector interface board](#) and Circuit board number 3.

The device is properly set up and operational only if both sensors named Liquid level sensor are inactive (their signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)

7.3 PCBs



7.3.1 Linear robot module control circuit board [S-UPA43041]

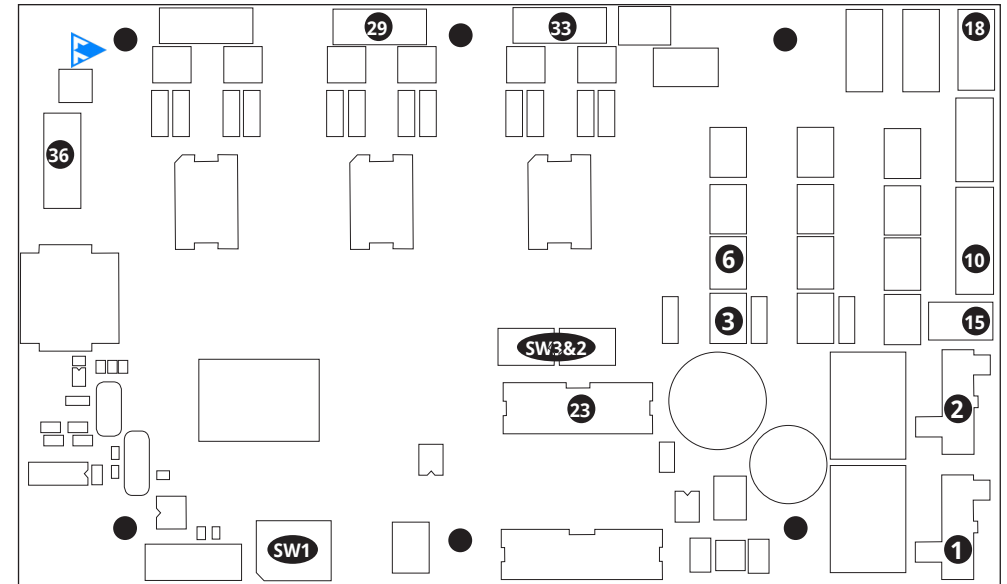


Figure 43: Board layout of Circuit board number 3.

Key to Figure 43

- 29. Linear robot vertical motor connector
- 33. Linear robot horizontal motor connector
- 18. Left door inductive sensor connector
- 36. PMC PCB (Circuit board number 8) board connector
- 6. Test tube position sensor connector
- 10. Horizontal motor interface board connector
- 3. Rinse well position sensor connector

15. Right door inductive sensor connector

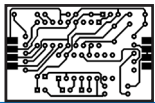
SW3&2. Two 8-way configuration DIP switches (On SW2, sliders 4 and 8 are on—in the down position; on SW3, only slider 2 is on.) (For the layout and numbering of the sliders [👉 Figure 40](#))

23. Bus cable connector to Circuit board number 1 (Feeder)

2. Power connector to Circuit board number 1 (Feeder)

SW1. PCB address DIP switch (sliders 1 and 2 are on—in the down position)

1. Power connector to the Oscillator interface board



7.3.2 Oscillator interface board [S-UPA42113]

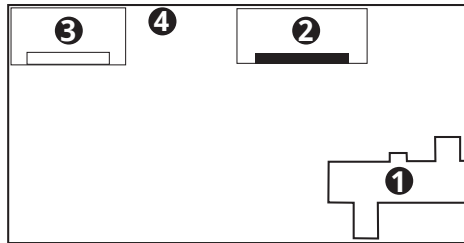
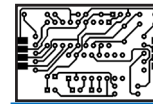


Figure 44: Board layout of the Oscillator interface board

Key to Figure 44

1. Power connector to Circuit board number 3
2. Right door inductive sensor signal transmitter connector
3. Right door inductive sensor connector
4. Oscillator



7.3.3 Horizontal motor interface board [S-UPA42022]

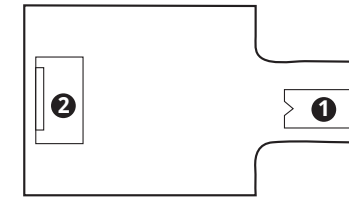
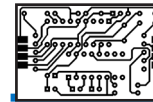


Figure 45: Board layout of the Horizontal motor interface board

Key to Figure 45

1. FFC connector to the Vertical motor interface board
2. Board-to-board connector to Circuit board number 3



7.3.4 Vertical motor interface board [S-UPA42033]



Figure 46: Board layout of the vertical motor interface board

Key to Figure 46

1. FFC connector to the Horizontal motor interface board
- DP1. High probe position sensor connector
- DP2. Low probe position sensor connector

7.4 Linear robot module Flow tests processes



7.4.1 Initialization

- If the pipetting probe is inside the rinse well when the process is started: the pipetting probe is moved into the 'High probe position', then into the 'Go in washer' position
- If the pipetting probe is not inside the rinse well when the process is started: the pipetting probe is moved into the 'High probe position', then into the 'Rinse well position' horizontal position, and finally the 'Go in washer' vertical position



7.4.2 Robot test

⚠ You can run a Robot test either with or without any liquid being dispensed on the test strip pads. If you want to run the test with liquid, place a liquid-filled test tube in the rack under the test tube aperture, and make sure that the waste tank is connected to the device before you start a Robot test.

⚠ Even if you are running the test without liquid, make sure that there is a LabStrip U11 Plus GL test strip on the pipetting stage before you run the test.

- 1 The pipetting probe is moved from the rinse well above the test tube aperture then to the height where sample aspiration takes place.
- 2 The pipetting syringes perform the movements necessary for sample aspiration (see [Syringe pump movements overview](#)).

3a The pipetting probe is moved into the position that corresponds to where sample pipetting starts, then, in turn, into the positions that correspond to each of the pads on a test strip sitting on the pipetting stage. The probe pauses in each pad position longer than during normal operation.

3b The pipetting syringes perform the movements necessary for dispensing the amount of urine samples specified for each test pad (see [Syringe pump movements overview](#)).

4 The pipetting probe is moved inside the rinse well.

⚠ If you have placed a liquid-filled test tube in the rack under the test tube aperture before running the Robot test, make sure that you run a Washing test at this point.



7.4.3 Level detection test

⚠ Make sure that there is at least one test tube in the rack under the test tube aperture filled with tap water before running a Level detection test.

⚠ Make sure that you use tap water. Distilled water is not suitable for the test.

The pipetting probe is moved inside the test tube until the tip of the probe touches the surface of the liquid in the test tube, then it is retracted. The process is repeated with every test tube the device detects in the rack. If there are further racks on the rack conveyor, the device will draw them in and perform the level detection test with each test tube it detects in them.



7.4.4 Shutdown test

The Linear robot module performs the same operations during Shutdown as during Initialization.

7.5 Service actions—Adjustments



7.5.1 Adjusting the linear robot alignment

1 Open the top cover of the device. Loosen the four (4) 3-millimeter hex socket screws that secure the linear robot module to the device chassis (2 at the front and 2 at the back of the crossbeam). Unscrew the 3-millimeter hex socket screws that secure the triangular support brackets on either end of the linear robot unit.

2 Grasp the linear robot module firmly and shift it around until the module moves quite freely, hanging on the device chassis by the securing screws.

3 Find the aligner screw (the screw on the right in [Figure 4](#)) screwed into the crossbar of the device chassis. Sink the aligner screw into the test tube aperture so that its semicircular knob fits snugly into the opening.

4 Moving the pipetting probe and the linear robot module manually, position the tip of the probe so that it is perfectly aligned with the hole on the top of the aligner screw (see [Figure 47](#)). Gently lower the pipetting probe by pulling on the

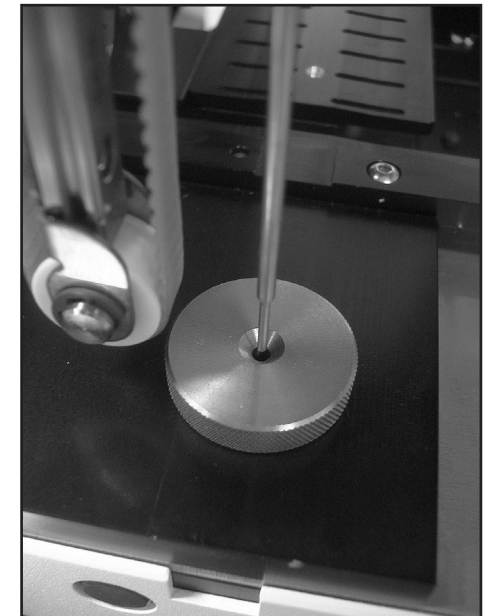


Figure 47: Using the aligner screw to check the alignment of the linear robot

vertical robot timing belt and carefully insert the pipetting probe into the aligner screw's hole. Adjust the robot module until you can slide the probe in and out of the central hole freely.

5 Place a LabStrip U11 Plus GL test strip in the way of the test strip clamps, and push the test strip onto the pipetting stage by manually moving the clamps along their gear shaft. Slowly move the pipetting probe manually over the test strip and adjust the robot module until the probe is parallel with the strip.

i It may be easier to check whether the linear robot moves the pipetting probe parallel with the test strip if you position the test strip on the pipetting stage so that the probe is above the leading edge of the strip, not above the pads on the strip as usual. If you choose to manually position the strip on the pipetting stage, however, you will need to pay particular attention that the strip itself is parallel with the pipetting stage.

6 Make sure that the linear robot is in perfect alignment with the test tube aperture and that it is completely parallel with the test strip on the pipetting stage, then re-tighten the securing screws and screw the triangular support brackets back to the device chassis.

7 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel. Loosen the four (4) 2-millimeter button head hex socket screws around the rinse well aperture that secure the rinse well to the bottom plate.

8 Place the aligner screw inside the rinse well aperture and manually move the pipetting probe over and inside the central hole of the aligner screw. Adjust the position of the loosened rinse well until the center hole in the aligner screw is perfectly aligned with the tip of the

pipetting probe.

10 Re-tighten the four (4) 2-millimeter button head hex socket screws around the rinse well.

11 Unless you are making this adjustment as part of device installation, in the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.5.2 Adjusting the front-back axis alignment of the linear robot horizontal motor

See [Adjusting the linear robot alignment](#).

⚠ The linear robot horizontal motor adjustments modify the following parameters in the Systest software on the **System parameter setup** menu on the **Robot** card tab: **Strip X correction**, **Strip Y front corr.**, and **Strip Y rear corr.** Do not modify these values directly.



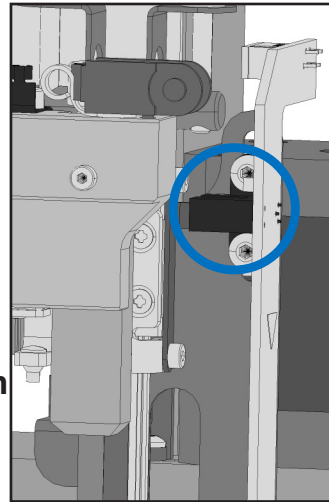
7.5.3 Adjusting the High- and Low probe position sensors

i The Low probe position sensor is currently not in use as a detector; you will not need to adjust it.

1 Complete steps 1–2 in [6.6.7](#).

2 Loosen the four (4) 2-millimeter hex socket button head screws of the High probe position sensor and the Low probe position sensor in their adjustable brackets (see Figure 48).

Figure 48: The High probe position sensor and its securing screws on the Vertical motor interface board



3 Adjust the High probe position sensor (see Figure 48) in its adjustable bracket so that there is a clearance of a couple of millimeters between the linear motion bearing that carries the pipetting probe and the pipetting probe block when the sensor's switch flag passes inside its slot. Figure 49 indicates the recommended clearance with the solid arrows. The dashed arrows indicate that the interface board and the sensor flag switch have been translated for a better view of the clearance.

4 Switch on the device and launch the Systest software. Under **Card tests** on the **Robot card** tab, tap the **High probe position** button in the **Robot Y motor** row. Check that the linear motion bearing that carries the pipetting probe stops short of the pipetting probe block.

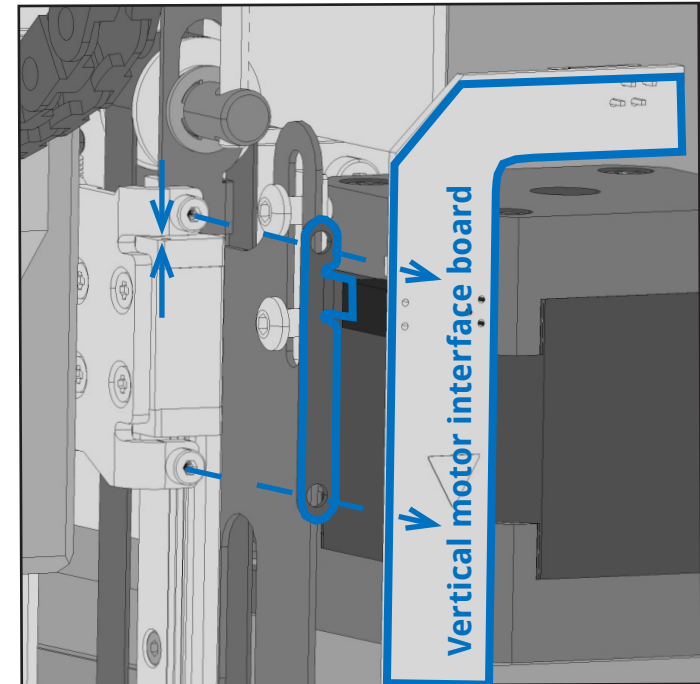


Figure 49: Recommended clearance between the probe carriage and the pipetting probe block



7.5.4 Preset system parameters

⚠ The following parameters are included in this manual for information purposes only. Do not modify any of these values without explicit prior permission from 77Elektronika service personnel.

i You can access these parameters in the Systest software on the **System parameter setup** menu on the **Robot card** tab by tapping the **Read parameters** button.

Key to the system settings in the Systest software on the **System parameter setup** on the **Robot card** tab:

- Nr. of disinf. fillup

The number of times the pipetting probe is moved above and inside the test tube to aspirate disinfectant solution, then above and inside the rinse well to deposit the solution during daily disinfection.

i When the pipetting probe aspirates the disinfectant for the last time that it is programmed to, it will move above and inside the rinse well, but will retain the disinfectant solution inside the probe for the duration of the daily disinfection.

- Level limit

See [1.5.6 Setting up the test tube minimum sample level](#).

- Nr. of mixing

The number of times sample mixing (see the [Aspiration and dispensing cycle](#)) is performed.

- Upper mixing

The number of motor steps that the pipetting probe is moved by down inside the test tube prior to sample mixing. The steps are counted off from the level of the fluid inside the test tube (see [1.5.6 Setting up the](#)

[test tube minimum sample level](#)).

- Lower mixing

A percentage value of the distance between the **Upper mixing** level and the lowermost point in the vertical motor's range. The percentage is a variable in the formula that determines the position inside the test tube where the sample aspirated during the mixing process is expelled back into the test tube.

- Aspiration point

A percentage value of the distance between the **Upper mixing** level and the lowermost point in the vertical motor's range. The percentage is a variable in a formula that determines the position inside the test tube where the sample for dispensing is aspirated from following the sample mixing process.

- Drop height

The number of motor steps that the pipetting probe is moved by vertically between each pad on the test strip during sample dispensing.

7.6 Service actions—Replacements



7.6.1 Replacing the linear robot module [S-UPA4403007]

1 Launch the Systest software. On the **Flow tests** menu, tap the **Shutdown test** button, then switch off the device and disconnect the AC power cable.

2 Open the top cover of the device. Remove the pipetting probe retainer plate from the crossbeam of the device chassis (see [Figure 4](#)), and reattach it in its original location so that it keeps the pipetting probe steady (see [Figure 3](#)).

3 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect the inductive sensor connectors (15 and 18), the Bus cable connector (23), and the Power connector (1) from Circuit board number 3 (see [Figure 43](#)). Unscrew the threaded plastic ring that secures the tube labeled IN4 to the manifold. Unscrew the 3-millimeter hex socket screws that secure the triangular support brackets to either end of the linear robot unit. Unscrew the four (4) 3-millimeter hex socket screws that secure the linear robot module to the device chassis (2 at the front and 2 at the back of the crossbeam).

⚠ Hold the module steady when removing its securing screws.

4 Remove the linear robot module and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ Make sure that the pipetting probe does not bump into anything that might bend it when you move the module.

5 Complete the steps in [Adjusting the linear robot alignment](#) to check and adjust the linear robot alignment.

6 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.2 Replacing the linear robot vertical motor [S-ALT4510071]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect connector 29 from Circuit board number 3.

3 Unscrew the four (4) 2.5-millimeter hex socket screws that secure the vertical motor to the linear robot vertical plate. Shift the motor to dislodge its timing belt.

4 Remove the motor and replace it with a new one. Replace the timing belt around the axle guide roller. Loosely screw back the securing screws and tighten them in a position where the timing belt is taut enough for normal operation.

⚠ The linear robot vertical motor timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Switch on the device, and launch the Systest software. Under **Card tests** on the **Robot card** tab, tap **Test tube position** in the Robot X motor row, then all the buttons in the Robot Y motor row in turn. If the motor seems to be particularly noisy because the timing belt twitches as it is moved, you will need to re-adjust the tautness of the timing belt by re-positioning the motor.

6 In the Systest software on the **Flow tests** menu, tap **Initialization**. Perform a test pipetting using a test tube with tap water or distilled water and a single LabStrip U11 Plus GL test strip. If the pipetting probe splatters the water when it is dispensing it on the test pads, you will need to readjust the tautness of the timing belt by re-positioning of the motor.

7 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.3 Replacing the linear robot vertical motor timing belt [S-56T25080]

1 Complete steps 1–3 of [6.6.2](#).

2 Remove the timing belt and replace it with a new one. Loosely screw back the vertical motor securing screws and tighten them in a position where the timing belt is taut enough for normal operation.

⚠ The linear robot vertical motor timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

3 Complete steps 5–7 in [6.6.2](#).



7.6.4 Replacing the linear robot horizontal motor [S-ALT4510071]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect connector 33 from Circuit board number 3.

3 Unscrew the four (4) 2.5-millimeter hex socket screws that secure the horizontal motor to the linear robot horizontal plate. Shift the motor to dislodge its timing belt.

4 Remove the motor and replace it with a new one. Replace the timing belt around the axle guide roller. Loosely screw back the securing screws and tighten them in a position where the timing belt is taut enough for normal operation.

⚠ The linear robot horizontal motor timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Switch on the device, and launch the Systest software. Under **Card tests** on the **Robot card** tab, tap **High probe position** in the **Robot Y motor** row, then all the buttons in the **Robot X motor** row in turn. If the motor seems to be particularly noisy because the timing belt is twitching as it is moved, you will need to re-adjust the tautness

of the timing belt by re-positioning the motor.

6 When the new motor is in place, adjust the pad-reading positions (see [1.5.3 Adjusting the pad-reading position](#)).

7 In the Systest software on the **Flow tests** menu, tap **Initialization**. Perform a test pipetting using a test tube with tap water or distilled water and a single LabStrip U11 Plus GL test strip. If the pipetting probe dispenses the water in the middle of some of the test pads, but not all of them, you will need to re-adjust the tautness of the timing belt by re-positioning of the motor.

8 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.5 Replacing the linear robot horizontal motor timing belt [S-56T25240]

1 Complete steps 1–3 in [6.6.4](#).

2 Unscrew the two (2) 1.5-millimeter screws that clamp the horizontal motor timing belt to the casing of the linear bushing that moves along the ball spline shaft (Figure 50 indicates the position of the screws when you look inside the device through the front doors).

3 Remove the timing belt and replace it with a new one. Loosely screw back the vertical motor securing screws and tighten them in a position where the timing belt is taut enough for normal operation.

⚠ The linear robot horizontal motor timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is

not too taut.

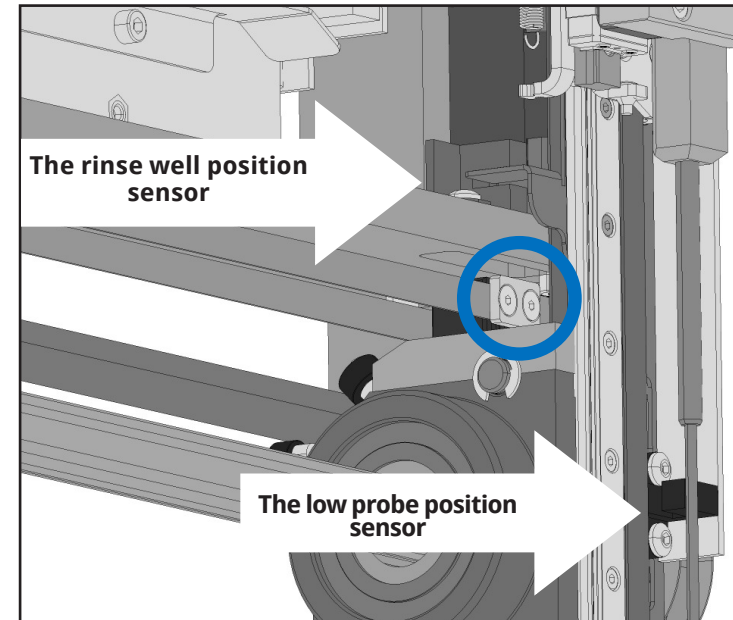


Figure 50: The location of the horizontal motor timing belt clamp screws and two of the linear robot position sensors

4 Complete steps 5–7 in [6.6.4](#)



7.6.6 Replacing the linear robot horizontal motor position sensors [S-ALT4560041]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 From Circuit board number 3, disconnect connector 6 if you are replacing the Test tube position sensor, and connector 3 if you are

replacing the Rinse well position sensor.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor you are replacing to the linear robot horizontal bracket.

i You can see the Rinse well position sensor in Figure 50. You can find its counterpart, the Test tube position sensor at the other end of the same bracket.

4 Remove the sensor you are replacing and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Complete the steps in [1.5.1](#) to check and, if necessary, adjust the position of the sensors.

6 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.7 Replacing the linear robot vertical motor position sensor [S-UPA42033]

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Disconnect the Flexible Flat Cable's connector from the Vertical motor interface board (see [Figure 46](#)).

3 Unscrew the four (4) (two for each sensor) 2-millimeter hex socket button head screws that secure the motor positions sensors, thereby the interface board itself, to the pipetting probe timing belt mount plate.

4 Remove the interface board and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Complete the steps in [6.5.3](#) to check and, if necessary, adjust the High probe sensor.

6 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.8 Replacing the pipetting probe / the pipetting block [S-UPA4403291 / S-UPA2634043]

1 Switch off the device, open the device doors, and bring the pipetting probe towards the front for easier access.

2 Unscrew the threaded plastic ring that secures the tube labeled IN4 to the pipetting block.

- if you are replacing the pipetting probe only, carefully push the probe upwards in its socket to remove it
- if you are replacing the pipetting probe with its mounting block, press its release button on the pipetting head (see Figure 51) and push the block upwards to remove it

3 Remove the part you are replacing, and insert a new one. Complete the relevant previous steps in reverse order and direction.

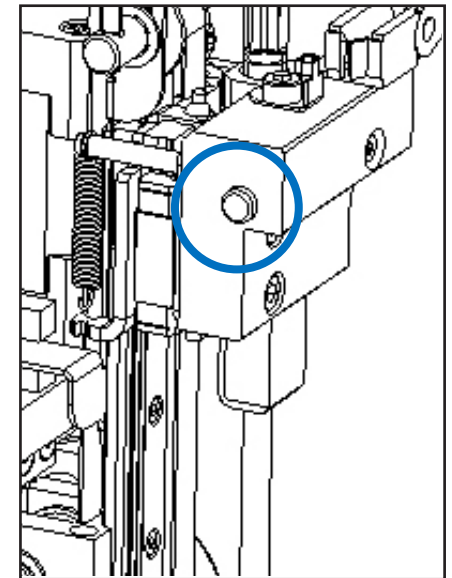


Figure 51: The pipetting probe carriage with the probe block release button

⚠ Make sure that you do not overtighten the threaded plastic ring that secures the IN4 tube. If you have a torque wrench available, apply 0.45 N·m torque when tightening the ring.

4 See [5.6.1](#), and check that the water line is filled up properly.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



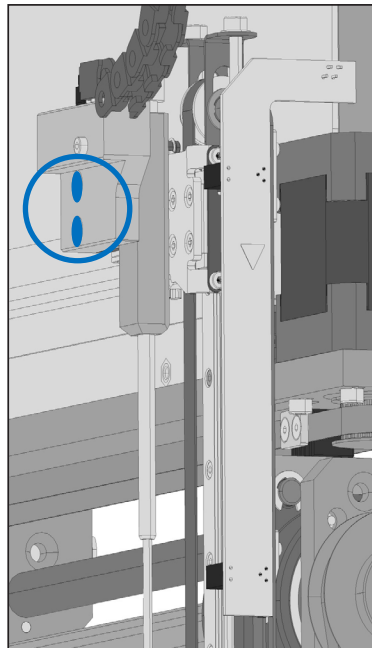
7.6.9 Replacing the pipetting head [S-UPA4403164]

1 Switch off the device, open the device doors, and bring the pipetting probe towards the front for easier access.

2 Unscrew the threaded plastic ring that secures the tube labeled IN4 to the pipetting block.

< 3 Unscrew the Phillips screw that secures the plastic energy chain to the top of the pipetting head.

Figure 52: The screws securing the pipetting head to the linear motion bearing and the pipetting block



4 Unhook the top hook of the coil spring (seen in Figure 51) from the linear motion bearing.

5 Unscrew the two (2) 2.5-millimeter hex socket screws that secure the pipetting head to the linear motion bearing and the pipette mounting block (see Figure 52).

6 Remove the pipette head and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ Make sure that you do not overtighten the threaded plastic ring that secures the IN4 tube. If you have a torque wrench available, apply 0.45 N·m torque when tightening the ring.

7 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.10 Replacing the Flexible Flat Cable (FFC) [S-UPA07131]

1 Switch off the device and open the device doors.

2 Disconnect the Flexible Flat Cable connector from both the Horizontal- and the Vertical motor interface boards (see [Figure 45](#) and [Figure 46](#)). **>**

3 Remove the FFC and replace it with a new one, making sure that it sits inside its dedicated narrow guard plate and that it cannot come into contact with any moving part.

4 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.11 Replacing Circuit board number 3

1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect all the cable connectors from Circuit board number 3 (see [Figure 43](#)).

3 Unscrew the four (4) 2.5-millimeter button head hex socket screws securing the board to the linear robot module guard plate.

4 Replace the circuit board with a new one and complete the previous steps in reverse order and direction. Refer to [Figure 43](#) when reconnecting the cables and pay particular attention to the configuration of the DIP switches.

< 5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.



7.6.12 Replacing the Oscillator interface board

1 See [4.2.2 Removing the front casing panel](#) and remove the back casing panel.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect all the connectors from the Oscillator interface board (see [Figure 44](#)).

3 Unscrew the two (2) 2.5-millimeter hex socket screws that secure the board to the linear robot module guard plate.

4 Remove the interface board and replace it with a new one. Complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.13 Replacing the Horizontal motor interface board

1 See [4.2.2 Removing the front casing panel](#) and remove the back casing panel.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure a protective plate with a grille over the module and remove the plate. Disconnect all the connectors from the Horizontal motor interface board (see [Figure 45](#)).

3 Unscrew the two (2) 2.5-millimeter hex socket screws that secure the board to the linear robot module guard plate.

4 Remove the interface board and replace it with a new one. Complete the previous steps in reverse order and direction.

5 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



7.6.14 Replacing the Vertical motor interface board

See [6.6.7](#)

8 The Rack transport module

 For a detailed overview of the module, see ["4.1.4 The Rack transport module"](#).

8.1 Motors



8.1.1 Rack positioning motor [S-ALT4510311]

A stepper motor that moves the rack catch along the rack passage, thereby pulling or pushing the racks delivered at the rack passage opening. plugged in to connector 29 on Circuit board number 4

home position: 'Pull in'

Significant positions:

- Pull in

The rack catch is moved into its innermost catch position towards the back end of the device along the rack guide plates.

sensor-determined motor position

- Push out

The rack catch is moved into its outermost catch position extending out of the device, ready to catch an arriving rack.

sensor-determined position

- Go to next tube

The rack catch is moved towards the back of the device until the Rack position sensor detects the next test tube position along the bottom of the rack in the rack passage

sensor-determined motor position



In the Systest software on the **System** parameter setup menu on the **Rack** card tab, the **Blank distance** value modifies this position. Do not modify this value without consulting 77Elektronika service personnel.



8.1.2 Rack mover motor [S-ALT4520231]

A DC motor that drives the conveyor belts on the rack conveyor via rotating a pin that catches the a spoke in the belt drive pulley axle.

plugged in to connector 27 on Circuit board number 4

Preprogrammed motor cycles:

- Long moving

The conveyor belts are moved long enough for a rack at the far right end of the rack conveyor to reach the rack aligner pins.

- Short moving

The conveyor belts are moved long enough for a rack at the rack aligner pins to align with the rack catch.

- Relieve the mover

The drive pulley is shifted slightly to the left to relieve the tension on the conveyor belts. During normal operation this happens after a rack is drawn inside the rack passage.

- Back moving

The conveyor belts are moved to the left by about the width of a rack. During normal operation, this happens before a processed rack is pushed out of the rack passage, so that there is always room on the rack conveyor in front of the passage for the rack about to be pushed out.

- STAT position

The conveyor belts are moved to the left for slightly longer than in case of a Back moving cycle. During normal operation, this happens when a statim measurement is required; this motor cycle ensures that there is room in front of the rack passage for the operator to place a rack with the statim sample. See [Send to STAT position](#).

8.1.3 Rack aligner motor [S-ALT4510131]

A linear stepper motor that tips the rack aligner platform to allow or prevent racks from moving in front of the rack passage and to steady the rack in front of the passage opening.

plugged in to connector 33 on Circuit board number 4 via connector 11 on the Rack aligner interface board

home position: 'Aligner close'

Significant positions:

- Aligner close

< The rack aligner platform is tipped so that its pins stop arriving racks from moving in front of the rack passage.

sensor-determined motor position

- Aligner open

The rack aligner platform is tipped so that its pins let arriving racks be moved through to the rack passage but not further.

sensor-determined motor position

8.2 Sensors

8.2.1 Rack delivery sensor [S-ALT4550071]

type: BOS 5K photoelectric sensor

plugged in to connector 18 on Circuit board number 4

Systest sensor signal		
Rack delivery	A rack is ready to be moved in front of the rack passage.	Sensor not activated.

The device is properly set up and operational only if the Rack delivery sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)

8.2.2 Conveyor maximum capacity sensor [S-ALT4550071]

type: BOS 5K photoelectric sensor

plugged in to connector 22 on Circuit board number 4

Systest sensor signal		
Conveyor maximum capacity	The rack conveyor has reached its maximum capacity of processed racks.  Until the racks are removed, no more racks will be pushed out of the rack passage.	Sensor not activated.

The device is properly set up and operational only if the Conveyor maximum capacity sensor is inactive (its signal is green) after system

initialization. (See [2.3.2 The Sensor tests menu](#).)



8.2.3 Tube presence sensor [S-ALT4550071]

type: BOS 5K photoelectric sensor

plugged in to connector 21 on Circuit board number 4

Systest sensor signal		
Tube presence	There is a test tube in the rack hole that is below the test tube aperture.	Sensor not activated.

The device is properly set up and operational only if the Tube presence sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)



8.2.4 Rack position sensor [S-ALT4570061]

type: passive IR LED proximity sensor

plugged in to connector 31 on Circuit board number 4

Systest sensor signal		
Rack position	Nothing is blocking the through-beam of the sensor; either the rack passage is empty, or there is a rack hole exactly below the test tube aperture	Sensor not activated.

The device is properly set up and operational only if the Rack position sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu](#).)

 A real-time value is also displayed next to the sensor signal. This dimensionless value should alternate between 0 and around 900 depending on whether the sensor is not activated

(the signal is green) or activated (the signal is red) respectively. If the real-time reading does not change when the sensor signal does, the sensor has malfunctioned.



8.2.5 Rack aligner sensors [S-URS42084]

type: transmissive photointerrupter

plugged in to connector 15 of Circuit board number 4 via connector 10 on the Rack aligner interface board

Systest sensor signal		
Aligner open	The set of pins on the left side of the rack delivery zone are extended, to stop and steady racks ready to be drawn into the rack passage.	Sensor not activated.
Aligner close	The set of pins on the right side of the rack delivery zone are extended, stopping racks from reaching the rack delivery zone.	Sensor not activated.

 These two sensors are not supplied separately, only as part of the Rack aligner interface board, as their screws secure the board to the Rack aligner unit bottom plate. The article number indicated therefore refers to both the sensors and the Rack aligner interface board.

The device is properly set up and operational only if the Aligner open sensor is inactive (its signal is green), and the Aligner close sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu](#).)



8.2.6 Rack catch sensors [S-ALT4560091]

type: transmissive photointerrupter

connectors (on Circuit board number 4)	corresponding Rack puller motor position	corresponding systest sensor signal
3	'Pull in'	Innermost catch
6	'Push out'	Outermost catch

Systest sensor signal		
		
Innermost catch	The rack catch is in the innermost catch position.	Sensor not activated.
Outermost catch	The rack catch is in the outermost catch position, ready to draw in the rack in the rack delivery zone.	Sensor not activated.

The device is properly set up and operational only if the Innermost catch sensor is active (its signal is red), and the Outermost catch position sensor is inactive (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu](#).)



8.2.7 Liquid level sensor [S-UPA42121]

type: an antenna picking up the oscillator's signal

Systest sensor signal		
		
Liquid level	The amplitude of the oscillator's signal is above the cutoff value. The amount of the sample in the test tube is enough to make a measurement. A real-time reading is also indicated.	The amplitude of the oscillator's signal is below the cutoff value: The amount of the sample in the test tube is enough for a measurement.



This sensor is synchronized with the sensor of the same name in the Robot module (see [Liquid level sensor](#)).



8.2.8 Barcode reader [S-90ICR803]

type: image code reader for reading 1D/2D codes

plugged in to connector 24 on Circuit board number 4

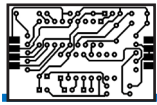


You can trigger the reader with the push-button on top.



Readings are displayed in the Systest software on the Messages screen.

8.3 PCBs



8.3.1 Circuit board number 4 [S-UPA43031]

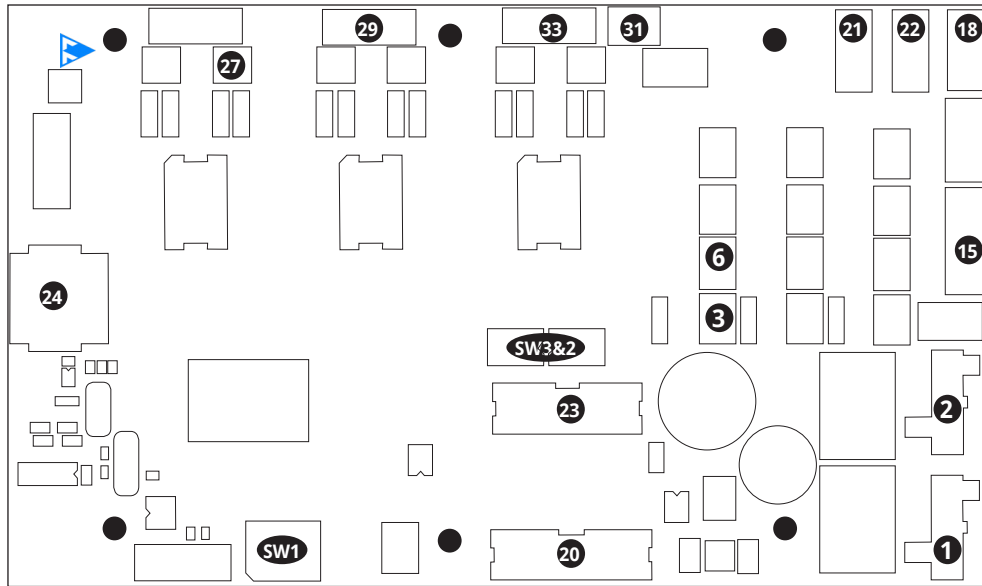


Figure 53: Board layout of Circuit board number 4.

Key to Figure 53

- 24. Ethernet connector used by the barcode reader
- 27. Rack mover motor connector
- 29. Rack positioning motor connector
- 33. Rack aligner motor interface board connector
- 31. Rack position sensor connector
- 21. Tube presence sensor connector
- 22. Rack delivery sensor connector
- 18. Conveyor maximum capacity sensor

- 6. Outermost catch position sensor connector
- 15. Rack aligner sensor connector
- 3. Innermost catch position sensor connection

SW3&2. Two 8-way configuration DIP switches (slider 2 of SW3 is on—in the down position, all other sliders are off) (For the layout and numbering of the sliders [Figure 40](#))

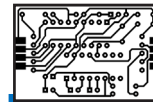
23. Bus cable connector from Circuit board number 2 (Washer module)

2. Power connector to the Washer module

SW1. PCB address DIP switch (only slider 3 is on—in the down position)

20. Level detector interface board connector

1. Power connector for the power supply



8.3.2 Rack aligner interface board [S-URS42084]

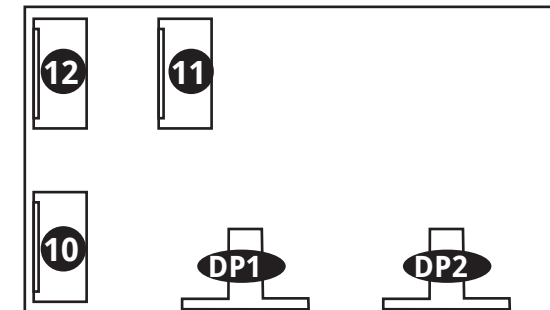


Figure 54: Board layout of the Rack aligner interface board

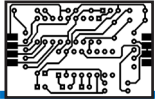
Key to Figure 54

- 12. Rack aligner motor board-to-board connector
- 11. Rack aligner motor connector

10. Rack aligner sensors board-to-board connector

DP1. Aligner close sensor connector

DP2. Aligner open sensor connector



8.3.3 Level detector interface board [S-UPA42121]

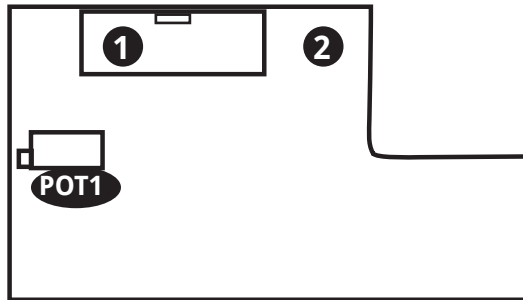


Figure 55: Board layout of the Level detector interface board

Key to Figure 55

1. 16-pin board-to-board connector

2. Level detector antenna pin connector

POT1. Potentiometer controlling the antenna

8.4 Rack transport module Flow test processes



8.4.1 Initialization

1

The Rack positioning motor moves the rack catch into its **Outermost catch** position

2

The Rack mover motor performs a **Back moving** cycle

The Rack aligner motor is moved into its **Aligner close** position

3

The Rack positioning motor moves the rack catch into its **Innermost catch** position

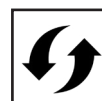


8.4.2 Tube change test

The device performs whatever motor movements are necessary to bring the next test tube present on the rack conveyor or inside the device below the test tube aperture so that the Tube presence sensor is activated. Depending on whether the next test tube is on a rack that is already inside the rack passage or on a rack that is on the rack conveyor, this Flow test process can involve a brief Rack positioning motor movement or all the steps in [7.4.4](#) several times over.



The barcode reader is also triggered whenever the Tube presence sensor is activated.



8.4.3 Push rack out

The rack catch is moved into its **Outermost catch** motor position



8.4.4 Rack change test

1

The Rack mover motor performs a **Back moving** cycle

2

The Rack positioning motor moves the rack catch into its **Outermost catch** position

3

The Rack mover motor performs a **Long moving** cycle

4

The Rack aligner motor is moved into its **Aligner open** position

5

The Rack mover motor performs a **Short moving** cycle

6

The Rack positioning motor moves the rack catch to bring the first tube hole of the rack at the rack delivery zone below the test tube aperture

The Rack aligner motor is moved into its **Aligner close** position



8.4.7 Shutdown test

The Rack transport module performs the same operations during Shutdown as during Initialization.



8.4.5 Read barcode

The barcode reader is triggered. Its readout is displayed in the Systest software on the **Messages** screen.



8.4.6 Level HW test

You can toggle the oscillator (see the [Oscillator interface board](#)) on and off by tapping this button. The button will turn green to signal that the oscillator is on.

8.5 Service actions—Adjustments



8.5.1 Tuning the oscillator antenna

1 See [Removing the back casing panel](#) and remove the back casing panel.

2 Switch on the device and launch the Systest software. Enter the **Flow tests** screen.

3 Tap the **Level HW test** button. Using a small-size precision slotted screwdriver, carefully turn the screw on the back of the potentiometer clockwise (see Figure 56). In the Systest software, enter the **Sensor tests** screen and check that the real-time value next to the **Liquid level** sensor signal is rising.

4 Stop turning the screw when the value next to the **Liquid level** sensor signal levels off (does not rise any further). Set the antenna to the highest possible value, then enter the **Flow tests** screen and tap **Level HW test** to turn the oscillator off.

5 Tap **Initialization**, then see the relevant section of the Linear robot module overview to run a **Level detection test**. Check that the tip of the pipetting probe is stopped at the surface of the tap water inside the test tubes and is not submerged in the liquid.

6 Tap **Initialization** before resuming normal operation.

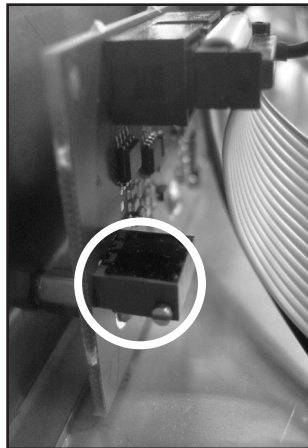


Figure 56: The potentiometer of the level detector



8.5.2 Adjusting the rack catch corrections

Complete the steps in [1.5.3 Adjusting the pad-reading position](#).



8.5.3 Adjusting the Rack delivery and the Conveyor maximum capacity sensors

1 See [“4.2.2 Removing the front casing panel”](#), and remove the front casing panel.

2 Reattach the rack conveyor unit and switch the device on.



You do not need to launch the User or the Systest software to adjust the sensors.

3 Put a solid object in front of the sensor you are adjusting at decreasing distances from the sensor. Check that the orange LED on the top of the sensor does not light up until the object you put in front of the sensor is in line with the rack aligner pins that are closest to the sensor (at a distance of 4 centimeters (1.5 inches).

4 Using a small-size precision slotted screwdriver, carefully turn the screw on the top

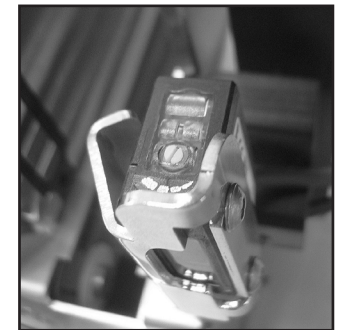


Figure 57: The Rack delivery sensor's potentiometer and securing screws

of the sensor (see Figure 57) until it is activated (its orange LED lights up) by a solid object coming within a range of 4 centimeters (1.5 inches),

but not by objects farther away.

5 Reattach the front casing panel.



8.5.4 Adjusting the Tube presence sensor

1 See [“4.2.2 Removing the front casing panel”](#), and remove the front casing panel.

2 Switch the device on and put your outstretched palm right in front of the slotted plastic plate that focuses the barcode reader’s beam so that the back of your hand touches the plate (see Figure 58).

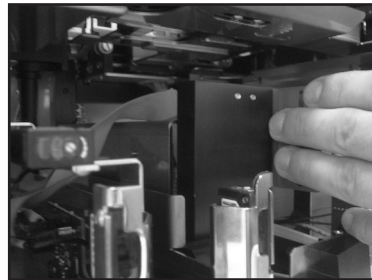


Figure 58: Checking the sensitivity of the Tube presence sensor

3 Check that the orange LED on the top of the sensor does not light up while your hand is touching the slotted plate, but that the sensor is activated whenever you place

both your touching outstretched palms next to the slotted plastic plate.

4 Using a small-size precision slotted screwdriver, carefully turn the screw on the top of the sensor (see Figure 57) until it is activated (its orange LED lights up) by a solid object coming within a range of two finger-width away from the barcode reader’s focusing plate, but not by objects farther away.

5 Reattach the front casing panel.



8.5.5 Adjusting the test tube alignment

1 Turn the device on, launch the Systest software, and open the device doors. Place a rack fully loaded with test tubes to the right of the rack delivery zone.

2 In the Systest software on the **Flow tests** menu, tap the **Tube change test** button to make the first test tube on the rack align with the test tube aperture.

3 Place the aligner screw into the test tube aperture (see [Figure 47](#)).

4 In the Systest software under **Card tests** on the **Robot card** tab, tap **STOP** in both motor rows. Manually move the pipetting probe over the test tube aperture and lower it into the hole in the aligner screw.

5 Check that the pipetting probe moves down in the middle of the test tube below the test tube aperture. If the test tube seems misaligned, tap the **Read parameters**, then the **Tube correction** button in the Systest software on the **System parameter setup** menu on the **Rack card**. Adjust the value with the **Increment(+)** or **Decrement(-)** button as necessary. Tap the **Save parameters** button to store the adjusted value. Remove the aligner screw from the test tube aperture, then tap **Initialization** on the **Flow tests** menu.



Each motor step translates to a 0.25 millimeter movement of the rack.

6 Repeat steps 2–5 until the first test tube is perfectly aligned with the pipetting probe, then tap the **Tube change test** button once for

every other test tube on the rack to check whether there is global test tube alignment.



8.5.6 Preset system parameters

⚠ The following parameters are included in this manual for information purposes only. Do not modify any of these values without explicit prior permission from 77Elektronika service personnel.

i You can access these parameters in the Systest software on the **System parameter setup** menu on the **Rack card** tab by tapping the **Read parameters** button.

Key to the system settings in the Systest software on the **System parameter setup** menu on the **Rack card** tab:

- Transfer time

This parameter is relevant only if your LabUMat 2 device is connected to a UriSed or UriSed 2 urine analyzer. The parameter indicates the maximum time in multiples of 0.01 seconds that is allowed to elapse between a test tube rack leaving the rack passage and the rack conveyor belts of the UriSed or UriSed 2 device.

- Barcode reader type

A variable that indicates the type of barcode reader installed in the device.

8.6 Service actions—Replacements



8.6.1 Replacing the Rack positioning unit [S-UPA4451003]

1 See [“4.2.2 Removing the front casing panel”](#), and remove the front casing panel.

2 Complete steps 2–3 in [7.6.2](#) to remove the Rack aligner unit from the device.

3 Carefully tip the device backwards to access its bottom casing panel and unscrew the two (2) 3-millimeter hex socket head screws that secure the Rack positioning unit to the device chassis at the front.

4 Unscrew the two (2) 3-millimeter hex socket head screws that secure the unit to the device chassis at the back of the device.

5 Disconnect connectors

- 24 (the Ethernet cable)
- 23 and 20 (the two 16-pin ribbon cables)
- 1 and 2 (the two power cables)

from Circuit board number 4 so that you can safely and comfortably remove the unit from the device.

6 Firmly grab the unit from the front by its housing plate, dislodge it, and carefully withdraw it from the device.

⚠ Take care not to damage the pipetting stage presence sensor or let the elements of Circuit board number 4 get caught in the Rack aligner unit’s tipper while you remove the unit.

- 7 Complete steps 2–5 in reverse order and direction with the new unit. Perform the adjustments in [8.5 Service actions—Adjustments](#) before resuming normal operation.



8.6.2 Replacing the rack aligner unit [S-URC4469001]

- 1 See [“4.2.2 Removing the front casing panel”](#) and remove the front casing panel.
- 2 Carefully tip the device backwards to access its bottom casing panel and unscrew the two (2) 3-millimeter hex socket head screws that secure the unit to the device chassis at the front.
- 3 Disconnect connectors 33 and 15 from Circuit board number 4, and carefully remove the unit from the device.
- 4 Replace the rack aligner unit with a new one. Complete the previous steps in reverse order and direction.
- 5 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.3 Replacing the Rack positioning motor

- 1 Complete steps 1–5 in [7.6.1](#) to remove the rack positioning unit.
- 2 Disconnect connector 29 from Circuit board number 4.
- 3 Reach inside the unit housing plate from the side to access the motor and unscrew the two (2) 2.5-millimeter hex socket screws that

secure the motor to two of the support bars.

- 4 Reach inside the unit housing plate through the bottom and take off the timing belt from the motor axle.
- 5 Remove the rack positioning motor through the bottom of the unit housing plate and replace with a new one. Complete the previous steps in reverse order and direction.

⚠ When replacing the Rack positioning motor, shift its position within its adjustable brackets so that the motor timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

- 6 Switch on the device, launch the Systest software, and tap **Initialization** on the **Flow tests** menu before resuming normal operation.



8.6.4 Replacing the Rack positioning motor timing belt [S-56T25240]

- 1 Complete steps 1–2 in [7.6.3](#) to remove the Rack positioning unit and disconnect its motor connector.
- 2 Unscrew the six (6) Phillips head screws that secure the three supporting crossbars to the bottom of the unit housing plate, and remove the crossbars.
- 3 Reach inside the unit housing plate from the side to access the motor plate and unscrew the two (2) 2.5-millimeter hex socket screws that secure the motor to two of the support bars. Take off the motor timing belt from the motor axle.

4 Disconnect all the connectors from Circuit board number 4. Unscrew the four (4) 2-millimeter hex socket button head screws that secure the PCB. Remove Circuit board number 4.

5 Unscrew the five (5) Phillips screws that secure the rack catch rail to the inside of the unit housing plate. Take out the rail with the timing belt attached.

6 Unscrew the two (2) 1.5-millimeter hex socket screws that secure the Rack positioning motor timing belt to the rack catch rail. Remove the timing belt and replace with a new one.

7 Complete the previous steps in reverse order and direction. Make sure that you cut the new piece of timing belt to the same length as the one you are replacing.

⚠ When re-tightening the Rack positioning motor screws, shift the motor's position within its adjustable brackets so that the motor timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

8 Switch on the device, launch the Systest software, and tap **Initialization** on the **Flow tests** menu before resuming normal operation.



8.6.5 Replacing the Rack mover motor

1 Complete steps 1–5 in [7.6.1](#) to remove the Rack positioning unit.

2 Remove the retaining ring that secures the rotating disk that drives the drive pulley of the rack conveyor to the flattened axle of the Rack

mover motor. Remove the rotating disk from the motor axle.

3 Disconnect connector 27 from Circuit board number 4.

4 Unscrew the three (3) 2-millimeter countersunk hex socket screws around the axle that secure the Rack mover motor to the unit housing plate.

5 Remove the motor through the bottom of the unit housing and replace it with a new one. Complete the previous steps in reverse order and direction.

6 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.6 Replacing the rack aligner motor

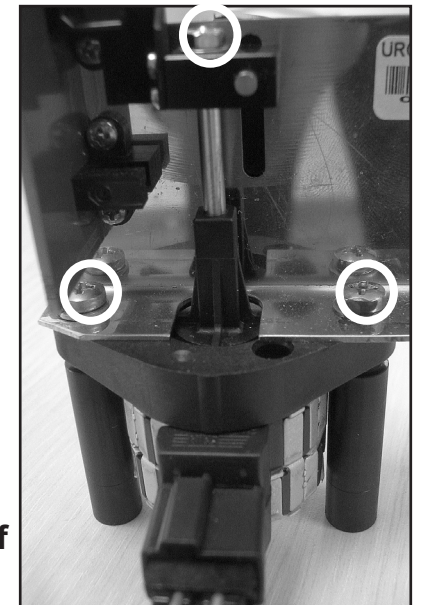
1 Complete steps 1–3 in [7.6.2](#) to remove the Rack aligner unit.

2 Unscrew the two (2) Phillips screws and the hex nut with a flat washer that secure the rack aligner motor to the rack aligner bottom plate (see Figure 59).



Save the hex nut and the flat washer. These parts are not supplied when you order a Rack aligner motor as a spare part.

Figure 59: The screws of the Rack aligner motor



3 Remove the Rack aligner motor and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ **Make sure that the switch flag of the Rack aligner sensors stays between the sensors when you secure the new motor's threaded axle to it.**

4 Switch on the device, launch the Systest software, and on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.7 Replacing the Rack delivery sensor

1 See ["4.2.2 Removing the front casing panel"](#) and remove the front casing panel.

2 Disconnect connector 22 from Circuit board number 4.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the rack passage plate.

4 Remove the sensor and replace with a new one. Complete the previous steps in reverse order and direction.

5 Complete the steps in [7.5.3](#) to check, and, if necessary, adjust the sensor's sensitivity.

6 Launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.8 Replacing the Conveyor maximum capacity sensor

1 See ["4.2.2 Removing the front casing panel"](#) and remove the front casing panel.

2 Disconnect connector 18 from Circuit board number 4.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the rack passage plate.

4 Remove the sensor and replace with a new one. Complete the previous steps in reverse order and direction.

5 Complete the steps in [7.5.3](#) to check, and, if necessary, adjust the sensor's sensitivity.

6 Launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.9 Replacing the Tube presence sensor

1 See ["4.2.2 Removing the front casing panel"](#) and remove the front casing panel.

2 Disconnect connector 21 from Circuit board number 4.

3 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the bracket at the top of the rack passage plate.

4 Remove the sensor and replace with a new one. Complete the previous steps in reverse order and direction.

5 Complete the steps in [7.5.4](#) to check, and, if necessary, adjust the sensor's sensitivity.

6 Launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.10 Replacing the Rack position sensor

1 See "[4.2.2 Removing the front casing panel](#)" and remove the front casing panel.

2 Disconnect connector 31 from Circuit board number 4.

3 Unscrew the 1.5-millimeter hex socket set screw on the side of the sensor block and gently pull the IR LED out of the block by its connector cables.

4 Insert the new IR LED into the sensor block and repeat the previous steps in reverse order and direction.

5 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.11 Replacing the Rack aligner sensors

These sensors are not supplied as separate parts. See [7.6.16](#) to replace the sensors together with their PCB.



8.6.12 Replacing the barcode reader

1 Complete steps 1–5 in [7.6.1](#) to remove the Rack positioning unit.

2 Disconnect connector 24 from Circuit board number 4.

3 Unscrew the two (2) 2.5-millimeter hex socket screws that secure the barcode reader to the barcode reader plate that is fixed to the rack passage plate.

4 Remove the barcode reader and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Place a rack filled with barcode-labeled test tubes (or use the single sample test tube with the barcode attached) on the delivery side of the rack conveyor. Switch on the device, launch the Systest software, and on the **Flow tests** menu, tap the **Tube change test** button as many times as the number of test tubes in the rack. In the messages screen, check that each test tube barcode is properly scanned and read, and adjust the position of the barcode reader in its adjustable bracket if the barcode scanner is not reading reliably.

6 See [Load defaults](#) and download the barcode parameters to the new device. Tap **Initialization** before resuming normal operation.



8.6.13 Replacing the Rack catch sensors

1 Complete steps 1–5 in [7.6.1](#) to remove the Rack positioning unit.

2 Disconnect connector 6 or 3 from Circuit board number 4, depending on whether you are replacing the Outermost or the Innermost catch sensor, respectively.

3 Unscrew the two (2) 1.5-millimeter hex socket set screws that secure the sensor to the unit housing plate. Save the washer plate.

4 Remove the sensor and replace it with a new one. Complete the relevant previous steps in reverse order and direction.

5 If you have replaced the Outermost catch position sensor, complete the steps in [1.5.3 Adjusting the pad-reading position](#).

6 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.15 Replacing the rack conveyor timing belts

1 Snap the rack conveyor unit off the device. Unscrew the four (4) leveller screws from the bottom of the unit and remove the hex nuts that secure them to the unit bottom.

2 Unscrew the two (2) Phillips head screws that secure the plastic bottom panel of the unit and remove the panel.

3 Unscrew the two (2) Phillips head screws that secure the rack conveyor side panels (see [Figure 7](#)), and remove the panels.



If you have a sediment analyzer connected to your LabUMat 2 device, you will need to remove only one side panel.

4 Unscrew the four (4) 2.5-millimeter hex socket cap screws that secure the unit front panel to the inside of the chassis, and remove the panel.

5 Unscrew the four (4) Phillips flat head screws on the bottom of the unit that secure the plastic spacer over the cogged timing belt rollers, and remove the spacer.

6 Loosen the two (2) Phillips flat head screws on the outside of the chassis that secure the block holding the cogged timing belt roller for the timing belt you are replacing. If you are replacing the timing belt that has two instead of a single cogged roller, you will also need to loosen the 1.5-millimeter hex socket set screw that secures the top cogged roller inside the roller block (see Figure 60).

7 Loosen the two (2) Phillips flat head screws on the bottom of the chassis that secure the block with the central stretcher bar and rollers for the timing belt you are replacing.

8 Wiggle the timing belt you are replacing free from the loosened rollers, and lift it off the rack conveyor.

9 Replace the timing belt with the new one. Complete the previous steps in reverse order and direction. Before you snap the rack conveyor unit back on the device, test the smooth operation of the timing belts by placing a rack with test tubes on the conveyor and manually rotating the belt-moving pin that the Rack mover motor engages.

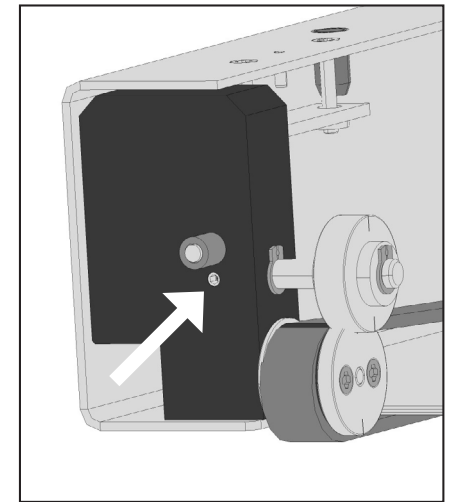


Figure 60: The set screw that secures the top cogged timing belt roller



8.6.16 Replacing Circuit board number 4

- 1 Complete steps 1–5 in [7.6.1](#) to remove the Rack positioning unit.
- 2 Disconnect all the connectors from Circuit board number 4 (see [Figure 53](#)).
- 3 Unscrew the four (4) 2.5-millimeter hex socket button head screws that secure the circuit board to the PCB plate. Save the serrated washers and the metal spacer rods.
- 4 Replace the circuit board with a new one and complete the previous steps in reverse order and direction. Refer to [Figure 53](#) when reconnecting the cables and pay particular attention to the configuration of the DIP switches.
- 5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.



8.6.17 Replacing the Rack aligner interface board

- 1 Complete steps 1–3 in [7.6.2](#) to remove the Rack aligner unit from the device.
- 2 Disconnect all the connectors from the interface board (see [Figure 54](#)).
- 3 Unscrew the four (4) Phillips head screws that secure the PCB and the rack aligner sensors to the unit bottom plate.
- 4 Remove the interface board and replace it with a new one. Complete the previous steps in reverse order and direction.

- 5 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



8.6.18 Replacing the Level detector interface board

- 1 Complete steps 1–3 in [7.6.2](#) to remove the Rack aligner unit from the device.
- 2 Disconnect all the connectors from the interface board (see [Figure 55](#)).
- 3 Unscrew the two (2) 2.5-millimeter hex socket button head screws that secure the interface board to the rack passage plate. Save the metal spacer bars.
- 4 Remove the interface board and replace it with a new one. Complete steps 2 and 3 in reverse order and direction.
- 5 Complete steps 2–6 in [7.5.1](#) to adjust the Oscillator antenna.

9 Strip timer and Photometer module

i For a detailed overview of the module, see [4.1.5 The Strip timer and photometer module](#).

! If the strip bin presence microswitch is not activated, the system disables all strip timer comb movements.

i The following chart describes the very first movement cycle the strip timer strip timer comb performs during normal operation. The initial **Pull to head** movement indicated is not performed in any of the subsequent cycles.

Motor movements during a single strip timer comb movement cycle	
Horizontal strip timer motor	Vertical strip timer motor
1 Pull to head	
	2 To strip position
3 Drop the strip	
	4 To home position
5 One step back	
	6 To strip position
7 Pull to head	
	8 To home position
9 To load position	
	10 To load position
11 Pull to head	
	12 To home position

 This is an interactive slideshow demonstration of a single cycle of the strip timer comb. Click the navigation button on the right to see the next image in the slideshow. Click the navigation button on the left to restart the slideshow. For demonstration purposes, the first five (5) cycles have been skipped: the slideshow illustrates a movement cycle that corresponds to a cycle that would be the sixth one during normal operation, when three (3) test strips are already stacked across the platform frame.



9.1 Motors



9.1.1 Horizontal strip timer motor [S-ALT4510011]

A stepper motor that moves the strip timer comb horizontally via a timing belt.

plugged in to connector 13 on Circuit board number 5

home position: the strip timer comb is at the pipetting stage

Significant motor positions:

- To home position

The conveyor comb is moved into its horizontal home position: to the end of its movement range at the pipetting stage where its leftmost prongs straddle the pipetting stage itself.

sensor-determined motor position

- To load position

Identical to the **To home position** in all respects

- One step back

The strip timer comb is moved horizontally so that a following rising movement of the conveyor high enough to engage the strips across the platform frame (**To strip position**, see [8.1.2](#)) will engage the strip that is the next in line (see Figure 61).

- Drop the strip

The strip timer comb is moved horizontally to the end of its range at the strip reading platform: its rightmost prongs extend over the right edge of the reading stage to push the test strip into the strip bin.

motor step number–determined motor position (counted off from the

To home position)

- Pull to head

The strip timer comb is moved horizontally in the direction of the strip reading platform so that its rightmost prongs push a test strip under the photometer for pad-reading.

motor step number–determined motor position (counted off from the

To home position)

- Comb check

The strip timer comb is briefly jerked left and right to make sure that the Strip timer comb presence sensor is activated.

motor step number–determined motor position

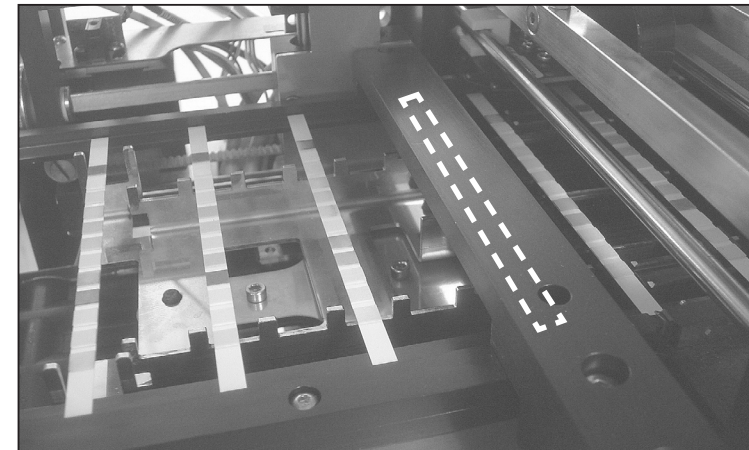


Figure 61: A full complement of test strips stacked across the platform support frame and the strip reading platform. The rightmost strip is below the photometer. The test strip concealed by the platform support crossbeam is indicated with a dashed outline



9.1.2 Vertical strip timer motor [S-ALT4510011]

A stepper motor that moves the strip timer comb vertically via a timing belt.

plugged in to connector 14 on Circuit board number 5

home position: 'To home position'

Significant positions:

- To home position

The strip timer comb is moved into its lowermost vertical position. None of the prongs engage any test strips across the pipetting stage, the platform frame, or the strip reading platform.

sensor-determined motor position

- To load position

The strip timer comb is moved into a vertical position where the large prongs on either end can engage the test strips, but the regular prongs along the middle edges of the conveyor cannot.

motor step number-determined motor position (counted off from the

To home position)

- To strip position

The strip timer comb is moved into a vertical position high enough for even the regular-size prongs in the middle edges of the conveyor to engage the strips across the platform frame.

motor step number-determined motor position (counted off from the

To home position)



9.1.3 'Head mover motor' [S-ALT4510011]

A stepper motor that moves the photometer over the test strip pads via a timing belt and three idler pulleys.

plugged in to connector 15 on Circuit board number 5

home position: 'To home position'

Significant positions:

- To home position

The photometer is moved to its home position at the end of its movement range towards the back of the device.

- To end position

The photometer is moved to its home position at the end of its movement range towards the front of the device: the LEDs are above the front leading edge of the test strip that is in pad-reading position.

motor step number-determined motor position (counted off from the

To home position)



You can trigger all the preprogrammed pad-reading movements (see [8.4.3](#)) and two (2) additional Head mover motor positions in the Systest software for adjustment purposes (see [1.5.3 Adjusting the pad-reading position](#))

9.2 Sensors



9.2.1 Horizontal strip timer home position sensor [S-ALT4560021]

type: transmissive photointerrupter

plugged in to connector 4 on Circuit board number 5

Systest sensor signal		
		
Strip puller home	The strip timer comb is in its horizontal home position.	Sensor not activated.

The device is properly set up and operational only if the Horizontal strip timer home position sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



9.2.2 Vertical strip timer home position sensor [S-ALT4560021]

type: transmissive photointerrupter

plugged in to connector 5 on Circuit board number 5

Systest sensor signal		
		
Comb lifter home	The strip timer comb is in its vertical home (that is, its lowest) position.	Sensor not activated.

The device is properly set up and operational only if the Vertical strip timer home position sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



9.2.3 Photometer home position sensor

type: transmissive photointerrupter

plugged in to connector 6 on Circuit board number 5

Systest sensor signal		
		
Head mover home	The photometer is in its home position.	Sensor not activated.

The device is properly set up and operational only if the Photometer home position sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



9.2.4 Strip timer comb presence sensor [S-ALT4550011]

type: BOS 5K photoelectric sensor

plugged in to connector 7 on Circuit board number 5

Systest sensor signal		
		
Comb present	The strip timer comb is in place.	Sensor not activated.

The device is properly set up and operational only if the Strip timer comb presence sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu.](#))



9.2.5 Strip bin microswitch sensor [S-UAZ42321]



If the strip bin presence microswitch is not activated, the system disables all strip timer comb movements.

type: microswitch

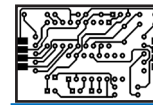
plugged in to connector 10 on Circuit board number 5

Systest sensor signal		
Drawer detection	Sensor not activated.	The Strip bin is in place.

The device is properly set up and operational only if the Strip bin presence sensor is active (its signal is green) after system initialization. (See [2.3.2 The Sensor tests menu.](#))

 **The sensor signals of this sensor are reversed relative to the other sensors: It is activated when its signal is green.**

9.3 PCBs



9.3.1 Strip timer and photometer control circuit board [S-UPA43061]

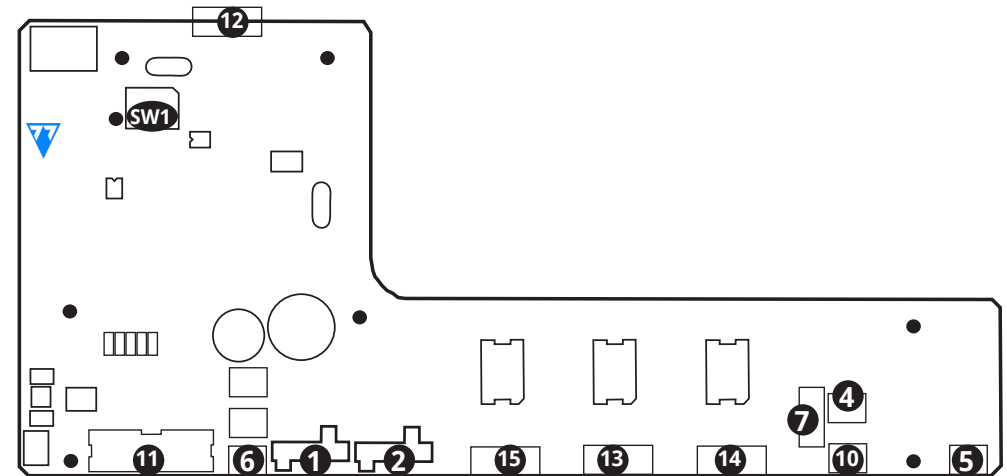
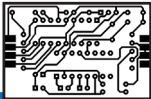


Figure 62: Board layout of Circuit board number 5

Key to Figure 62

- 12. Photometer interface board
- SW1. PCB address DIP switch (sliders 1 and 3 are on—in the down position)
- 11. 16-pin Bus ribbon cable connector
- 6. Photometer home position sensor connector
- 1. Power connector for the power supply
- 2. Power connector to the Dispenser module
- 15. Head mover motor connector
- 13. Horizontal strip timer motor connector
- 14. Vertical strip timer motor connector

- 7. Strip timer comb presence sensor connector
- 4. Horizontal strip timer home position sensor connector
- 10. Strip bin microswitch sensor connector
- 5. Vertical strip timer motor home position sensor connector



9.3.2 Photometer interface board [

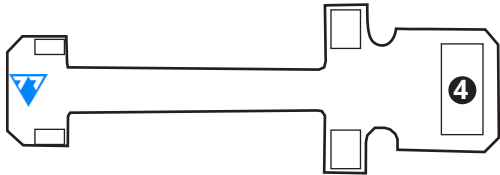


Figure 63: Board layout of the Photometer interface board

Key to Figure 63

4. Flexible Flat cable connector to Circuit board number 5

i There is a second, rectangular PCB that is part of the photometer assembly. The other circuit board is, however, not serviceable.

9.4 Strip timer and Photometer module Flow tests processes

! If the strip bin presence microswitch is not activated, the system disables all strip timer comb movements.



9.4.1 Initialization

1	The strip timer comb is moved into its vertical loading, then its vertical home position.	The Photometer is moved into its home position.
2	The strip timer comb performs three (3) cycles of the following movements:	
	Vertical strip timer motor	Horizontal strip timer motor
	To strip position	
		Drop the strip
	To home position	
		To home position



9.4.2 Strip comb test

The strip timer comb performs a complete movement cycle (see [Strip timer comb cycle](#)).



9.4.3 Head cycle test

The Photometer performs a complete pad-reading cycle. The LEDs light up above each pad as in normal operation, and reflectance values are displayed in the text box on the right-hand side of the **Flow tests** screen.



9.4.4 Go to next pad

The Photometer is moved into the position corresponding to above the next pad in the Bilirubin–Specific gravity direction. If you tap the button while the Photometer is in its home position, the unit will first be moved into its end position before stopping over the Bilirubin pad.



9.4.5 Shutdown test

The Strip timer and Photometer module performs the same operations during Shutdown as during Initialization, except that the strip timer comb is not moved into its vertical load and vertical home positions before starting the three movement cycles.

9.5 Service actions—Adjustments



9.5.1 Repositioning the module after replacement

- 1 See [Removing the top side casing panels](#) and remove the top side casing panel on the side of the strip timer module.
- 2 Put their serrated washers in place and loosely screw in the two (2) horizontal-position 3-millimeter hex socket screws that secure the module to the device chassis along the back of the module bottom plate.
- 3 Put their serrated washers in place and loosely screw in the two (2) vertical-position 3-millimeter hex socket screws that secure the module to the device chassis in the adjustable bracket at the front of the module.
- 4 Unscrew the four (4) 2.5-millimeter hex socket screws that secure the protective plate over the photometer bracket, and remove the protective plate. Looking in through the side of the device to check, grab and shift the module until the reading platform is level with, or, preferably, slightly lower than the platform frame (the frame that the test strips were placed across on the way to the reading platform.)
- 5 When you have properly aligned the module, tighten the vertical-position screws in the adjustable brackets first, then the two horizontal-position ones at the back of the device.
- 6 See [8.5.2](#) and adjust the pad-reading position if necessary.
- 7 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.5.2 Adjusting the pad-reading position

See [1.5.3 Adjusting the pad-reading position](#).

⚠ The pad-reading parameter adjustments modify the following parameters in the Systest software on the System parameter setup menu on the Strip-Head moving card tab: Head pos. correction, Checking position, and Strip start position. Do not modify these values directly on the System parameter setup menu.



9.5.3 Adjusting the strip reading platform clearance and alignment

1 Switch off the device and unscrew the four (4) 2.5-millimeter screws that secure the protective plate over the photometer bracket.

2 See [Removing the top side casing panels](#) and remove the top side casing panel.

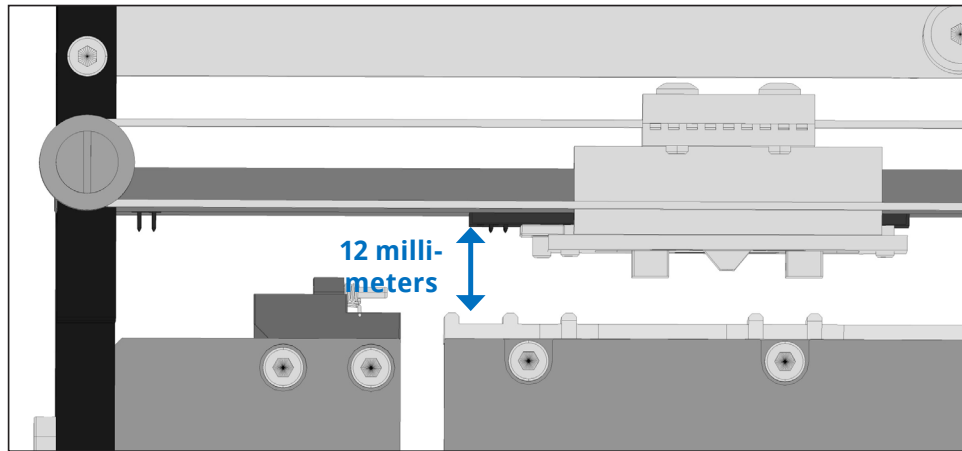


Figure 64: Partial cutaway of the photometer unit showing proper clearance between the photometer and the reading platform

3 Insert a clean and demagnetized 12-millimeter thick gauge block or equivalent measuring tool between the ribs on the reading platform and the lower (rectangular) PCB of the photometer (see Figure 64). The 12-millimeter gauge block should fit snugly between the platform ribs and the rectangular PCB. If the gap is too wide or too narrow, adjust the reading platform accordingly with the four (4) 2.5-millimeter hex socket screws that secure it to the unit housing plate (see Figure 65).

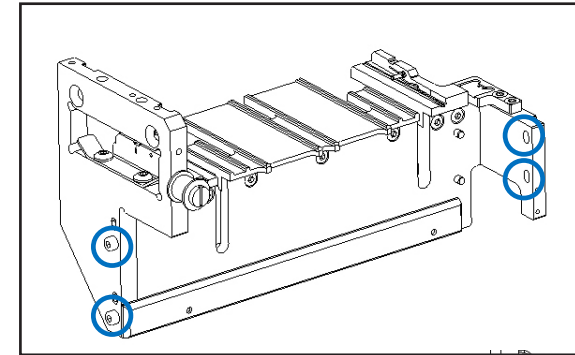


Figure 65: The strip reading platform adjusting screws

4 Repeat step 2 in at least one more place along the reading platform to ensure parallel alignment.

5 Check whether the reading platform is level with, or, preferably, slightly lower than the platform frame. If it is not, see [8.5.2](#) to reposition the unit.

6 See [1.5.3 Adjusting the pad-reading position](#) and adjust the pad-reading position if necessary.

7 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.5.4 Adjusting the strip timer comb front-back axis parallelism

- 1 Loosen one or both of the 2-millimeter hex socket screws that secure the strip timer comb's bottom plate to the bracket plate connecting the conveyor and the strip timer train (see Figure 66).

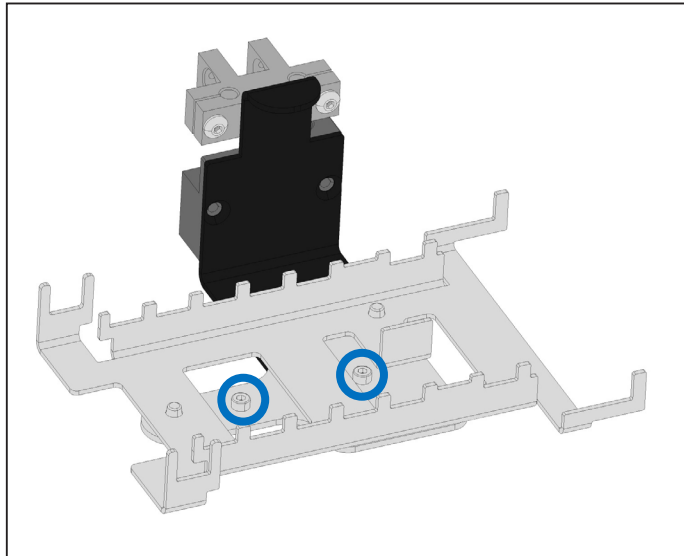


Figure 66: The Strip timer comb adjusting screws

- 2 Rotate the strip timer comb in the adjustable bracket of the screws you loosened until the rightmost large prongs of the conveyor are perfectly parallel with the reading platform. Re-tighten the screws when you are done.
- 3 Switch on the device. Place a LabStrip U11 GLtest strip on the left edge of the reading platform. Launch the Systest software. Under **Card tests** on the **Strip-Head moving card** tab, tap the **Pull to**

head button in the **Strip puller motor** row to make the Strip timer comb push the test strip below the photometer.

- 4 On the **Flow tests** menu, run a **Head cycle test** and check whether the LEDs on the photometer light up consistently above the center of the pads along the whole length of the test strip.
- 5 Repeat steps 1–4 if necessary until the Strip timer comb pushes the test strip under the photometer perfectly in line with the photometer.
- 6 On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.5.5 Preset system parameters

⚠ The following parameters are included in this manual for information purposes only. Do not modify any of these values without explicit prior permission from 77 Elektronika service personnel.

i You can access these parameters in the Systest software on the System parameter setup menu on the Robot card tab by tapping the Read parameters button.

Key to the system settings in the Systest software on the **System parameter setup** menu on the **Strip-Head moving card** tab:

- Incubation time

The maximum time, in multiples of 0.1 seconds, that is allowed to elapse between sample dispensing and the photometric measurement.

- Comb cycle

The time, in milliseconds, that elapse between two consecutive movement cycles of the strip timer comb during normal operation.

- Fine step in scan modes

This parameter is relevant only to development engineers fine-tuning the device.

- Back step

The number of motor steps the strip timer comb is moved by horizontally towards the pipetting stage before every vertical To load movement as a security measure to avoid accidentally knocking the test strip off the stage.

9.6 Service actions—Replacements



9.6.1 Replacing the Strip timer and photometer module [S-UPA4408007]

- 1 Remove the strip timer comb. See [Removing the back casing panel](#) and remove the back casing panel.
- 2 Cut the cable tie that secures the bundle of the Bus ribbon cable and the two power cables to the fastener clip screwed to the strip timer unit bottom plate, then disconnect their connectors (11, 1, and 2), along with connectors 10 and 7 (see [Figure 62](#)).

⚠ When reconnecting the cables, you will need to bundle the Bus ribbon cable and the power cables together with a cable tie and secure the bundle to the fastener clip before pulling it through the service opening in the module's bottom plate. Besides the Bus cable and the power connectors, the Strip timer comb presence sensor's cable also runs through the service opening.

- 3 Unscrew the two (2) 3-millimeter hex socket screws that secure the module to the device chassis at the back and the two (2) 3-millimeter hex sockets that secure the module to the device chassis in an adjustable bracket at the front.
- 4 Remove the module through the back and replace it with a new one. Complete the previous steps in reverse order and direction.
- 5 Complete the steps in [8.5.1](#) to properly reposition the module. See [1.5.3 Adjusting the pad-reading position](#) and adjust the pad-reading positions if necessary.

6 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.6.2 Replacing the Horizontal strip timer motor

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Disconnect connector 13 from Circuit board number 5 (see [Figure 62](#)) and pull the strip timer comb bracket plate away from the photometer bracket.

3 Unscrew the four (4) 2-millimeter hex socket button head screws, accessible behind the shaft and timing belt of the Horizontal strip timer motor, that secure the motor to the central motor plate that is attached to the module bottom plate (see [Figure 67](#)). Save the serrated washers and the flat washers.

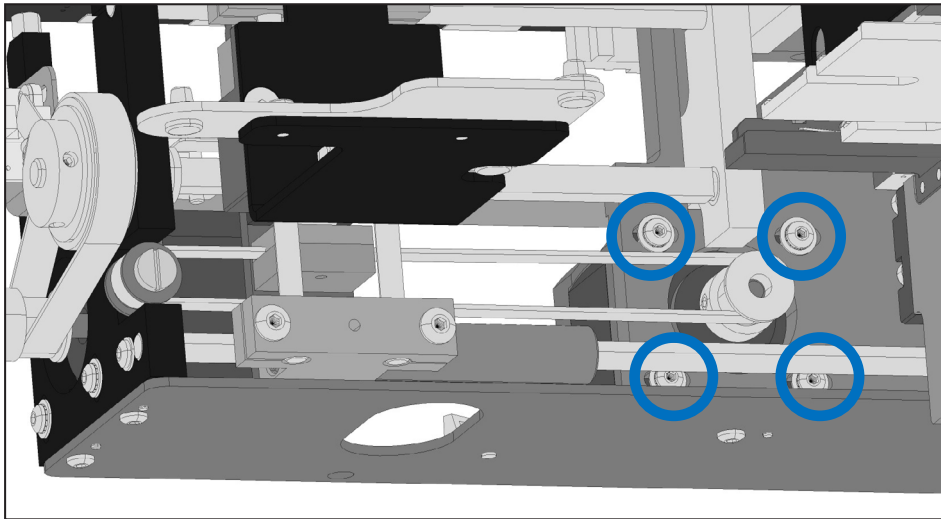


Figure 67: The securing screws of the Horizontal strip timer motor as seen from below the Strip timer unit

4 Dislodge the timing belt from the motor axle. Remove the motor and replace with a new one. Complete the previous steps in reverse order and direction.

⚠ When replacing the Horizontal strip timer motor, shift its position within its adjustable brackets so that the motor timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Complete steps 5–6 in [8.6.1](#) before resuming normal operation.



9.6.3 Replacing the Horizontal strip timer motor timing belt [S-56T25122]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the two (2) 2.5-millimeter hex socket countersunk head screws that secure the strip timer comb bracket plate to the strip timer carriage and remove the bracket plate.

3 Unscrew the two (2) 2-millimeter hex socket screws that secure the timing belt clamp to the strip timer carriage (see [8.6.1](#)).

4 Loosen the four (4) 2-millimeter hex socket button head screws, accessible behind the shaft and timing belt of the Horizontal strip timer motor, that secure the motor to the central motor plate that is attached to the module bottom plate (see [Figure 67](#)).

5 Shift the motor enough to dislodge its timing belt. Remove the timing belt and unscrew the size T10 Torx timing belt clamp screw (see Figure 68).

6 Screw the clamp on the new timing belt, pass it through behind the strip timer carriage and pull it over the camshaft and the idler pulley.

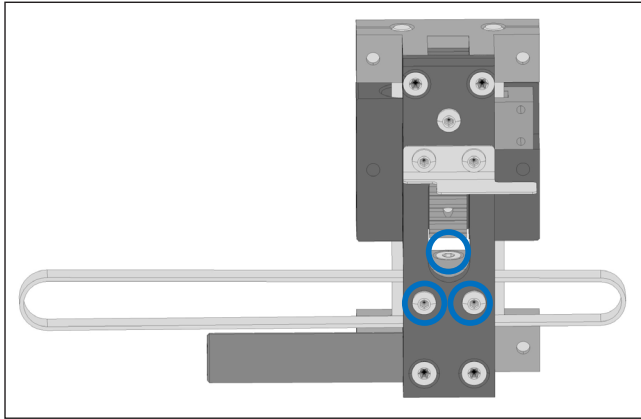


Figure 68: The Strip timer carriage seen from the back of the device with the timing belt clamp screw and the clamp securing screws indicated

7 Shift the motor to pull the timing belt taut and re-tighten the Horizontal strip timer motor screws.

⚠ The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

8 Complete steps 1–3 in reverse order and direction. Switch on the device and launch the Systest software.

9 Place a LabStrip U11 Plus GL test strip on the pipetting stage. Under **Card tests** on the **Feeder card** tab, tap the **Go to comb**, then

the **Go to feeder** button in the **Strip pusher motor** row. On the **Strip-Head moving card** tab, tap **To load position** in the **Comb lifter motor** row. Check that the strip timer comb engages the test strip reliably. If not, loosen the Horizontal strip timer motor home position sensor screws (see 8.6.8), and reposition the sensor, thereby adjusting the Horizontal strip timer motor's **To home position** motor position. Repeat this step and the adjustment as many times as necessary.

10 Complete steps 1–6 in [1.5.3 Adjusting the pad-reading position](#).

11 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.6.4 Replacing the Vertical strip timer motor

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Disconnect connector 14 from Circuit board number 5 (see [Figure 58](#)).

3 Unscrew the four (4) 2-millimeter hex socket button head screws that secure the motor to the plate at one end of the strip timer motor shafts. (The plate with the small drive axle and the large idler pulley coming through, on the left in [Figure 67](#).) Save the serrated washers and the flat washers.

4 Dislodge the timing belt from the motor axle. Remove the motor and replace with the new one. Complete the previous steps in reverse order and direction.

⚠ When replacing the Vertical strip timer motor, shift its position within its adjustable brackets so that the motor timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Complete steps 5–6 in [8.6.1](#) before resuming normal operation.



9.6.5 Replacing the Vertical strip timer motor timing belt [S-56T25064]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Disconnect connector 14 from Circuit board number 5 (see [Figure 62](#)). Loosen the four (4) 2-millimeter hex socket button head screws that secure the motor to the plate at one end of the strip timer motor shafts. (The plate with the small drive axle and the large idler pulley coming through, on the left in [Figure 67](#).)

3 Rotate the sensor flag on the idler pulley out of the way and shift the motor to dislodge the timing belt.

4 Remove the timing belt and replace it with a new one. Tighten the motor's securing screws only halfway so that you can shift its position. Move the strip timer comb manually and check that the sensor flag on the idler pulley enters the slit of the sensor only when the strip timer comb is in its lowermost position. Tighten the the securing screws completely when the sensor flag, the timing belt, and the strip timer

comb's position are coordinated.

⚠ When replacing the Vertical strip timer motor timing belt, shift the motor within its adjustable brackets so that the timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Complete steps 1–3 in reverse order and direction. Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.6 Replacing the Head mover motor

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Disconnect connector 14 from Circuit board number 5 (see [Figure 62](#)).

3 Unscrew the four (4) 2-millimeter hex socket button head screws that secure the motor to the plate at one end of the strip timer motor shafts. (The plate with the drive axle and two idlers coming through.) Save the serrated washers and the flat washers.

4 Unscrew the four (4) and six (6) 2.5-millimeter hex socket screws that secure the protective plates over the photometer bracket and the strip timer unit, respectively, and remove the plates. Dislodge the timing belt from the motor axle. Remove the motor and replace with the new one. Complete the previous steps in reverse order and direction.

⚠ When replacing the Head mover motor, shift its position

within its adjustable brackets so that the motor timing belt is taut enough. The timing belt is taut enough when you can turn the cogged surface towards yourself only half-way by hand. Make sure that the timing belt is not too taut.

5 Complete steps 5–6 in [8.6.1](#) before resuming normal operation.



9.6.7 Replacing the Head mover motor timing belt [S-56T25216]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the two (2) 2-millimeter hex socket button head screws that secure the timing belt to the Photometer assembly.

3 Loosen the four (4) 2-millimeter hex socket button head screws that secure the motor to the plate at one end of the strip timer motor shafts. (The plate with the drive axle and two idlers coming through.)

4 Shift the motor to dislodge the timing belt. Remove the timing belt and replace with the new one. Complete the previous steps in reverse order and direction.

5 See [1.5.3 Adjusting the pad-reading position](#) and in the Systest software under **Position setup**, check and adjust the **Head strip position** and the **Head reference position**.



9.6.8 Replacing the Horizontal strip timer home position sensor



Before removing the sensor, it is recommended that you mark its position inside its adjustable brackets with a marker pen or

a boxcutter. When replacing the sensor, position it to align with your markings.

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device. Push the strip timer comb towards the reading platform.

2 Unscrew the six (6) 2.5-millimeter hex socket screws that secure the protective plate over the strip timer unit, and remove the protective plate.

3 Disconnect connector 4 from Circuit board number 5 (see [Figure 62](#)) and unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the sensor plate above the Vertical strip timer motor.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ The sensor should be positioned in its adjustable brackets so that the right leading edge of its sensor flag runs along the middle of the sensor slot.

5 Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.9 Replacing the Vertical strip timer home position sensor



Before removing the sensor, it is highly recommended that you mark its position inside its adjustable brackets with a marker pen or a boxcutter. When replacing the sensor, position it to align with your markings.

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device. Bring the strip timer comb to its topmost position to get the sensor flag out of the way.

2 Unscrew the six (6) 2.5-millimeter hex socket screws that secure the protective plate over the strip timer unit, and remove the protective plate.

3 Disconnect connector 5 from Circuit board number 5 and unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the upright flap of the sensor plate above the Vertical strip timer motor.

4 Remove the sensor and replace it with a new one. Tighten the sensor's securing screws only halfway so that you can shift its position. Move the strip timer comb manually and check that the sensor flag on the idler pulley enters the slit of the sensor only when the strip timer comb is in its lowermost position. Tighten the the securing screws completely when the sensor is in its right place.

< 5 Complete steps 1–3 in reverse order and direction. Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.10 Replacing the Head mover home position sensor

i Before removing the sensor, it is recommended that you mark its position inside its adjustable brackets with a marker pen or a boxcutter. When replacing the sensor, position it to align with your markings.

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device. Move the photometer towards the front of the device to get it out of the way.

2 Unscrew the six (6) 2.5-millimeter hex socket screws that secure the protective plate over the strip timer unit, and remove the protective plate.

3 Disconnect connector 6 from Circuit board number 5 (see [Figure 62](#)). Reaching in through the opening in Circuit board number 5, unscrew the two (2) 2-millimeter hex socket button head screws that secure the sensor to the sensor plate above the Head mover motor.

4 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ The sensor should be positioned in its adjustable brackets so that its two securing screws are in line with the screw securing the sensor plate to the motor plate next to the sensor.

5 Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation. **>**



9.6.11 Replacing the Strip timer comb presence sensor

1 See [Removing the back casing panel](#) and remove the back casing panel.

2 Unscrew the two (2) 3-millimeter hex socket screws that secure the power supply to the device chassis at the back of the device. Save

the serrated washers.

3 Shift the power supply so that you gain access to the Strip timer comb presence sensor securing screws (see Figure 69).

4 Disconnect connector 7 from Circuit board number 5 (see [Figure 62](#)) and unscrew the two (2) 2.5-millimeter hex socket screws that secure the sensor to its bracket plate.

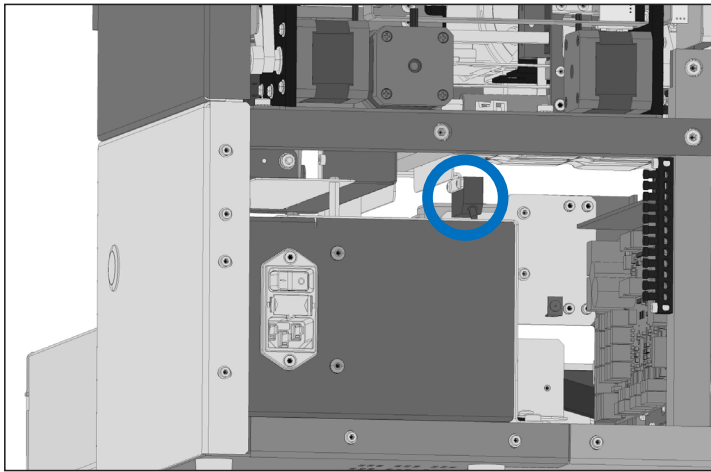


Figure 69: The Comb presence sensor behind the power supply as seen from the back of the device

5 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

⚠ When reconnecting the sensor cable, make sure that you pass it through the service opening in the Strip timer unit bottom plate on the way to Circuit board number 5.

6 Switch on the device and launch the Systest software. Enter **Sensor tests**, and check that the sensor is activated (its signal is red) when the Strip timer comb is above it. If not, carefully adjust the potentiometer

of the sensor with a small-size precision slotted screwdriver.

7 Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.12 Replacing the Strip bin microswitch [S-UAZ42321]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the two (2) 1.5-millimeter hex socket screws that secure the microswitch to the Head mover motor motor plate.

3 Remove the microswitch and replace it with a new one. Complete the previous steps in reverse order and direction.

4 Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.13 Replacing the Photometer assembly [S-UPA4408231]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the four (4) 2.5-millimeter hex socket screws that secure the protective plate over the photometer bracket and remove the plate.

3 Disconnect the flexible flat cable from connector 4 on the Photometer interface board by opening the slider clip and pulling the FFC free (see [Figure 62](#)). Unscrew the four (4) 2-millimeter hex socket screws

that secure the photometer assembly to its mount (see Figure 70) and carefully lift the photometer assembly off the mount.

4 Replace the photometer with a new one and complete the previous steps in reverse order and direction.

5 Complete the steps in [8.5.3](#) to adjust the photometer clearance and alignment.

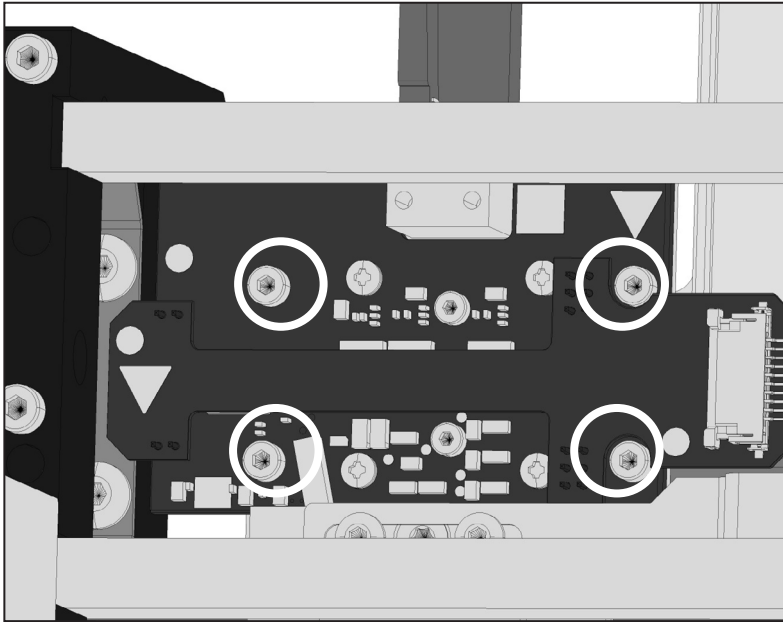


Figure 70: The Photometer assembly and its securing screws, seen from above

6 Complete the steps in [1.5.3 Adjusting the pad-reading position](#) to adjust the pad-reading position.

7 In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.



9.6.14 Replacing the photometry reference pad [S-UPA4408182]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the six (6) 2.5-millimeter hex socket screws that secure the protective plate over the strip timer unit, and remove the plate. Disconnect the flexible flat cable from connector 12 on Circuit board number 5 (see [Figure 62](#)).

3 Using a 6-millimeter (SAE size 7/32) wrench, unscrew the six (6) threaded spacers that secure Circuit board number 5 to the module bracket plate, and carefully slide the PCB to the side without removing it, to access the photometry reference pad.

4 Unscrew the two (2) size T10 Torx screws that secure the off-white photometry reference pad to the reading platform.

5 Remove the reference pad and replace it with a clean and unscratched new one. Complete the previous steps in reverse order and direction.

6 Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.15 Replacing the strip-retaining springs: [S-UPA0931396] right/[S-UPA0931386] left

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

Unscrew the four (4) 2.5-millimeter hex socket screws that secure the protective plate over the photometer bracket and remove the plate.

2 Unclasp the spring you are replacing from its slot in the reading platform, and pull its coiled center off the pin in the platform block.

3 Replace the spring with a new one and complete the previous steps in reverse order and direction.

4 See [1.5.3 Adjusting the pad-reading position](#) and check the adjustment of the pad-reading position to see whether the springs are symmetrical and similarly tight, and that they do not affect the pad-reading position.

 **It is recommended that you always replace both springs at the same time to avoid uneven strip positioning.**



9.6.16 Replacing the Photometer FFC [S-UPA07141]

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

1 Unscrew the four (4) 2.5-millimeter hex socket screws that secure the protective plate over the photometer bracket and remove the plate.

< 2 Disconnect the flexible flat cable from both the Photometer interface board and Circuit board number 5 by opening the slider clip and pulling the FFC free.

3 Remove the FFC and replace it with a new one. Complete the previous steps in reverse order and direction.

4 Switch the device on and launch the Systest software. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.



9.6.17 Replacing Circuit board number 5

1 Complete steps 1–3 in [8.6.1](#) and remove the module from the device.

2 Unscrew the six (6) 2.5-millimeter hex socket screws that secure the protective plate over the strip timer unit, and remove the plate. Disconnect all the connectors from Circuit board number 5 (see [Figure 62](#)).

3 Using a 6-millimeter (SAE size 7/32) wrench, unscrew the six (6) threaded spacers that secure Circuit board number 5 to the module bracket plate.

4 Remove the PCB and replace it with a new one. Complete the previous steps in reverse order and direction.

 **When reconnecting the cables, you will need to bundle the Bus ribbon cable and the power cables together with a cable tie and secure the bundle to the fastener clip before pulling it through the service opening in the module's bottom plate. Besides the Bus cable and the power connectors, the Strip timer comb presence sensor's cable also runs through the service opening.**

5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.

10 Sample dispenser module

 For a detailed overview of the module, see [The Sample dispenser module](#)

10.1 Syringe pump movements overview

Aspirating syringe movement	 Suction	 Discharge	Mixing syringe movement
 Bubble size	Air gap creation		
	Sample mixing: 1. Unmixed sample aspiration 2. Discharge to stir sample		 Aspirate for mixing  Discharge to mix
 Aspirate  Pre-tense	Sample aspiration		 Aspirate  Pre-tense
10X  Dispense  Retain sample	Sample dispensing		
	PMC measurement		 Back suction
 To home position	Rinsing		 Mixing home

10.2 Motors



10.2.1 Aspirating syringe motor [S-UPA4404053]

A stepper motor that moves the plunger inside the Aspirating syringe via a gear rack.

plugged in to connector 33 on Circuit board number 6

home position: 'To home position': the syringe plunger is in its topmost position.

Significant motor positions:

- Aspirate

The plunger of the Aspirating syringe is moved down to aspirate 100µL of sample from the test tube during normal operation.

motor step number–determined motor position, counted off from the **Bubble size** position

- Dispense

During normal operation, the plunger of the Aspirating syringe dispenses preset amounts of sample onto each pad, varying according to how much is needed for the particular reaction to take place.

motor step number–determined motor position, counted off from the **Pre-tense** motor position

 When you manually trigger this movement in the Systest software under Card tests on the Sampler card tab, the syringe will only form a small drop of fluid at the end of the pipetting probe.

- Bubble size

The Aspirating syringe draws up 10µL of air to form the air bubble that isolates the distilled water and the urine sample in the fluidic system.

motor step number–determined motor position, counted off from the

To home position.

- Pre-tense

The motor axle is moved 20 motor steps upwards to eliminate backlash in the gear rack. During normal operation, this movement is performed just before pipetting starts to ensure precise sample dispensing.

motor step number–determined motor position, counted off from the

Aspirate motor position

- Retain sample

The plunger is moved 30 motor steps downward to prevent any fluid from dripping from the tip of the pipetting probe. This movement is performed during normal operation after the last portion of sample has been dispensed and the pipetting probe is on its way above and inside the rinse well.



10.2.2 Mixing syringe motor [S-UPA4404053]

A stepper motor that moves the plunger inside the Mixing syringe via a gear rack.

plugged in to connector 29 on Circuit board number 6

home position: 'Mixing home': The syringe plunger is in its topmost position.

Significant motor positions:

- Aspirate for mixing

The Mixing syringe plunger is moved down to aspirate 600 µL of fluid. During normal operation this is the amount of sample initially drawn up, part of which is then discharged to stir up the sample.

motor step number–determined motor position, counted off from the

Mixing home position

- Discharge to mix

The Mixing syringe plunger is moved up to discharge 400 µL of fluid. During normal operation, this is the amount of sample used to stir up and homogenize the sample in the test tube.

motor step number–determined motor position, counted off from the

Aspirate for mixing position

- Aspirate

The Mixing syringe plunger is moved down to aspirate 350 µL of fluid. During normal operation, this amount of sample is used as a reserve for pipetting.

motor step number–determined motor position, counted off from the

Discharge to mix position

- Disinfection

The Mixing syringe plunger is moved down to aspirate 1.4 milliliters of fluid. During normal operation, this is a single measure of disinfectant solution (see [5.5.4 Disinfection test](#)) used in the daily disinfection sequence.

- Pre-tense

The motor axle is moved 10 motor steps upwards to eliminate backlash in the gear rack. This movement is performed just before the PMC

measurement to ensure that the Drive sample to PMC motor movement makes the required amount of sample reach the PMC.

motor step number–determined motor position, counted off from the **Aspirate for mixing** position

- Drive sample to PMC

The Mixing syringe plunger is moved down 300 motor steps to aspirate 500µL of air to force the aspirated sample from the pipetting probe to the PMC.



10.2.3 Aspiration valve [S-ALT4590161]

The electromagnetic valve attached to the manifold that seals the Water line.

plugged in to connector 27 on Circuit board number 6

The valve is open when it is turned on.

10.3 Sensors



10.3.1 Aspirating syringe home position sensor [S-ALT4560251]

type: transmissive photointerrupter

plugged in to connector 6 on Circuit board number 6

Systest sensor signal		
Asp. pump home	The Aspirating syringe plunger is in home (top-most) position.	Sensor not activated

The device is properly set up and operational only if the Aspirating pump home sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu](#))



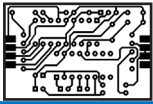
10.3.2 Mixing syringe home position sensor [S-ALT4560251]

type: transmissive photointerrupter

Systest sensor signal		
Mixing home	The Mixing syringe plunger is in home (topmost) position.	Sensor not activated

The device is properly set up and operational only if the Aspirating pump home sensor is active (its signal is red) after system initialization. (See [2.3.2 The Sensor tests menu](#))

10.4 PCBs



10.4.1 Sample dispenser module circuit board [S-UPA43051]

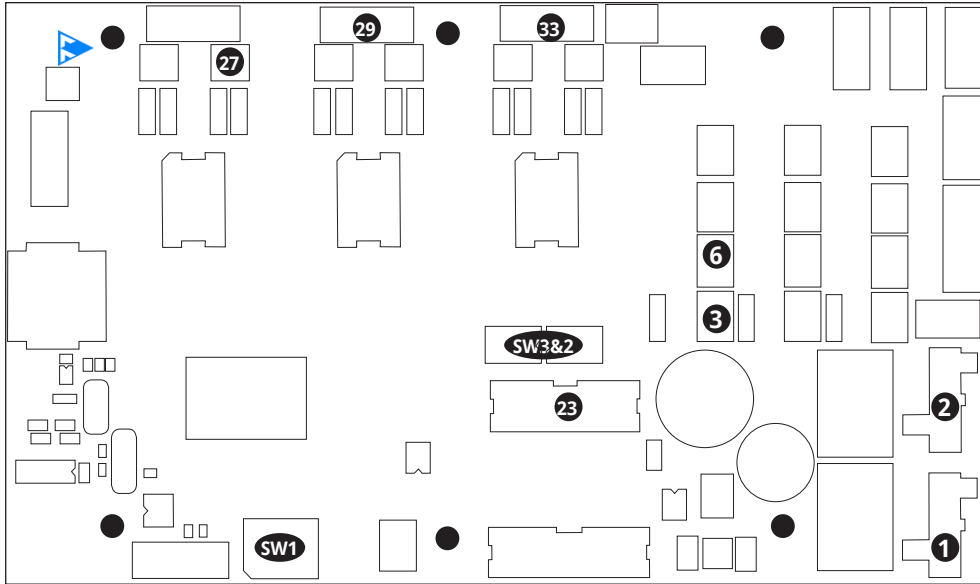


Figure 71: Board layout of Circuit board number 6.

Key to Figure 71

- 27. Aspiration valve motor connector
- 29. Mixing syringe motor connector
- 33. Aspiration syringe motor connector
- 6. Aspirating syringe home position sensor connector
- 3. Mixing syringe home position sensor connector
- SW3&2. Two 8-way configuration DIP switches (only sliders 4,5, & 8 on SW2 are on—in the down position)
- 23. Bus cable connector from the Strip timer and photometer PCB

2. Power supply connector going to the PC module

SW1. PCB address DIP switch (sliders 2 and 3 are on—in the down position) (For the layout and numbering of the sliders [Figure 40](#))

1. Power supply connector coming from the Strip timer and photometer unit

10.5 Sample dispenser module Flow test processes

There are no Flow test processes specific to the Sample dispenser module. The Washer and the Sample dispenser modules together constitute the fluidic system of the device. For details of the Flow test processes involving the fluidic system as a whole, see [Figure 37](#).

10.6 Service actions—Adjustments



10.6.1 Preset system parameters

⚠ The following parameters are included in this manual for information purposes only. Do not modify any of these values without explicit prior permission from 77Elektronika service personnel.

i You can access these parameters in the Systest software on the System parameter setup menu on the Sampler card tab by tapping the Read parameters button.

Key to the system settings in the Systest software on the **System parameter setup** menu on the **Sampler card** tab:

- Mixing delay

The time, in milliseconds, that the system idles after the mixing process

has finished, before starting aspiration.

- Aspiration delay

The time, in milliseconds, that the system idles after sample aspiration has finished, before moving the pipetting probe out of the test tube.



10.6.2 Performing a fluid retention test

- 1 Perform all the checks listed in [5.6.1](#).
- 2 Place a rack with a test tube filled with tap water on the rack conveyor to the right of the rack delivery zone. Launch the Systest software and on the **Flow tests** menu, tap the **Rack change test** button to align the test tube on the rack with the test tube aperture.
- 3 On the Card tests menu on the **Robot card** tab in the **Linear robot vertical motor** row, tap **High probe position**, then in the **Linear robot horizontal motor** row tap **Test tube position**. When the pipetting probe is above the test tube, tap **Aspirating position** in the **Linear robot vertical motor** row.
- 4 On the **Card tests** menu on the **Sampler card** tab in the **Mixing syringe motor** row, tap **Aspirate** to draw the water inside the pipetting probe. On the **Card tests** menu on the **Robot card** tab in the **Linear robot vertical motor** row, tap **High probe position**. Wait half a minute and check that the aspirated water does not form a bubble at the end of the pipetting probe.
- 5 On the **Card tests** menu on the **Robot card** tab in the **Linear robot horizontal motor** row, tap **Rinse well position**, then

Test tube position. Repeat a few times. Check that no water drips off the end of the pipetting probe during the horizontal movements.

6 If the device failed the previous checks, some air is leaking into the fluidic system. Check and re-tighten all the tube fittings (see [Figure 37](#)).

10.7 Service actions—Replacements

⚠ Except for sensor replacement, every procedure described in this section involves breaching the fluidic system. Because of this, you should drain all liquids from the system before performing these replacements by running a Shutdown test in the Flow tests menu of the Systest software (see [5.5.5 Shutdown test](#)). For the same reason, once you have completed the replacements, perform the checks detailed in [5.6.1](#).



10.7.1 Replacing the Sample dispenser unit [S-UPA4425005]

- 1 Switch off the device. Open the doors and the top cover. Disconnect all connectors from Circuit board number 6 (see [Figure 71](#)).
- 2 On the manifold side, unscrew and disconnect the fluid tube that runs into (labeled IN2), and the tube that runs out of the manifold (labeled IN3).
- 3 Unscrew the two (2) 3-millimeter hex socket screws that secure the module bottom plate to the crossbeam of the device chassis. Save the serrated washers.
- 4 Remove the module and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Switch the device on and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.2 Replacing the Aspirating syringe plunger [S-55201120]

- 1** Switch off the device and open the doors and the top cover.
- 2** Unscrew the aspirating syringe from the manifold and dislodge the base of the plunger from the gear rack bracket.

⚠ Take care when handling the threaded cap of the pipetting syringe not to break or crack the glass barrel.

- 3** Remove the complete syringe. Make sure that the O-ring between the manifold and the syringe stays in place.

i An O-ring is also supplied when you order a new plunger in case it is lost during replacement. However, unless the O-ring is damaged, you will not have to replace it when replacing the plunger.

- 4** Remove the plunger from the barrel by pulling it out and replace it with a new one. Push the plunger inside the barrel as far as it can go then move it up and down in the barrel a few times to establish proper traction.

- 5** Replace the complete syringe, making sure that the base of the plunger is sitting snugly in the slot of the gear rack bracket. Screw the aspirating syringe back onto the manifold.

⚠ When replacing the syringe, only tighten the threaded cap to the manifold by hand to avoid damaging the glass barrel.

- 6** Switch on the device and launch the Systest software. On the **Card tests** menu on the **Sampler card** tab, tap the **Aspirate** and the **To home position** buttons in the **Aspiration pump** row a few times to settle the plunger firmly inside the barrel. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.3 Replacing the Mixing syringe plunger [S-55201270]

- 1** Switch off the device and open the doors and the top cover.
- 2** Unscrew the mixing syringe from the manifold and dislodge the base of the plunger from the gear rack bracket.

⚠ Take care when handling the threaded cap of the pipetting syringe not to break or crack the glass barrel.

- 3** Remove the complete syringe. Make sure that the O-ring between the manifold and the syringe stays in place.

i An O-ring is also supplied when you order a new plunger in case it is lost during replacement. However, unless the O-ring is damaged, you will not have to replace it when replacing the plunger.

- 4** Remove the plunger from the barrel by pulling it out and replace it with a new one. Push the plunger inside the barrel as far as it

can go then move it up and down in the barrel a few times to establish proper traction.

5 Replace the complete syringe, making sure that the base of the plunger is sitting snugly in the slot of the gear rack bracket. Screw the mixing syringe back onto the manifold.

⚠ When replacing the syringe, only tighten the threaded cap to the manifold by hand to avoid damaging the glass barrel.

6 Switch on the device and launch the Systest software. On the **Card tests** menu on the **Sampler card** tab, tap the **Aspire for mixing** and the **To home position** buttons in the **Mixing pump** row a few times to settle the plunger firmly inside the barrel. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.4 Replacing the Aspiration valve [S-ALT4590161]

1 Switch off the device and open the doors and the top cover. Disconnect connector 27 from Circuit board number 6 (see [Figure 71](#)).

2 Unscrew the two (2) 2.5-millimeter hex socket screw that secure the Aspiration valve to the manifold. Save the O-ring.

3 Remove the valve and replace it with a new one. Complete the previous steps in reverse order and direction.

4 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.5 Replacing the Aspirating syringe home position sensor

1 Switch off the device and open the doors and the top cover. Disconnect connector 6 from Circuit board number 6 (see [Figure 71](#)).

2 Move the sensor flag of the Aspirating syringe out of the sensor slit. Unscrew the two (2) 2.5-millimeter hex socket screws that secure the sensor to the module bottom plate.

3 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

4 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.6 Replacing the Mixing syringe home position sensor

1 Switch off the device and open the doors and the top cover. Disconnect connector 3 from Circuit board number 6 (see [Figure 71](#)).

2 Move the sensor flag of the Mixing syringe out of the sensor slit. Unscrew the two (2) 2.5-millimeter hex socket screws that secure the sensor to the module bottom plate.

3 Remove the sensor and replace it with a new one. Complete the previous steps in reverse order and direction.

4 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



10.7.7 Replacing Circuit board number 6

- 1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2 Disconnect every connector from Circuit board number 6.
- 3 Unscrew the four (4) 2-millimeter hex socket button head screw that secure the circuit board to the embedded PC bottom casing panel. Save the serrated washers and the metal spacers.
- 4 Remove the PCB and replace it with a new one. Complete the previous steps in reverse order and direction.
- 5 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.



11 The thermal printer

i For a detailed overview of the module, see [4.1.7 The Printer module](#).

11.1 Disabling the thermal printer

The printer is an optional module of the LabUMat 2 device, and you can choose to order your analyzer without it. If you have the thermal printer included in your device but wish to run LabUMat 2 without it, you can disable it to prevent error messages and failed system initializations.

1 Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.

i On some older devices, the administrator password is 'masteryoda'.

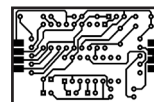
< 2 On the Windows desktop, tap **Start**, and then **My Computer**. Double-tap volume E:, and find the **LabUMat 2** folder.

3 Double-tap **LabUMat 2** to open it and double-tap the **comDriver.ini** file.

4 In the Notepad file that pops up, find the line **PRINTER**, and change it to 0 (zero).

5 Close the program, log off from and restart the device. Launch the Systest software, and on the **Flow tests** menu, tap **Initialization** to save your changes.

11.2 PCBs



11.2.1 The printer module control circuit board [S-UPA4407043]

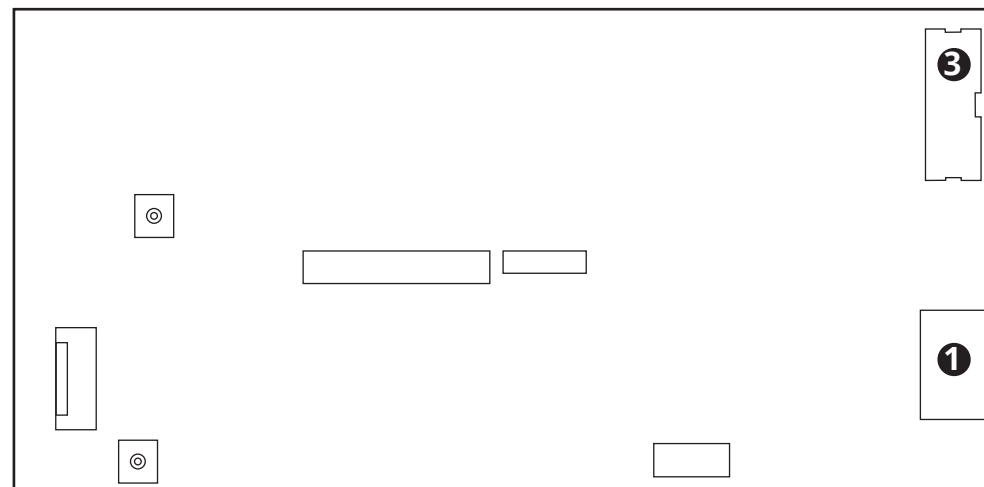


Figure 72: Board layout of Circuit board number 7

Key to Figure 72

- 1. 16-pin bus ribbon cable connector
- 3. Power connector from the embedded PC

11.3 Thermal printer Flow tests processes



11.3.1 Initialization

The printer performs the movements necessary to move the printer paper, if there is any loaded, by a few lines in both directions, .



11.3.2 Line feed test

The printer pushes out an amount of printer paper determined by the **Line space** parameter, if there is any printer paper loaded. See 10.4.1.



11.3.3 Printer test

The printer prints out a randomized full test result.



11.3.4 Print character table

The printer prints out a complete character set—every graphical character it is capable of printing.



11.3.5 Stop printing

Tapping this button stops ongoing printing.

11.4 Service actions—Adjustments



11.4.1 Adjusting the preset printer parameters

- Line space

This parameter indicates the number of lines the printer leaves blank between the individual lines of a result printout. The default value is 6.

- Result separator

This parameter indicates the number of lines the printer leaves blank between two sets of results on the printout. The default value is 70.

11.5 Service actions—Replacements



11.5.1 Replacing the thermal printer

- 1 Remove the paper from the printer if there is any loaded. Switch off the device and complete the steps in [4.2.5 Removing the top casing panel](#) to remove the lid. Disconnect the connectors on Circuit board number 7. Unscrew the four (4) 2-millimeter hex socket button head screws that secure Circuit board number 7 to the reverse side of the device lid.
- 2 Unscrew the three (3) 1.5-millimeter hex socket button head screws that secure the printer to Circuit board number 7 and remove the thermal printer block.
- 3 From the lever on the side of the thermal printer block, remove the pin that the printer cover pushbutton depresses when opening the printer cover. Push the pin into the lever of the new thermal printer unit.
- 4 Replace the thermal printer block. Make sure that the printer cover pushbutton engages with the pin pressed into the lever. Complete the previous steps in reverse order and direction.
- 5 Load a roll of printer paper into the printer. Switch on the device. Launch the Systest software, and on the Flow tests menu, tap the Line feed test and the Printer test buttons to check that the new printer is functioning properly.



11.5.2 Replacing Circuit board number 7

- 1 Complete steps 1–2 in [10.5.1](#).
- 2 Unscrew the two (2)-millimeter hex socket button head screws that secure the printer cover pushbutton and its mount to Circuit board number 7.
- 3 Remove the circuit board and replace it with a new one. Complete the previous steps in reverse order and direction.
- 4 Complete the steps in [2.3.5 The Card setup menu](#) to set up the newly installed circuit board.

12 The Physical Measurement Cell module

 For a detailed overview of the module, see [4.1.1 The Feeder and strip pusher module](#).



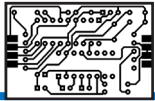
12.1 Aspiration, mixing, and PMC measurement cycle

The interactive slideshow on the following page describes the flow of air, distilled water, and urine inside the section of tubing from the tip of the pipetting probe up to the manifold (the probe, the tubes labeled 'IN4' and 'IN3', plus the PMC unit) during a single sample aspiration and mixing—dispensing—PMC measurement cycle. Use the navigation button to the right of the images to load the next image, and the navigation button to the left of the images to restart the sequence.



12.2 PCBs

⚠ The diagrams in this section are included for information purposes only. The single connector you are allowed to manipulate is the one marked number 1 in Figure 70.



12.2.1 PMC control circuit board

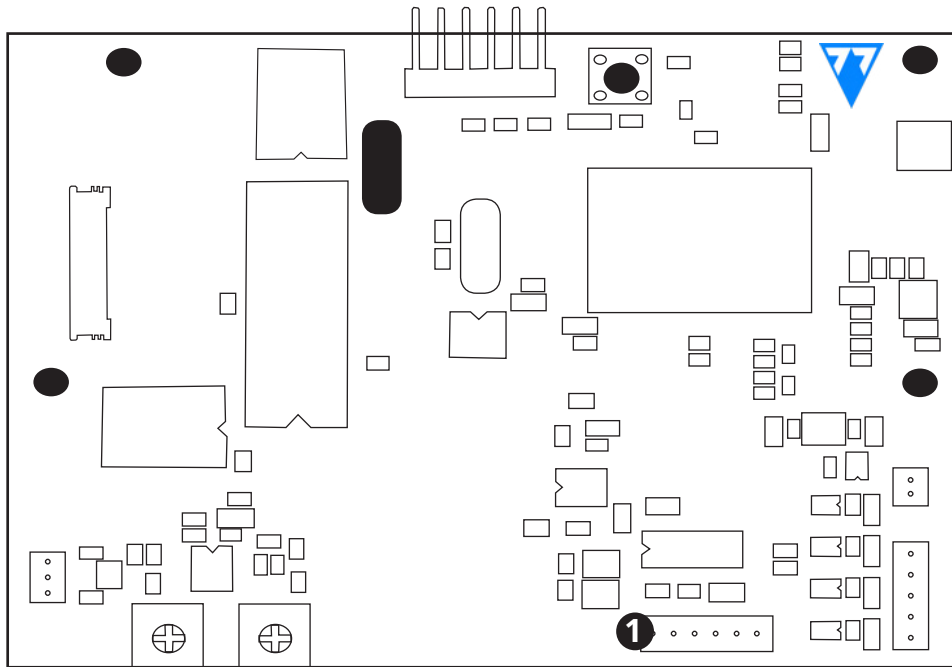
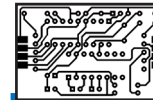


Figure 73: Board layout of Circuit board number 8

Key to Figure 73

1. Ribbon cable connector to Circuit board number 3 (the linear robot module circuit board)



12.2.2 PMC LED circuit board

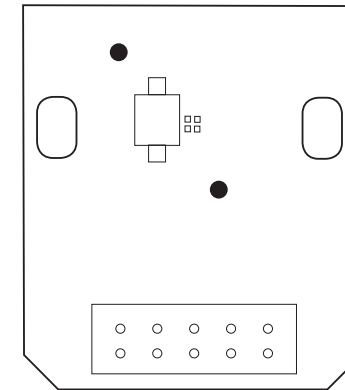
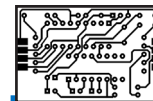


Figure 74: Board layout of the PMC LED circuit board



12.2.3 PMC CCD circuit board

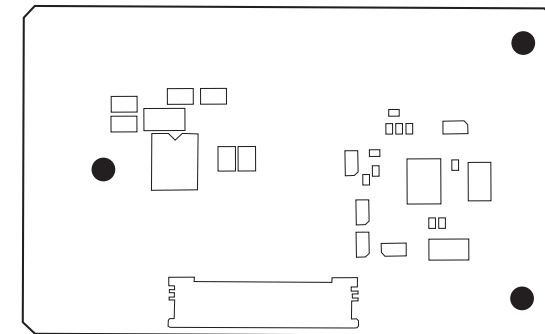


Figure 75: Board layout of the PMC CCD circuit board

12.3 The PMC module Flow tests processes



12.3.1 Initialization

The calibration parameters that control the PMC measurement processes are downloaded to Circuit board number 8 from the embedded PC.



12.3.2 PMC user calibration

⚠ Make sure that both the distilled water and the waste tanks are connected to the intact fluidic system before you perform a PMC user calibration.

The device pumps distilled water from the distilled water tank into the PMC unit, and measures its specific gravity. The system then compares the result to the PMC factory calibration and the last user calibration result (if any). If the difference between the current measurement peak and the stored default is within the defined limits, the currently measured result is stored as the last user calibration result. If the calibration fails, the last successful calibration data will not be overwritten.

ⓘ If the calibration fails, a stylized PMC symbol will be displayed next to all subsequent database records as long as a successful PMC user calibration is performed.



ⓘ The user calibration only affects the specific gravity measurement, and neither the turbidity, nor the color measurements.



12.3.3 PMC test

The processes that are necessary for all three of the PMC measurements

are performed in turn inside the module whether there is any liquid inside the measurement cell or not.



No motor movement that drives the fluidic system is performed during a PMC test.

12.4 Service actions—Replacements



The PMC module can only be ordered and is only supplied as a whole unit. Do not remove or tamper with any of its individual parts.



12.4.1 Replacing the Physical Measurement Cell module [S-UW64417001]

1 Switch off the device and open the front doors. To remove all liquids from the fluidic system, run a **Shutdown test** on the **Flow tests** menu in the Systest software, and disconnect the liquid tanks and their sensors from the device.

2 Disconnect the tubes labeled IN3 and IN4 from the PMC module (the tubes between the manifold and the PMC and the PMC and the pipetting probe, respectively) with their fittings.

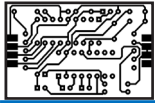
3 Unscrew the four (4) size T10 Torx screws that secure the PMC module to the PMC bracket plate.

4 Remove the module and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Complete [5.6.1](#) to check the integrity of the fluidic system. In the Systest software on the **Flow tests** menu, tap **Initialization** before resuming normal operation.

13 The PC and LED touchscreen module

13.1 PCBs



13.1.1 The embedded PC motherboard
[S-9A945SE2]

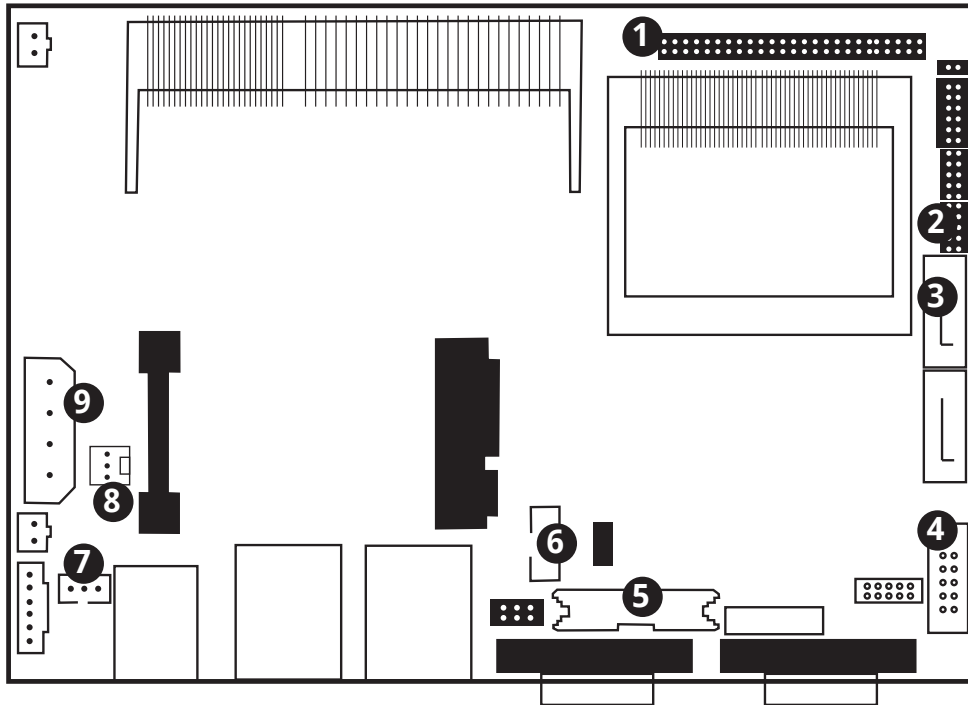
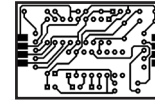


Figure 76: Board layout of the embedded PC motherboard

Key to Figure 76

1. Communication port connector
2. Touchscreen USB connector
3. HDD SATA connector
4. Audio connector

5. Display LVDS connector
6. Motherboard inverter connector
7. Standby signal controller
8. Connector for the CPU fan
9. Motherboard power connector



13.1.2 The embedded PC interface board
[S-UPA42096]

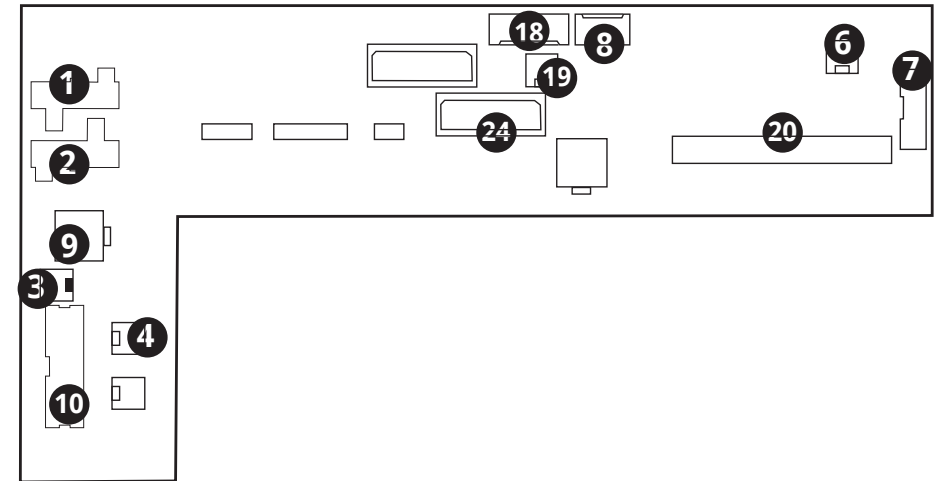


Figure 77: Board layout of the PC module interface circuit board

Key to Figure 77

1. Power connector to the Printer module
2. Power connector from the Sample dispenser module
3. 12V power connector from the power supply
4. Board-to-board connector to the motherboard standby signal controller connector

6. Speaker connector
7. Board-to-board connector to the motherboard audio connector
8. Fan connector
9. Connector for the power-on signal to the power supply
10. 16-pin Bus ribbon cable connector
18. Display backlight control connector
19. Board-to-board connector to the motherboard inverter
20. Communication port
24. Board-to-board connector to the motherboard power connector

13.2 Service actions—Adjustments

 To enter text in on-screen boxes, use a connected keyboard or tap inside the text box to access the virtual on-screen keyboard.



13.2.1 Performing a Disk Cleanup

The Disk Cleanup utility helps you free up disk space. It searches your drive for unnecessary files that you can safely delete.

1 Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.

 On some older devices, the administrator password is 'masteryoda'.

2 On the Windows desktop, tap **Start**, then tap **Run**. In the **Open** box, enter 'cleanmgr' (without the inverted commas), and tap **OK**.

3 Restart the device for the changes to take effect.



13.2.2 Performing disk error checking

Chkdsk (Chkdsk.exe) is a command-line utility that checks volumes on your hard disk drive for problems. The tool then tries to repair any problems that it finds. You must be logged in as administrator to use the tool.

1 Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.



On some older devices, the administrator password is 'masteryoda'.

2 On the Windows desktop, tap **Start**, then tap **Run**. In the **Open** box, enter 'cmd' (without the inverted commas), and tap **OK**.

3 Enter the 'chkdsk volume:/r' command (without the inverted commas), and tap **OK**.



If one or more of the files on the hard disk are open, you will receive the following message:
Chkdsk cannot run because the volume is in use by another process. Would you like to schedule this volume to be checked the next time the system restarts? (Y/N)

Type Y, and then press **ENTER** to schedule the disk check, and then switch off the device and complete the previous steps to start

the disk check.

4 Restart the device for the changes to take effect.



13.2.3 Performing Defragmentation

Disk Defragmenter is a system administrator utility that can increase system access speed by rearranging files stored on the disk to occupy contiguous storage locations.

1 Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.

 On some older devices, the administrator password is 'masteryoda'.

2 On the Windows desktop, tap **Start**, then tap **Run**. In the **Open** box, enter 'cmd' (without the inverted commas), and tap **OK**.

3 Enter 'defrag C: -f' without the inverted commas to defragment your C: drive.

 Depending on how fragmented the drive is, defragmentation may take up to several hours to complete.

4 Enter 'defrag E: -f' without the inverted commas to defragment your E: drive.

5 Restart the device for the changes to take effect.



13.2.4 Reinstalling the device software

 It is highly recommended that you make a backup copy of the complete 'LabUMat' folder on a removable drive before completing the following steps.

1 Download the .msi-extension LabUMat 2 software installer file from 77Elektronika's dedicated FTP server, or contact 77Elektronika service personnel to receive the installer file via e-mail. Save the installer file on the embedded PC.

2 Uninstall the device software you are currently running: Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator. On the desktop, tap **Start**, tap **Control Panel**, and then double-tap **Add or Remove Programs**. Find and tap **LabUMat 2** among the currently installed programs, and then tap **Remove**. If the operating system prompts you to confirm the removal of the program, tap **Yes**.

3 Find and double-tap the .msi-extension software installer file you saved in Step 1. Follow the instructions that appear on the screen.

 Make sure that you install the device software on volume E: in a folder named LabUMat 2. The installer will offer this location by default.

13.3 Service actions—Replacements



13.3.1 Replacing the PC and touchscreen module [S-UPA4411005]

- 1 Open the lid of the device. See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2 See [Figure 71](#) and remove Circuit board number 6.
- 3 Disconnect connectors 1, 2, 3, 9, and 10 from the PC interface board (see [Figure 77](#)).
- 4 Loosen the four (4) 2-millimeter hex socket button head screws in the device chassis that the PC and touchscreen module is suspended from.
- 5 Carefully lift the module out of the chassis and replace it with a new one. Complete the previous steps in reverse order and direction.
- 6 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



13.3.2 Replacing the PC motherboard [S-9A945SE2]

- 1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel. Disconnect all the connectors from Circuit board number 6 on the bottom side of the PC and touchscreen module.

- 2 Disconnect connectors 1, 2, 3, 9, and 10 from the PC interface board (see [Figure 77](#)).
- 3 Unscrew the four (4) 2-millimeter hex socket button head screws that secure the module's top casing panel, and remove the casing panel. Save the serrated washers.
- 4 Disconnect all of the cables and connectors from the PC motherboard (see [Figure 76](#)).
- 5 Unscrew the four (4) 2-millimeter hex socket button head screws that secure the motherboard to the module bottom plate.
- 6 Remove the motherboard and replace it with a new one. Complete the previous steps in reverse order and direction.
- 7 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



13.3.3 Replacing the embedded PC interface board

- 1 Complete steps 1–3 in 12.3.2.
- 2 Disconnect all the cables and connectors from the PC interface board. (see [Figure 77](#)).
- 3 Unscrew the ten (10) 2-millimeter hex socket button head screws that secure the interface board to the module bottom plate.
- 4 Remove the interface board and replace it with a new one. Complete the previous steps in reverse order and direction.

5 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.



13.3.4 Replacing the Hard Disk Drive [S-3EHDD004]

1 Complete steps 1–3 in [12.3.1](#).

2 From the reverse side of the module bottom casing panel, unscrew the four (4) 2-millimeter hex socket button head screws that secure the hard disk and its rubber feet to the module bottom casing panel.

3 Remove the HDD. Remove its rubber feet and attach them to the new HDD. Insert the new HDD and complete the previous steps in reverse order and direction.

4 Contact 77Elektronika's service staff at the **service@e77.hu** e-mail address for information about how you can reinstall the operating system and the LabUMat 2 software on the disk.



13.3.5 Replacing the LCD touchscreen [S-UPA4411015]

1 Complete steps 1–4 in [12.3.1](#), and remove the module from the device.

2 Unscrew the four (4) 2-millimeter hex socket button head screws that secure the module's top casing panel, and remove the casing panel. Save the serrated washers.

3 Disconnect connectors 2 and 5 from the PC motherboard, and connector 18 from the PC interface board.

4 Unscrew the four (4) touchscreen securing screws: two (2) 2-millimeter hex socket screws accessible through the back side of the touchscreen casing, and two (2) 2-millimeter hex socket screws at the back of the touchscreen unit inside the module housing (see Figure 78).

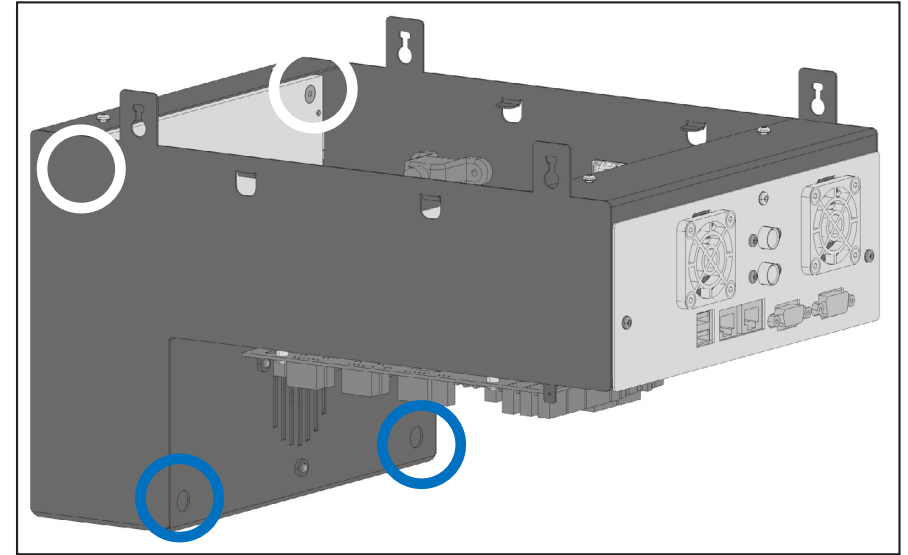


Figure 78: The PC and touchscreen module seen from the back, with the securing screws of the touchscreen indicated

5 Carefully reach inside the module housing and, steadying the display from the outside, snap it out of the module.

6 Replace the touchscreen display with a new one, and complete the previous steps in reverse order and direction.

7 Switch on the device. In the user software interface, tap the **Log off system** button. In the login window that appears, enter 'administrator' as user name and 'A1b2C3' as password (both without the inverted commas) to log in as administrator.

 On some older devices, the administrator password is 'master-yoda', without the inverted commas.

8 On the Windows desktop, tap **Start**, then **My computer**. Double-tap **Local Disk (C:)**.

9 Navigate to **C:/Windows/system32/xAuto4PtsCal.exe** and double-tap the executable file to launch the touchscreen calibration program.

10 Tap the calibration symbols that appear in the four corners of the LCD screen as they are displayed. Try to tap as close to the center of the symbols as possible.

11 See [4.2.1 Removing the back casing panel](#) and launch the Systest software. On the **Flow tests** menu, tap **Initialization** to store the changes.



14 The Power supply

 For an overview of the module, see [4.1.10 The Power supply](#).

14.1 Replacing the power supply [S-UPA4410005]

 **The power supply is not serviceable. Do not try to disassemble it.**

- 1 See [4.2.1 Removing the back casing panel](#) and remove the back casing panel.
- 2 Disconnect all the cables from the power supply module. Unscrew the two (2) 3-millimeter hex socket screws that secure the power supply to the device chassis.
- 3 Unscrew at least one of the two 2-millimeter hex socket button head screws that hold the protective plate over the top grille in the power supply housing plate. Save the metal spacer bar. Move the protective plate out of the way so that you can remove the power supply block without having to disconnect the various cables that run under the protective plate.
- 4 Carefully remove the power supply and replace it with a new one.
- 5 Complete the previous steps in reverse order and direction.
- 5 Switch on the device and launch the Systest software. On the **Flow tests** menu, tap **Initialization** before resuming normal operation.

14.2 Replacing the power supply fuses [S-36T8000M]

- 1 In the Systest software on the **Flow tests** menu, tap the **Shutdown test** button, then switch off the device and disconnect the AC power cable.
- 2 Flip open the cover of the fuse holder (marked number 2 in Figure 79), and remove the fuses.
- 3 Replace the fuses with the new ones.
- 3 Complete the previous steps in reverse order and direction. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.

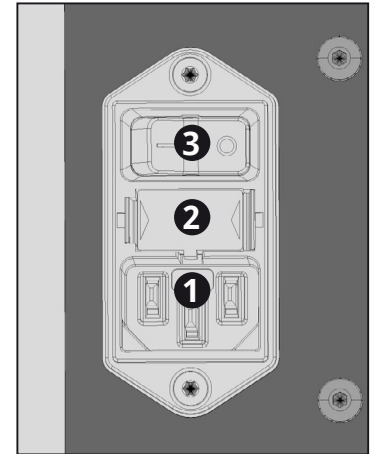


Figure 79: The AC power, fusebox, and power switch unit

14.3 Replacing the standby switch [S-UW642171]

- 1 In the Systest software on the **Flow tests** menu, tap the **Shutdown test** button, then switch off the device and disconnect the AC power cable.
- 2 See [4.2.4 Removing the bottom side casing panels](#), and remove the bottom side panel on the side of the device where the standby switch is located.
- 3 Disconnect the switch and the three (3) cables that connect it to the power supply.

4 Replace the standby switch and complete the previous steps in reverse order and direction. Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.

14.4 Replacing the power switch [S-33D1DD12]

1 In the Systest software on the **Flow tests** menu, tap the **Shut-down test** button, then switch off the device and disconnect the AC power cable.

2 Flip open the cover of the fuse holder (marked number 2 in [Figure 79](#)), and remove the fuses.

3 Unscrew the two (2) size T10 Torx screws that secure the power switch to the power supply chassis. Carefully pull the power switch out of the power supply as far as the cables connecting the power switch to the power supply allow.

4 Disconnect the cables from the power switch. Replace the power switch with the new one, and complete the previous steps in reverse order and direction. Replace the original fuses you removed in step 2.

5 Enter the **Flow tests** menu and tap **Initialization** before resuming normal operation.

Appendix 1 Maintenance kit

S-UPAPMKIT	Quantity
Photometry reference pad	1
Strip-retaining spring, left	1
Strip-retaining spring, right	1
Plastic tubing labeled GRA1	1
Plastic tubing labeled DIR1	1
Plastic tubing labeled DIR2	1
Plastic tubing labeled DIR3	1
Aspirating syringe plunger	1
Mixing syringe plunger	1
Peristaltic pump flexible silicone tubing	2
Water filter	1

Appendix 2 Recommended spare parts

 Sensors	
Flipped strip sensor	Strip progress sensor
Tube presence sensor	Rack position sensor
Unassigned transmissive photointerrupter without cables	Rack catch sensors
 PCBs	
Circuit board number 2	Oscillator interface board
Strip pusher motor interface board	Circuit board number 1
Circuit board number 4	Circuit board number 6
Level detector interface board	Circuit board number 5
Circuit board number 3	Circuit board number 7
 Cables	
Internal data bus cabling	Flexible Flat Cable (Linear robot module)
Flexible Flat Cable (Strip timer and Photometer module)	
 Actuators	
Waste pump	Aspiration valve
Waste line sealing valve	
 Miscellaneous	
Strip-retaining spring (left and right)	Rack positioning motor timing belt
Physical Measurement Cell	Strip timer comb
Photometer assembly	Hard Disk Drive
Pipetting probe	Maintenance kit (at least 3)

Appendix 3 Technical specifications

⚠ All connected devices should conform to IEC 60950 on Information technology equipment and its extensions relevant to the type of device.

General	
Evaluated parameters	Bilirubin, Urobilinogen, Ketones, Ascorbic acid, Glucose, Protein (Albumin), Blood, pH, Nitrite, Leucocytes (via LabStrip U11 Plus GL test strips) Specific gravity, Color, Turbidity (via the Physical Measurement Cell)
Technology used	Reflectance photometry (wavelengths: 505, 530, 620, 660 nm)
Throughput	240 tests per hour
Batch size	100 test tubes
Memory capacity	10 000 results
Physical dimensions	
Size	600 x 650 x 635 millimeters (Width, Depth, Height)
Weight	55 kilograms (121 lbs)
Interfaces	USB, RS 232 serial ports, PS/2, VGA
Display	800x600 TFT color touch screen
Printer	built-in thermal printer
Power	
Main unit	100–250 VAC 50–60 Hz/maximum 200 W
Fuse	2 x T 8A L
Circumstances of operation	
Operational temperature	15–30°C (59–86°F)
Operational humidity	20–80 % relative humidity
Circumstances of storage	
Storing temperature	-20–80°C (-4–176°F)
Storing humidity	20–80 % relative humidity

Barcode scanner	
Barcode types recognized	CODE 39, CODE 128, EAN-13, EAN-8, Interleaved 2 of 5, CODABAR
Minimum height of recognized barcodes	20 millimeters
Test tube rack	10-tube rack (Only the rack supplied by the manufacturer may be used!)
Test tube specifications	
Minimum sample volume in tube	2mL (checked by a liquid level sensor)
Urine homogenization	Stirring by mixing up the sample
Height (conical-bottom tubes)	70–110 millimeters
Height (straight-bottom tubes)	70–105 millimeters
Diameter at the top of the tube	16–17.5 millimeters 14–16 millimeters when using rack adapter clips
Maximum diameter at the top of the rack (56 millimeters above the bottom of the tubes)	16.5 millimeters
Test strips	
Type	LabStrip U11 Plus GL
Analyte pads	Bilirubin, Urobilinogen, Ketones, Ascorbic acid, Glucose, Protein (Albumin), Blood, pH, Nitrite, Leucocytes
Package	150 pieces per vial
Maximal strip load	150 pieces (a single vial)
Rinsing system	
Rinsing liquid in tank	Distilled or ion-exchanged water
Tank volume	5 or 10 liters
Rinsing liquid consumption	5 liters of distilled water is enough for at least 300 measurements
Disinfectant	At least 6 milliliters of 2:100 Sodium hypochlorite
Waste tank size	holds the waste products of up to 200 measurements

Appendix 4 Disposal information

⚠ You must NOT dispose of your used LabUMat 2 device or any of its parts as municipal solid waste.

⚠ Under no circumstances will any product be approved or accepted for return without carrying a completed Disinfection Certificate. Furthermore, there will be no reimbursement for such spare parts.

Preparation prior to shipment

1 If the return product did not come into contact with any body fluid, check in the **The products listed below have not come into contact with biohazardous body fluids.** checkbox in the Disinfection Certificate, and skip to step 3 below.

2 If the return product is potentially contaminated, complete the steps in [Disinfecting products \(instruments and parts\)](#) below. In the Disinfection Certificate, check the **I completed the instructed disinfection steps for the below mentioned products return.** checkbox.

3 Fill out and sign the Disinfection Certificate, place it in the outer box of the return package.

List of accepted disinfectants

Bacilol plus

Mikrozid AF Liquid

Mikrozid AF Liquid

Sanisol S003

Dispatch (Cal. Tech. Industries, Inc.) Staphasept (Correlated Products, Inc.).

similar locally available disinfecting agents 1% sodium hypochlorite solution/bleach

Disinfecting products (instruments and parts)

Prerequisite equipment and materials

- 1% sodium-hypochlorite solution/bleach
- Distilled water
- 70% EtHO
- Lint free cloths
- Lab coat
- Lab gloves (powder free)
- Safety goggles

Instructions

1 Put on lab coat, gloves, and safety goggles.

2 If you would like to return an assembled device, complete the lettered steps below. If you would like to return one or more parts, skip to step 3 below.

- 1 Remove dust, splashes, and dry residue from the surface of the device.
- 2 Rinse all the tubing inside the device with a 1% sodium-hypochlorite solution. Make sure that no liquid remains inside the tubes when you pack the device for shipment.

- 3 Wipe the complete surface of the device with a lint-free cloth moistened with a 1% sodium-hypochlorite solution, and then dry it off with a lint-free cloth dipped in water. Skip to step 4 below.

3 If you would like to return one or more parts, complete the lettered steps below:

- 1 Separate electronic and sensitive optical components, and de-energize any PCBs.
- 2 Wipe the surfaces of electronic and optical components with a lint-free cloth moistened with a 70% ethanol solution.

⚠ Do not use sodium-hypochlorite on these parts, and never submerge PCBs in the ethanol solution.

- 3 Wipe the surfaces of non-electronic or non-optical components with a lint-free cloth moistened with a 1% sodium-hypochlorite solution, and then dry it off with a lint-free cloth dipped in water.

4 Dry all surfaces completely before you pack up the return shipment. Use the original packaging or the packaging of the replacement product. Pack PCBs in antistatic bags.

5 In the Disinfection Certificate, check the **I have completed the instructed disinfection steps for the return of the parts listed below.** checkbox. Fill out the Disinfection Certificate, and place it in the outer box of the return shipment.

Disinfection Certificate

I, _____

(First Name / Last Name) hereby certify that:

- I have completed the instructed disinfection steps for the return of the products listed below. ☐
- The products listed below have not come into contact with biohazardous body fluids. ☐

RMA No.:	S/N or part number:
	S/N or part number:
	S/N or part number:
	S/N or part number:
	S/N or part number:
	S/N or part number:

By accepting authorization to return this product, I assume all responsibility for disinfection and cleaning.

I accept that under no circumstances will any returned spare part or product be approved or accepted for return, without carrying a completed Disinfection Certificate. Furthermore there will be no reimbursement for such spare parts or products.

Date: _____ Signature: _____

Appendix 5 Support and ordering

77 Elektronika offers full service support for its products. Feel free to contact us if you encounter any problem with the LabUMat 2 device that consulting this manual does not or only partially solve. There are several channels of communication available, listed below.

By phone or e-mail

You can reach qualified 77 Elektronika service staff during office hours at the service hotline and the service staff e-mail address:

+36 1 371 0546

service@e77.hu

The number and the address are also listed on our website (www.e77.hu) in the 'For Distributors' section.

Through the Helpdesk

1 Sign in to the Helpdesk in the **For Distributors** section of our website (www.e77.hu) using your 77 Elektronika account user name and password.

2 Press the **New Issue** button in the top left corner.

3 In the **Description** text box, provide as much information about the problem as you can. You can also include photos or video clips that highlight the problem as attachments.

4 When you are done, press **Send** at the top of the screen. You will get status update notification e-mails as the reported issue is processed.

Ordering

You can order any replaceable part of the device directly from 77 Elektronika.



Do not use the phone to place orders. Use the written forms of contact listed below and always include the article number of the part or parts you are requesting.

Send your order by fax to

+36 1 206 1481

or by e-mail to

service@e77.hu.

Our service staff will get in touch with you to confirm your order as soon as possible.

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Appendix 7 Module firmware requirements

Device software version	Compatible Module PCB firmware
2.1.76 (Release date: April 2014)	Feeder: 2.4.17.0
	Washer: 2.3.14.0
	Linear robot: 2.6.9.0
	Rack transport: 2.3.19.0
	Strip timer & Photometer: 2.3.6.0
	Sample dispenser: 1.0.20.0
	Thermal printer: 2.0.2.0
	PMC: 2.1.18.0
	Measuring head: 2.1.0.20
	PC: 1.5.18.0



Laboratory					
Device serial number		Average throughput of the device:	1–150 measurements/day☐	150–300 measurements/day☐	300+ measurements/day☐
Servicing engineer					
Date					
Maintenance action		1–150 measurements	150–300 measurements	300+ measurements	✓
Feeder module Recommended frequency of maintenance					
Clean the feeder drum and the strip turner ramp	3 min.	every 6 months	every 4 months	every 2 months	
Grease the side of the feeder drum	2 min.				
Clean the strip orientation sensor	3 min.				
Adjust the feeder blade	5 min.	every 6 months	every 4 months	every 2 months	
Set up correct pipetting pos.	3 min.				
Test motor and sensor functions	5 min.	every 12 months	every 12 months	every 6 months	
Washer module Recommended frequency of maintenance					
Perform all listed adjustments	5 min.	every 6 months	every 4 months	every 2 months	
Replace the components in the maintenance kit	15 min.	every 12 months	every 12 months	every 6 months	
Linear robot module Recommended frequency of maintenance					
Check probe straightness and clear way inside rinse well	3 min.	every 6 months	every 4 months	every 2 months	
Adjust linear robot alignment	3 min.				
Test motor and sensor functions	5 min.	every 12 months	every 12 months	every 12 months	
Rack transport module Recommended frequency of maintenance					
Level the rack conveyor	2 min.	every 6 months	every 4 months	every 2 months	
Adjust the rack catch corrections	5 min.				
Clean the rack positioning unit timing belt	5 min.				
Clean the rack conveyor inside and outside	8 min.				
Clean and lubricate the rack conveyor timing belt	3 min.				
Test motor and sensor functions	5 min.	every 12 months	every 12 months	every 12 months	
Strip timer and photometer Recommended frequency of maintenance					
Adjust the pad-reading positions	6 min.	every 6 months	every 4 months	every 2 months	
Replace the components in the Maintenance kit	10 min.	every 12 months	every 12 months	every 12 months	
Test motor and sensor functions	5 min.				
Sample dispenser module Recommended frequency of maintenance					
Perform a fluid retention test	3 min.	every 6 months	every 4 months	every 2 months	
Replace the components in the Maintenance kit	2 min.	every 12 months	every 12 months	every 12 months	
Test motor and sensor functions	5 min.				
Power supply Recommended frequency of maintenance					
Clean the inside of the power supply with pressurized air	2 min.	every 12 months	every 12 months	every 6 months	
Embedded PC Recommended frequency of maintenance					
Perform a Disk cleanup, error checking, and defragmentation	10 min.	every 12 months	every 12 months	every 6 months	